Quantum Chaos to Thermodynamics

A course companion on eigenstate thermalization

Ze-Yu Xing

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Table of contents

Welcome 5
- The question that organizes this book . . . 5
- Who this is for . . . 5
- Reading routes . . . 5
- Conventions and the strength of claims . . . 6
- How to use the calculations . . . 6

I From dynamics to thermal states 7

1 States, observables, and equilibrium ensembles 8
- 1.1 The quantum state as a prediction rule . . . 8
- 1.2 Three ensembles and their assumptions . . . 9
- 1.3 Thermodynamic functions from a partition function . . . 9
- 1.4 Typicality: a kinematic fact with a dynamical limitation . . . 10
- 1.5 Exercises and worked answers . . . 11

2 Classical mechanics: regular motion, ergodicity, and chaos 12
- 2.1 Hamiltonian flow preserves volume . . . 12
- 2.2 Integrability and action-angle coordinates . . . 13
- 2.3 Ergodicity, mixing, and observable relaxation . . . 13
- 2.4 Instability and Lyapunov exponents . . . 14
- 2.5 Coarse observations and reversible dynamics . . . 15
- 2.6 Quantization changes the diagnostic . . . 15
- 2.7 Exercises and worked answers . . . 15

3 Random matrices and the fingerprints of quantum chaos 17
- 3.1 The three Gaussian ensembles . . . 17
- 3.2 Level repulsion from a two-dimensional calculation . . . 18
- 3.3 Unfolding and symmetry resolution . . . 19
- 3.4 Eigenvectors and matrix elements . . . 19
- 3.5 From classical dynamics to wavefunctions . . . 20
- 3.6 Exercises and worked answers . . . 20

4 Physical many-body Hamiltonians and eigenstate structure 22
- 4.1 Two useful model families . . . 22
- 4.2 Why symmetry reduction is physics . . . 23
- 4.3 Eigenstates in a physically chosen basis . . . 23
- 4.4 Local density of states and survival probability . . . 24
- 4.5 Entanglement adds a different diagnostic . . . 24
- 4.6 Exercises and worked answers . . . 25

5 Eigenstate thermalization and quantum equilibration 26
- 5.1 Exact unitary dynamics and the diagonal ensemble . . . 26
- 5.2 A rigorous variance bound . . . 26

5.3 The ETH ansatz . . . 27
5.4 How diagonal ETH produces thermal values . . . 27
5.5 Weak ETH, strong ETH, and observable choice . . . 28
5.6 The quantum ergodic theorem and its limits . . . 28
5.7 Exercises and worked answers . . . 29

6 Quantum quenches and numerical tests of thermalization 30
6.1 What a quench fixes . . . 30
6.2 Three stationary predictions . . . 30
6.3 What to measure in an exact-diagonalization study . . . 31
6.4 A reproducible small-chain illustration . . . 32
6.5 Recurrences, times, and the order of limits . . . 32
6.6 Why integrable quenches differ . . . 33
6.7 Exercises and worked answers . . . 33

II Thermodynamics from microscopic dynamics 35

7 Entropy, work, and thermodynamic structure 36
7.1 From amplitudes to populations . . . 36
7.2 What a master equation assumes . . . 37
7.3 Uniform populations: fixed point versus attractor . . . 37
7.4 Three entropies and a rigorous inequality . . . 38
7.5 Kelvin's principle from passivity . . . 38
7.6 When diagonal entropy becomes thermodynamic entropy . . . 39
7.7 Fundamental relation and first law . . . 40
7.8 Exercises and solutions . . . 40

8 Fluctuation theorems and energy diffusion 42
8.1 Measuring work in a closed quantum system . . . 42
8.2 Microscopic reversibility and Crooks' relation . . . 42
8.3 Work cumulants and dissipation . . . 44
8.4 A microcanonical relation and the eigenstate question . . . 44
8.5 Why a bath produces detailed balance . . . 45
8.6 Drift, diffusion, and a finite-size Einstein relation . . . 45
8.7 Two conserved quantities . . . 46
8.8 Exercises and solutions . . . 47

9 Response, fluctuations, and reciprocity 48
9.1 Linear response from the von Neumann equation . . . 48
9.2 Causality and dispersive response . . . 49
9.3 Spectral noise and detailed balance . . . 49
9.4 The fluctuation-dissipation theorem . . . 50
9.5 How an eigenstate can satisfy thermal FDT . . . 51
9.6 Two-observable ETH requires joint information . . . 52
9.7 Onsager–Casimir reciprocity . . . 52
9.8 Nonlinear relations from a fluctuation symmetry . . . 53
9.9 Exercises and solutions . . . 53

10 Driven systems and finite reservoirs 55
10.1 A particle in a moving chaotic cavity . . . 55
10.2 Solving the cavity diffusion model . . . 56
10.3 A kicked harmonic probe inside a finite system . . . 57

10.4 A transition in fluctuation growth . . . 58
10.5 Two finite bodies exchanging energy . . . 59
10.6 An exact finite-reservoir distribution . . . 60
10.7 Exercises and solutions . . . 61

III Constraints and long-lived memory **62**

11 Integrability and constrained equilibrium **63**
11.1 Maximum entropy with many constraints . . . 63
11.2 Free fermions: occupations, correlations, and dephasing . . . 64
11.3 Hard-core bosons and the Jordan–Wigner string . . . 65
11.4 Generalized eigenstate thermalization . . . 66
11.5 A transverse-field Ising quench . . . 66
11.6 Interacting integrability: the XXZ completeness problem . . . 67
11.7 Exercises and worked answers . . . 68

12 Prethermalization and kinetic relaxation **70**
12.1 Fast relaxation and slowly changing charges . . . 70
12.2 What second-order perturbation theory actually gives . . . 71
12.3 From many-body transitions to a fermionic collision integral . . . 71
12.4 An entropy theorem and the destination of collisions . . . 72
12.5 Time scales and the interpretation of a plateau . . . 73
12.6 Exercises and worked answers . . . 73

IV Work through it **75**

13 Mathematical tools and two revealing examples **76**
13.1 The kicked rotor: from a map to localization . . . 76
13.2 Zeta zeros: unfolding, evidence, and conjecture . . . 77
13.3 When the uniform state is an attractor . . . 79
13.4 A constructive proof of Birkhoff's theorem . . . 80
13.5 A rigorous passive-work inequality . . . 80
13.6 Deriving the energy drift–diffusion relation . . . 81
13.7 Deriving Onsager reciprocity . . . 82
13.8 Exercises and worked answers . . . 83

14 A reproducible numerical laboratory **84**
14.1 Reproduce before extending . . . 84
14.2 Experiment 1: level statistics with a known ensemble . . . 84
14.3 Experiment 2: exact dynamics of a small spin chain . . . 85
14.4 Experiment 3: classical phase-space structure . . . 87
14.5 Experiment 4: distributions from a continuum model . . . 87
14.6 A project that tests a claim . . . 88

Guide to the review **89**
Introduction, chaos, and eigenstates . . . 89
Thermodynamic laws and response . . . 89
Integrability and auxiliary derivations . . . 90

References **92**

Welcome

The question that organizes this book

An isolated quantum state evolves reversibly. A cup of coffee cools irreversibly. How can the second statement emerge from the first? This book develops the answer through observables, energy eigenstates, dephasing, conservation laws, and controlled approximations. The central bridge is the **eigenstate thermalization hypothesis**, or ETH: in suitable interacting systems, individual energy eigenstates already encode thermal predictions for simple observables.

This is an independent course companion to the review by Luca D'Alessio, Yariv Kafri, Anatoli Polkovnikov, and Marcos Rigol [1]. Its topic sequence supplies our backbone. The explanations, worked calculations, exercises, code, and figures here are original. This is neither an edition of the review nor a claim that its authors endorse the book. Read the original review for its historical synthesis and extensive literature survey.

Who this is for

You should know vectors, matrices, derivatives, elementary probability, and the Schrödinger equation. Chapter 1 develops the density matrix, partial trace, entropy, and the ensembles used later. No prior course in quantum chaos or nonequilibrium statistical mechanics is assumed. Keep a notebook: reproducing a short derivation usually teaches more than reading a long one.

The book moves from classical chaos and random matrices to many-body eigenstates and quenches; then from population dynamics to entropy, work fluctuations, response, and heating; finally to integrability, generalized ensembles, and prethermalization. The mathematical appendix supplies the auxiliary proofs, while the numerical lab makes the principal diagnostics reproducible in Julia. A section-by-section coverage map includes every numbered subsection and appendix of the review.

Reading routes

Goal	Route
Learn why an isolated system can thermalize	Chapters 1–6, then the numerical lab
Understand microscopic thermodynamics	Chapters 1, 5, and 7–10, then appendix proofs
Study failure of ordinary thermalization	Chapters 1, 5–6, and 11–12
Run a reading course	Read in order; solve each worked checkpoint before viewing its solution

A twelve-week course can assign Chapters 1–2 together, Chapters 3–12 one per week, and the numerical lab as a final project. The appendix is consulted alongside the relevant chapter. The full book is also available through the PDF download in the navigation bar.

Conventions and the strength of claims

We use $k_B = 1$ unless restored explicitly, so entropy is dimensionless and inverse temperature is $\beta = 1/T$. Chapters may set $\hbar = 1$ for a calculation and say so. N counts particles or sites, D is a Hilbert-space dimension, and $S(E) = \ln[\rho(E)\delta E]$ is a shell entropy; the reference window δE must be specified when an absolute entropy matters. An observable is a Hermitian operator. "Local" means supported on a bounded number of nearby sites as the total system grows.

Three types of statements recur:

- **Exact statements** follow from linear algebra, unitary dynamics, or explicitly stated probability assumptions. Examples are conservation of von Neumann entropy under unitary evolution and the two-measurement Jarzynski identity for a Gibbs initial state.
- **Conditional results** follow once assumptions such as ETH, nondegenerate energy gaps, local equilibration, weak driving, or a Markov approximation hold. The assumptions belong to the result.
- **Numerical observations and conjectures** include Wigner–Dyson behavior in many physical models, strong ETH in generic systems, and the random-matrix description of zeta zeros. Finite-size plots do not turn these into theorems.

Thermalization concerns a chosen class of observables and a stated time window. It does not mean that a pure global state becomes a mixed global Gibbs state. Spectral chaos, entanglement, equilibration, and transport are related diagnostics with distinct definitions. The chapters develop those distinctions before using them.

How to use the calculations

Each chapter includes worked reasoning and exercises with solutions. Attempt an exercise first, then compare assumptions as well as the final formula. The numerical lab specifies the Hamiltonian, seed, observables, and output files. Its figures illustrate finite systems and analytically specified models; they are not copied data from the review. The repository preserves both source and figure-generating code.

This first edition is a substantive teaching draft. Its coverage map describes topical coverage, not an assertion that every original numerical dataset or literature citation has been reproduced. Corrections can be reported through the repository's issue link, ideally with a chapter, equation, and a proposed correction.

Part I

From dynamics to thermal states

1 States, observables, and equilibrium ensembles

A gas in a sealed box obeys reversible microscopic equations. Nevertheless, its pressure becomes predictable without knowing every particle's position or every amplitude of its wavefunction. The central question of this book is how a small set of observables can become insensitive to most microscopic information. We begin by defining exactly which information an observable accesses and what an equilibrium prediction means.

1.1 The quantum state as a prediction rule

A normalized vector $|\psi\rangle$ describes a pure state. The probability of obtaining the result a when measuring $A = \sum_a a P_a$ is $\langle\psi|P_a|\psi\rangle$. A density matrix packages both pure and mixed preparations:

$$\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|, \qquad p_j \geq 0, \qquad \sum_j p_j = 1, \qquad \langle A\rangle_\rho = \mathrm{Tr}(\rho A). \tag{1.1}$$

A density matrix is positive, Hermitian, and has unit trace. Its decomposition into preparation vectors is generally not unique; predictions depend on ρ, not on a chosen decomposition. Purity, $\mathrm{Tr}\,\rho^2$, equals one for a pure state and is smaller for a mixed state. The von Neumann entropy, in units with $k_\mathrm{B} = 1$, is

$$S_\mathrm{vN}(\rho) = -\,\mathrm{Tr}(\rho \ln \rho). \tag{1.2}$$

Zero eigenvalues contribute zero because $x \ln x \to 0$ as $x \to 0^+$. Unitary evolution, $\rho(t) = U(t)\rho(0)U^\dagger(t)$, preserves every eigenvalue of ρ, hence preserves this entropy. An isolated pure state therefore never becomes a mixed Gibbs state as a statement about its full density matrix.

Why, then, can a subsystem become thermal? For $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$, the reduced state is $\rho_A = \mathrm{Tr}_B\,\rho$. In bases $|a\rangle|b\rangle$,

$$(\rho_A)_{aa'} = \sum_b \rho_{ab,a'b}, \qquad \mathrm{Tr}(\rho A \otimes I_B) = \mathrm{Tr}_A(\rho_A A). \tag{1.3}$$

For the entangled vector $(|00\rangle + |11\rangle)/\sqrt{2}$, the global entropy is zero but $\rho_A = I_A/2$ has entropy $\ln 2$. Entanglement allows local ignorance within a completely specified global state. No averaging over an observer's ignorance is needed for this example.

1.2 Three ensembles and their assumptions

Suppose energy is the only extensive conserved quantity besides fixed particle number and volume. An isolated equilibrium system with energy known within a window $\mathcal{W} = [E - \Delta/2, E + \Delta/2]$ is assigned

$$\rho_{\mathrm{mc}} = \frac{P_{\mathcal{W}}}{D_{\mathcal{W}}}, \qquad P_{\mathcal{W}} = \sum_{E_n \in \mathcal{W}} |n\rangle\langle n|, \qquad D_{\mathcal{W}} = \operatorname{Tr} P_{\mathcal{W}}. \tag{1.4}$$

The window should contain many levels while remaining narrow on the energy scale over which the target observable changes appreciably. The microcanonical prescription is a prediction rule whose dynamical justification remains to be established; it is not an extra term in the Schrödinger equation.

For a system exchanging energy with a large reservoir, the canonical state is

$$\rho_\beta = \frac{e^{-\beta H}}{Z}, \qquad Z = \operatorname{Tr} e^{-\beta H}, \qquad \beta = 1/T. \tag{1.5}$$

If particles can also be exchanged, $\rho_{\beta,\mu} = e^{-\beta(H-\mu N)}/\mathcal{Z}$, provided $[H, N] = 0$. Other exact charges require corresponding constraints. Throughout this book, comparing states from different conserved-charge sectors without specifying their weights is an avoidable source of confusion.

Here is a derivation of the Boltzmann factor. Let $H = H_A + H_B + H_{AB}$ and initially neglect the boundary interaction when counting levels. If A has energy ϵ_a, the number of compatible reservoir states is $\Omega_B(E_{\mathrm{tot}} - \epsilon_a)$. Therefore

$$p_a \propto e^{S_B(E_{\mathrm{tot}} - \epsilon_a)} \simeq e^{S_B(E_{\mathrm{tot}}) - \beta\epsilon_a}, \qquad \beta = \left.\frac{\partial S_B}{\partial E_B}\right|_{E_{\mathrm{tot}}}. \tag{1.6}$$

The next term is $-\epsilon_a^2/(2T^2 C_B)$ after choosing a suitable reference energy. A reservoir of large heat capacity suppresses this correction. The approximation also requires a coupling weak enough that the subsystem energy remains meaningful. At appreciable coupling, the reduced equilibrium state need not be $e^{-\beta H_A}/Z_A$ for the bare subsystem Hamiltonian.

The same canonical distribution follows by maximizing $-\sum_n p_n \ln p_n$ with fixed normalization and mean energy. A variation gives $\ln p_n = -1 - \alpha - \beta E_n$. This is a useful inference principle, but maximizing entropy by itself does not prove that a particular Hamiltonian carries an arbitrary preparation to that distribution.

1.3 Thermodynamic functions from a partition function

Differentiating the finite-dimensional partition function gives exact identities:

$$U = -\partial_\beta \ln Z, \qquad \mathrm{Var}_\beta(H) = \partial_\beta^2 \ln Z, \qquad C = \frac{\mathrm{Var}_\beta(H)}{T^2}. \tag{1.7}$$

For finite-range systems away from a critical point, both U and C usually scale as the number of sites L. Thus the energy width scales as $\sqrt{L}$ while the mean energy is extensive. The width in energy density, $\Delta e = \Delta E/L$, vanishes as $L^{-1/2}$. This is the scale separation behind much ensemble equivalence.

More explicitly, write the number of states near $E = Le$ as $\exp[Ls(e) + o(L)]$. The canonical sum is approximated by

$$Z(\beta) \approx \int \mathrm{d}e \, \exp\{L[s(e) - \beta e]\}. \tag{1.8}$$

If $s(e)$ is smooth and strictly concave near a unique maximum of $s(e) - \beta e$, a saddle point gives $s'(e_*) = \beta$ and a narrow distribution around e_*. A local observable smooth in e then has the same leading prediction in microcanonical and canonical ensembles. This argument assumes the thermodynamic limit and thermodynamic stability. Long-range interactions, first-order coexistence, and nonconcave entropy can obstruct simple versions of equivalence. Even when local observables agree, the global density matrices and energy fluctuations can differ greatly.

For L independent two-level systems with energies 0 and ϵ, $Z = (1 + e^{-\beta\epsilon})^L$ and the excited fraction is $q = (1 + e^{\beta\epsilon})^{-1}$. The microcanonical count with $M = Lq$ excitations is $\binom{L}{M}$. Stirling's formula gives $s(q) = -q \ln q - (1-q)\ln(1-q)$, so $\partial S/\partial E = \epsilon^{-1} \ln[(1-q)/q]$, recovering the same temperature. This solvable counting problem illustrates ensemble equivalence without assuming chaos.

1.4 Typicality: a kinematic fact with a dynamical limitation

Let a constrained subspace $\mathcal{H}_R$ have dimension D_R and define $\Omega = P_R/D_R$. Choose a unit vector uniformly with the unitary invariant Haar measure in this subspace. For an observable $A_R = P_R A P_R$,

$$\mathbb{E}\langle A\rangle_\psi = \frac{\mathrm{Tr}_R\, A_R}{D_R}, \qquad \mathrm{Var}(\langle A\rangle_\psi) = \frac{\mathrm{Tr}_R(A_R^2)/D_R - [\mathrm{Tr}_R\, A_R/D_R]^2}{D_R + 1}. \tag{1.9}$$

One obtains this by using $\mathbb{E}[(|\psi\rangle\langle\psi|)^{\otimes 2}] = (I + \mathsf{F})/[D_R(D_R+1)]$, where F swaps two copies of the subspace. The variance is at most $\|A\|^2/(D_R+1)$. Most vectors therefore give nearly the same answer for any fixed bounded observable when D_R is large. The canonical-typicality results of [2] establish corresponding closeness statements for a sufficiently small subsystem under appropriate dimension conditions; see also [3] for the energy-shell formulation.

This is a statement about measure on a space of vectors. A time-independent Hamiltonian keeps the absolute values of its energy-basis coefficients fixed, so a physical orbit never samples all Haar-random vectors. Moreover, easily prepared low-entanglement states are atypical. Typicality explains why thermal-looking pure states are abundant; it leaves open whether physical dynamics reaches them, and how quickly. Those questions motivate quantum chaos and ETH.

> **i** What does "thermal" mean here?
>
> Specify an observable class and an accuracy. A subsystem is thermal when its reduced state is close enough to an equilibrium state to reproduce that class. Equilibration means small deviations from a stationary value for most times. Thermalization additionally identifies that value with an equilibrium ensemble fixed by the conserved macroscopic data.

1.5 Exercises and worked answers

1. Entropy without mixing. For $|\psi\rangle = \sqrt{p}|00\rangle + \sqrt{1-p}|11\rangle$, calculate the purity and entropy of A.

Answer

Tracing B removes the cross terms: $\rho_A = \mathrm{diag}(p, 1-p)$. Therefore $\mathrm{Tr}\,\rho_A^2 = p^2 + (1-p)^2$ and $S_A = -p\ln p - (1-p)\ln(1-p)$. The global state remains pure for every p.

2. Finite reservoir correction. Show that $S_B''(E_B) = -1/(T^2 C_B)$ and identify when the quadratic correction in Equation 1.6 matters.

Answer

Differentiate $S_B' = 1/T$: $S_B'' = -(1/T^2)(\partial T/\partial E_B) = -1/(T^2 C_B)$. Expanding around the mean subsystem energy gives a correction of order $(\epsilon_a - \bar{\epsilon})^2/(T^2 C_B)$. It is small when the reservoir heat capacity greatly exceeds the subsystem energy-variance scale measured in units of T^2.

3. A zero-temperature caveat. Why does the saddle-point argument not automatically justify a canonical description of an arbitrary low-energy pure state?

Answer

It compares equilibrium ensembles under smoothness and large-deviation assumptions. It says nothing about a pure state's overlaps with eigenstates. Near a spectral edge, the number of participating levels may be small and the entropy expansion can fail. Dynamical thermalization requires additional information about the eigenstates and the preparation.

2 Classical mechanics: regular motion, ergodicity, and chaos

Classical chaos provides an instructive route from microscopic trajectories to equilibrium averages. It also provides warnings: instability of trajectories, exploration of an energy surface, and relaxation of an observable are different mathematical properties. We separate them before asking what survives quantization.

2.1 Hamiltonian flow preserves volume

For f canonical coordinate-momentum pairs, let $z = (q_1, \dots, q_f, p_1, \dots, p_f)$. Hamilton's equations are $\dot{q}_i = \partial H/\partial p_i$ and $\dot{p}_i = -\partial H/\partial q_i$. The Poisson bracket,

$$\{A, B\} = \sum_i \left(\frac{\partial A}{\partial q_i}\frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial B}{\partial q_i} \right), \qquad \frac{\mathrm{d}A}{\mathrm{d}t} = \{A, H\} \tag{2.1}$$

for an observable without explicit time dependence, makes conserved quantities transparent: I is conserved if $\{I, H\} = 0$.

The divergence of the phase-space velocity vanishes because mixed derivatives cancel:

$$\nabla_z \cdot \dot{z} = \sum_i \left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \right) = 0. \tag{2.2}$$

A transported probability density consequently obeys Liouville's equation, $\partial_t \rho + \{\rho, H\} = 0$. Along each trajectory ρ is constant. The fine-grained Gibbs entropy $-\int \rho \ln \rho \, \mathrm{d}z$ is conserved. Classical relaxation therefore cannot mean that the exact density is crushed onto a thermal attractor in phase space. Hamiltonian systems have no dissipative attractor of that kind.

For bounded motion, the normalized microcanonical measure is

$$\mathrm{d}\mu_E(z) = \frac{\delta(E - H(z))\,\mathrm{d}z}{\int \delta(E - H(z))\,\mathrm{d}z}. \tag{2.3}$$

The delta function includes a geometric weight: in coordinates on the energy surface it is the area element divided by $|\nabla H|$. Uniform measure on a plotted surface is therefore not automatically the microcanonical measure. A trajectory spends more time where it moves slowly.

2.2 Integrability and action-angle coordinates

A system with f independent conserved quantities in involution, $\{I_i, I_j\} = 0$, is Liouville integrable under suitable regularity assumptions. A compact, connected, regular common level set of these quantities is an f-dimensional torus. The action variables are

$$I_i = \frac{1}{2\pi} \oint_{\gamma_i} \sum_j p_j \, \mathrm{d}q_j, \qquad H = H(I), \qquad \dot{I}_i = 0, \qquad \dot{\theta}_i = \omega_i(I). \tag{2.4}$$

Each angle is periodic modulo 2π. Thus $\theta_i(t) = \theta_i(0) + \omega_i t$. A trajectory can densely fill a torus when its frequencies are rationally independent, but this torus has dimension f, smaller than the $2f - 1$ energy surface for $f > 1$. Being dense on the accessible invariant torus does not imply energy-surface ergodicity.

For two uncoupled oscillators, $H = \omega_1 I_1 + \omega_2 I_2$. Fixing total energy still leaves the energy division between oscillators undetermined. If the first oscillator initially carries nearly all the energy, it keeps that energy. A microcanonical average over all divisions cannot describe every trajectory. This simple failure is the classical precursor of the need for additional charges in integrable quantum systems.

Weakly coupling an integrable system does not immediately destroy all invariant tori. For nondegenerate systems and sufficiently small smooth perturbations, KAM theory guarantees persistence of a set of nonresonant tori. Near resonances the dynamics can instead form islands, separatrices, and chaotic regions. A realistic phase portrait may therefore be mixed. A single parameter does not turn every region of phase space from regular to chaotic at the same moment.

2.3 Ergodicity, mixing, and observable relaxation

A measure-preserving flow Φ^t is ergodic when invariant sets have measure zero or one. Equivalently, for suitable observables and almost every initial point,

$$\lim_{\tau \to \infty} \frac{1}{\tau} \int_0^\tau A(\Phi^t z) \, \mathrm{d}t = \int A \, \mathrm{d}\mu_E. \tag{2.5}$$

"Almost every" excludes sets of measure zero and says nothing about how fast the average converges. Ergodicity is a statement about infinite-time averages, while an experiment observes finite-time behavior.

Mixing is stronger. For square-integrable observables it requires decay of connected correlations:

$$C_{AB}(t) = \int A(\Phi^t z) B(z) \, \mathrm{d}\mu_E - \left(\int A \, \mathrm{d}\mu_E \right) \left(\int B \, \mathrm{d}\mu_E \right) \longrightarrow 0. \tag{2.6}$$

An initial density of the form $\rho_0 = (1 + \epsilon B)\rho_{\mathrm{mc}}$, with $\int B \, \mathrm{d}\mu_E = 0$, gives $\langle A(t) \rangle - \langle A \rangle_{\mathrm{mc}} = \epsilon C_{AB}(t)$. This identity connects a statistical property of the equilibrium flow with relaxation of a weak perturbation. A stretched and folded density can become indistinguishable from equilibrium to a smooth observable even though its fine-grained entropy is unchanged.

A useful exactly solvable example is free motion on a circle, $\theta(t) = \theta_0 + \omega t$. Continuous-time averages equal the uniform angular average for $\omega \neq 0$. Yet $A = e^{i\theta}$ has a correlation proportional to $e^{i\omega t}$, whose magnitude never decays. The motion is ergodic on the circle under the continuous flow but is not mixing. This example prevents conflating an average with relaxation.

2.4 Instability and Lyapunov exponents

For nearby initial conditions, linearize Hamilton's equations:

$$\delta\dot{z} = JH''(z(t))\delta z, \qquad \lambda = \lim_{t\to\infty} \frac{1}{t} \ln \frac{\|\delta z(t)\|}{\|\delta z(0)\|}, \quad J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}. \tag{2.7}$$

The limit is taken for infinitesimal perturbations with appropriate repeated rescaling; two finite trajectories cannot separate exponentially forever in a bounded region. A positive maximal exponent characterizes exponential sensitivity in typical chaotic trajectories. Symplectic dynamics pairs exponents as $+\lambda$ and $-\lambda$, preserving phase-space volume. Autonomous Hamiltonian flows also have zero exponents associated with flow and energy conservation in the usual nonsingular setting.

An inverted oscillator has exponential instability, but its unbounded escape is not the equilibrium chaos of a confined gas. Conversely, global transport can be slow in a chaotic region because narrow passages and approximate conservation laws restrict motion. A Lyapunov time measures loss of trajectory predictability; it need not equal the thermalization time.

For the area-preserving standard map,

$$p_{n+1} = p_n + K \sin\theta_n, \qquad \theta_{n+1} = \theta_n + p_{n+1} \pmod{2\pi}, \tag{2.8}$$

the Jacobian has determinant one. Small K leaves extensive regular structures; larger K produces substantial chaotic regions. The map is a convenient teaching model of stretching and transport, but claims about its global ergodicity need careful parameter and phase-space qualifications.

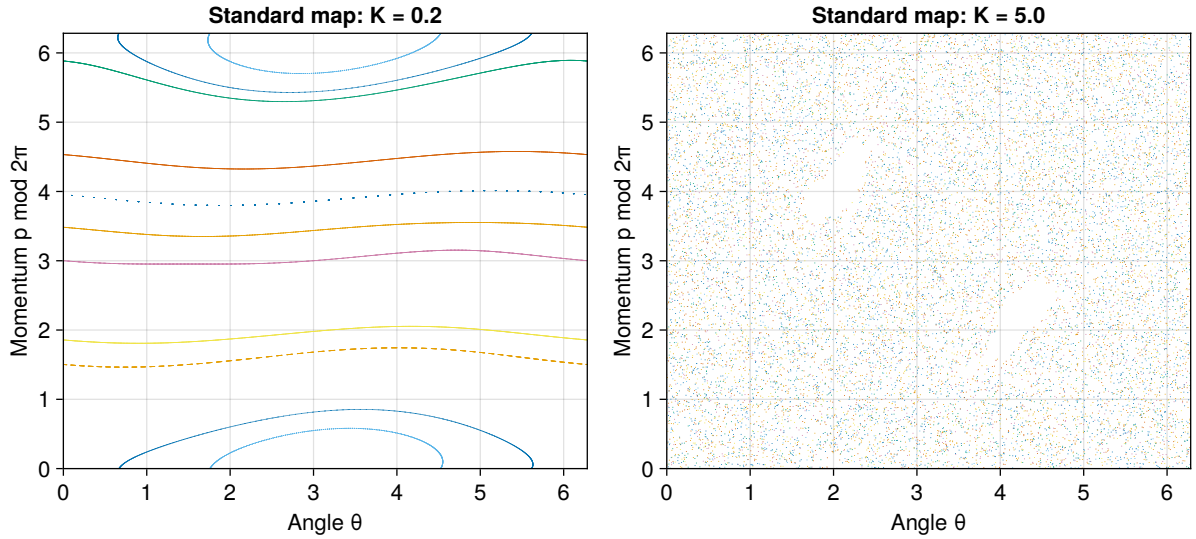


Figure 2.1: Original phase portraits of the standard map at $K = 0.2$ and $K = 5$. Nine initial conditions are iterated for 1,800 steps; the displayed momentum is reduced modulo 2π, giving a torus portrait. Regular islands and chaotic-looking regions can coexist. The code and preparation are documented in the numerical laboratory.

2.5 Coarse observations and reversible dynamics

Partition phase space into cells and retain only the probability in each cell. Stretching and folding can move a smooth initial distribution into narrow filaments crossing many cells. Cell averages then lose access to the detailed filaments. The entropy of that coarse description can increase while the fine-grained entropy remains constant. Its value depends on the resolution, and it need not increase monotonically for every initial distribution.

Reversing every momentum can retrace a time-reversal-invariant trajectory. Such a reversal also reconstructs the fine correlations needed to reverse the apparent entropy increase. This observation locates the origin of the arrow of relaxation in a special low-correlation preparation and restricted observations, together with dynamics that distributes information over inaccessible scales. It does not license replacing Hamilton's equations by irreversible evolution at the microscopic level.

2.6 Quantization changes the diagnostic

Semiclassical quantization of a regular torus gives, schematically, $I_i = \hbar(n_i + \alpha_i/4)$, where α_i is a Maslov index determined by turning points. Independent actions produce families of quantum numbers. In contrast, an ergodic classical orbit has no such global action labeling.

Two important proposals connect these structures to quantum spectra: generic integrable systems tend toward Poisson level statistics in the semiclassical limit [4], while fully chaotic systems tend toward the appropriate random-matrix statistics after resolving symmetries [5]. These are semiclassical universality proposals with assumptions and exceptions, not theorems covering every Hamiltonian. Arithmetic symmetries, degeneracies, mixed phase space, and the choice of limit matter.

Unitary dynamics preserves the overlap between two state vectors: $\langle\psi(t)|\phi(t)\rangle = \langle\psi(0)|\phi(0)\rangle$. Thus the classical definition based directly on exponentially separating state vectors cannot simply be copied. Quantum chaos instead uses spectra, eigenfunctions, matrix elements, and dynamical probes. The next chapter develops the spectral route.

2.7 Exercises and worked answers

1. Volume preservation. Verify that the standard map preserves area even though it can amplify a small perturbation.

Answer

Using input ordering (θ, p), the Jacobian is $\begin{pmatrix} 1 + K\cos\theta & 1 \\ K\cos\theta & 1 \end{pmatrix}$, whose determinant is one. An expanding direction can coexist with a contracting direction of reciprocal stretching factor. Area preservation does not bound the largest singular value by one.

2. An integrable microcanonical mismatch. For two identical harmonic oscillators at total energy E, compare the long-time value of the first oscillator's energy with the microcanonical prediction.

Answer

The trajectory preserves E_1, so its time average is the initially chosen E_1. Exchange symmetry gives $\langle E_1 \rangle_{\mathrm{mc}} = E/2$. The two agree only for a special energy division. Each invariant torus retains information absent from the energy-only ensemble.

3. Correlation versus average. Show explicitly why uniform rotation is not mixing.

Answer

With uniform measure and $A = e^{i\theta}$, $B = e^{-i\theta}$, both individual averages vanish. Their correlation is $\int e^{i(\theta+\omega t)} e^{-i\theta}\, \mathrm{d}\theta/(2\pi) = e^{i\omega t}$. It does not tend to zero, even though the time average of each nonzero Fourier harmonic vanishes.

3 Random matrices and the fingerprints of quantum chaos

A many-body Hamiltonian is highly structured: interactions are local, charges are conserved, and its entries are not independent random numbers. Random-matrix theory is useful because some fluctuations can forget that detailed structure. The aim is to identify which fluctuations are universal, after choosing the correct energy window and symmetry sector.

3.1 The three Gaussian ensembles

A Hermitian matrix can be diagonalized as $H = UEU^\dagger$. Gaussian invariant ensembles assign a probability density of the form $P(H) \propto e^{-c\,\mathrm{Tr}\,H^2}$, with c setting the energy scale. Invariance under basis rotations makes eigenvector orientations uniform and independent of the eigenvalues. The relevant rotation group depends on antiunitary symmetry:

Ensemble	Effective antiunitary structure in the sector	Dyson index β_{D}
Gaussian orthogonal (GOE)	An antiunitary symmetry with square $+1$ permits a real basis	1
Gaussian unitary (GUE)	No antiunitary symmetry acting within the sector	2
Gaussian symplectic (GSE)	An antiunitary symmetry squares to -1; count Kramers pairs once	4

The Dyson index is unrelated to inverse temperature. A physical time-reversal operator can exchange symmetry sectors instead of acting within one. Classification must therefore be done after the Hamiltonian is decomposed into irreducible symmetry blocks. Merely inspecting whether some terms look complex in a chosen basis is insufficient.

Changing variables from matrix entries to eigenvalues and eigenvectors produces a Jacobian:

$$P(E_1, \dots, E_D) \propto \exp\left[-c\sum_n E_n^2\right] \prod_{m<n} |E_m - E_n|^{\beta_{\mathrm{D}}}. \tag{3.1}$$

To see the origin of the product, make an infinitesimal change of eigenbasis, with $\mathrm{d}K = U^\dagger \mathrm{d}U$. In that basis, $U^\dagger \mathrm{d}H U = \mathrm{d}E + [\mathrm{d}K, E]$, so an off-diagonal variation has magnitude $|E_m - E_n|\,|\mathrm{d}K_{mn}|$. Each pair has one independent real component for GOE, two for GUE, and four for GSE after accounting for Kramers structure. The change of volume element therefore contributes one, two, or four powers of the gap for each pair. Invariance makes the remaining angular measure independent of E.

The product suppresses nearby eigenvalues. It can also be read as a logarithmically repelling gas confined by a quadratic potential. This effective gas is a probability model of spectra, not a claim that energy levels exert physical forces.

3.2 Level repulsion from a two-dimensional calculation

Consider a real symmetric 2×2 matrix,

$$H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \qquad E_+ - E_- = 2\sqrt{[(a-c)/2]^2 + b^2}. \tag{3.2}$$

GOE variances make $x = (a-c)/2$ and $y = b$ identically distributed independent Gaussians. In polar coordinates, the area element is $r\,\mathrm{d}r\,\mathrm{d}\phi$. Consequently the gap density is proportional to the gap near zero. After normalizing the mean gap to one, the result is the Wigner surmise,

$$P_{\mathrm{GOE}}(s) = \frac{\pi}{2} s\, e^{-\pi s^2/4}, \qquad P_{\mathrm{GUE}}(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}. \tag{3.3}$$

The GUE has three independent components controlling the gap, giving a volume element proportional to $r^2 \mathrm{d}r$. GSE has repulsion $P(s) \propto s^4$ at small spacing. The two-level formulas approximate large-matrix nearest-neighbor statistics very accurately but are not the exact large-D spacing laws.

For independent unfolded levels, the probability that an interval of length s contains no level is e^{-s}. Differentiating gives $P_{\mathrm{P}}(s) = e^{-s}$. Unlike GOE or GUE statistics, this Poisson process permits arbitrarily small spacings with finite probability density.

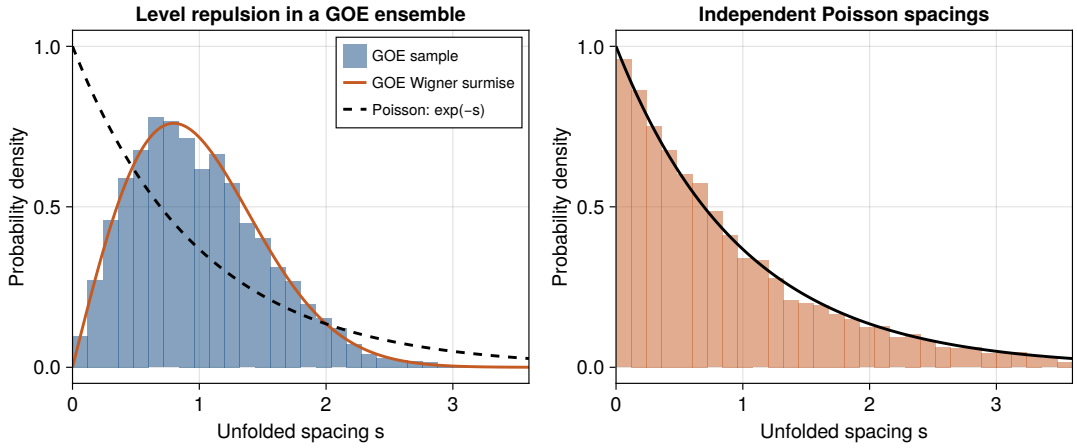


Figure 3.1: Original comparison of exact Poisson samples with 25 independently sampled GOE matrices of dimension 350, retaining the central 60 percent of each spectrum and unfolding with the semicircle density. The GOE curve is the Wigner surmise. Sampling and unfolding choices are documented in the numerical laboratory.

3.3 Unfolding and symmetry resolution

A spectrum generally has an energy-dependent mean density $\bar{\rho}(E)$. Define the smooth integrated counting function $\bar{N}(E) = \int^E \bar{\rho}(E')\,\mathrm{d}E'$ and unfolded levels $x_n = \bar{N}(E_n)$. Their mean spacing is approximately one. Using raw spacings across a broad band confuses the smooth density of states with fluctuations. Conversely, fitting the staircase too closely can remove the correlations one wants to measure.

Adjacent-gap ratios reduce this sensitivity:

$$\delta_n = E_{n+1} - E_n, \qquad r_n = \frac{\min(\delta_n, \delta_{n+1})}{\max(\delta_n, \delta_{n+1})} \in [0, 1]. \tag{3.4}$$

A slowly varying local density cancels between adjacent gaps. For independent exponential gaps the density is $P(r) = 2/(1+r)^2$, giving $\langle r \rangle = 2\ln 2 - 1$. GOE spectra instead have a larger mean, close to 0.53. [6] developed accurate random-matrix ratio distributions. Ratios still require symmetry resolution and an energy window: a sharp change of density over a few gaps can spoil the cancellation.

Independent symmetry sectors can cross without repulsion. Combining them produces excess small spacings even when each sector separately has GOE statistics. A correct numerical workflow first fixes particle number, momentum when translation applies, parity where compatible, and any additional exact charge. At special momenta, extra spatial symmetries often survive.

A further distinction is between the *mean* density and *fluctuations* around it. With entries normalized so that off-diagonal variance is v^2/D, the large-D Gaussian ensembles have the semicircle density

$$\bar{\rho}(E) = \frac{D}{2\pi v^2}\sqrt{4v^2 - E^2}, \qquad |E| < 2v. \tag{3.5}$$

Its integral is D, so it counts levels rather than probabilities. Many physical local Hamiltonians instead have an approximately Gaussian bulk density, reflecting sums of numerous local energy contributions. Agreement with GOE fluctuations does not require a semicircular physical density: unfolding removes that smooth profile. The figure uses semicircle unfolding because its matrices are themselves sampled from GOE. A physical-chain calculation requires a suitable smooth density for that chain.

3.4 Eigenvectors and matrix elements

For a uniformly distributed real unit vector with components u_i,

$$\mathbb{E}u_i^2 = \frac{1}{D}, \qquad \mathbb{E}u_i^4 = \frac{3}{D(D+2)}. \tag{3.6}$$

Thus a single component is approximately Gaussian with variance $1/D$ for large D. The scaled intensity $x = Du_i^2$ has the Porter–Thomas density $P(x) = e^{-x/2}/\sqrt{2\pi x}$. Complex Haar vectors instead have exponentially distributed scaled intensities in the large-D limit. Orthogonality and normalization correlate components; treating them as independent Gaussians is an approximation justified only for suitable low-order questions at large dimension.

Let A be a fixed real symmetric observable and remove its identity part, $A_c = A - (\operatorname{Tr} A/D)I$. Orthogonal invariance gives

$$\mathbb{E}A_{nn} = \frac{\operatorname{Tr} A}{D}, \qquad \operatorname{Var}(A_{nn}) = \frac{2\operatorname{Tr} A_c^2}{D(D+2)}. \tag{3.7}$$

The typical off-diagonal variance is asymptotically half this value. For a bounded local observable, $\operatorname{Tr} A_c^2$ is normally proportional to D, so fluctuations scale as $D^{-1/2}$ in amplitude. When D grows exponentially with system size, individual fluctuations are exponentially small. The trace part contributes equally to all diagonal elements and nothing off diagonal.

This calculation provides a seed for ETH, but a full random matrix loses the energy dependence of physical observables. It has no spatial geometry, propagation speed, transport law, or meaningful low-energy quasiparticles. A physical many-body observable typically has a frequency-dependent band profile in the energy basis, which the ETH ansatz restores.

3.5 From classical dynamics to wavefunctions

The Berry–Tabor proposal associates generic regular semiclassical systems with Poisson spectra, subject to nondegeneracy and genericity assumptions [4]. The Bohigas–Giannoni–Schmit proposal associates sufficiently chaotic semiclassical systems with the appropriate random-matrix correlations [5]. Neither says that a harmonic oscillator is Poisson or that every chaotic spectrum is GOE: exact degeneracies, arithmetic structure, and antiunitary symmetries matter.

Berry's random-wave picture provides a complementary eigenfunction hypothesis [7]. For a free particle well inside a two-dimensional chaotic billiard, build a wavefunction from many plane waves of equal wave number k with random phases and isotropically distributed directions. A central-limit argument makes the local amplitude approximately Gaussian. Angular averaging gives the illustrative correlator

$$\mathbb{E}[\psi^*(\mathbf{r})\psi(\mathbf{r}')] \simeq \frac{1}{V}\int_0^{2\pi} \frac{\mathrm{d}\phi}{2\pi} e^{ik\hat{\mathbf{n}}(\phi)\cdot(\mathbf{r}'-\mathbf{r})} = \frac{1}{V} J_0(k|\mathbf{r}'-\mathbf{r}|). \tag{3.8}$$

The energy constraint fixes the wavelength even though the phases are random. Boundary conditions, short periodic orbits, scars, and mixed dynamics produce corrections. A random wave is therefore an energy-constrained statistical model, richer than a vector whose position-space entries are all unrelated.

> **i** What the diagnostics establish
>
> Agreement with a spacing distribution is numerical evidence for a universality class over the tested scales. It does not prove eigenstate thermalization for every observable, nor set a relaxation time. Eigenvectors, operator matrix elements, and dynamics must be examined separately.

3.6 Exercises and worked answers

1. Normalize the GOE surmise. Starting from $P(s) = ase^{-bs^2}$, impose normalization and mean spacing one.

Answer

$\int_0^\infty s e^{-bs^2}\,\mathrm{d}s = 1/(2b)$ gives $a = 2b$. The mean condition gives $a\sqrt{\pi}/(4b^{3/2}) = 1$, hence $b = \pi/4$ and $a = \pi/2$.

2. Derive the Poisson ratio law. Take independent unit-mean exponential gaps x, y.

Answer

On the region $x \le y$, use $x = ry$, with Jacobian y. The two equivalent regions give $P(r) = 2\int_0^\infty y e^{-(1+r)y}\,\mathrm{d}y = 2/(1+r)^2$. Integrating $rP(r)$ from zero to one yields $2\ln 2 - 1$.

3. An identity operator test. Why must the identity part be subtracted in Equation 3.7?

Answer

For $A = aI$, every basis gives $A_{nn} = a$ and $A_{mn} = 0$ for $m \neq n$. A formula using $\operatorname{Tr} A^2$ alone would predict fluctuations that do not exist. Subtracting aI leaves $A_c = 0$, as required.

4 Physical many-body Hamiltonians and eigenstate structure

Random matrices explain universal fluctuations, but a many-body Hamiltonian has much more structure than an invariant ensemble. This chapter connects the two descriptions through concrete lattice models, symmetry sectors, and energy-constrained eigenstates. The distinction matters because a state may be spread over many basis vectors without giving thermal predictions.

4.1 Two useful model families

For spin-1/2 degrees of freedom on an open chain, consider

$$H_{\mathrm{Ising}} = J \sum_{j=1}^{L-1} \sigma_j^z \sigma_{j+1}^z + h_x \sum_{j=1}^{L} \sigma_j^x + h_z \sum_{j=1}^{L} \sigma_j^z. \tag{4.1}$$

Here σ^α has eigenvalues ± 1, and the Hilbert-space dimension is 2^L. The $h_z = 0$ transverse-field Ising chain is integrable: a transformation to fermions reduces it to independent quasiparticle modes. Generic nonzero transverse and longitudinal fields remove that free-particle solution. "Generic" excludes special limits and does not imply a theorem of ETH. With real couplings, complex conjugation is an antiunitary symmetry in the σ^z basis, so GOE is the relevant benchmark within a fully resolved real symmetry sector. This antiunitary operation need not coincide with physical reversal of every spin.

A particle-conserving example uses spinless fermions or hard-core bosons:

$$\begin{aligned} H = & -t \sum_j (c_j^\dagger c_{j+1} + \mathrm{h.c.}) + V \sum_j (n_j - \tfrac{1}{2})(n_{j+1} - \tfrac{1}{2}) \\ & - t' \sum_j (c_j^\dagger c_{j+2} + \mathrm{h.c.}) + V' \sum_j (n_j - \tfrac{1}{2})(n_{j+2} - \tfrac{1}{2}). \end{aligned} \tag{4.2}$$

For the usual uniform one-dimensional chain the nearest-neighbor limit connects to the integrable XXZ model; next-nearest-neighbor terms generically break integrability. Fixing particle number N gives dimension $\binom{L}{N}$, already exponential near half filling. Boundary conditions and particle statistics require care: fermionic hopping acquires occupation-dependent signs, and periodic hard-core bosons and fermions do not always share identical boundary sectors. Exact diagonalization studies such as [8] use these families to examine the crossover between integrability and chaos.

4.2 Why symmetry reduction is physics

A symmetry Q satisfying $[H, Q] = 0$ divides the Hilbert space into invariant subspaces. Diagonalization may be performed separately in each one. This both reduces computational cost and states which macroscopic information is being held fixed. Examples include total particle number, total magnetization, lattice momentum, and spatial reflection.

Two subtleties are common. First, noncommuting symmetry generators cannot in general all be fixed simultaneously. For example, reflection maps momentum k to $-k$ and preserves an individual momentum sector only at special momenta. Second, if the preparation occupies several sectors, its sector weights are conserved. An equilibrium comparison must keep those weights or justify why the chosen observable cannot distinguish them.

In the Ising chain, the spin-flip operator $\prod_j \sigma_j^x$ is a symmetry at $h_z = 0$ and is broken by a generic longitudinal field. Reflection remains for a spatially uniform open chain. A boundary perturbation can remove reflection, but one must then describe the perturbed Hamiltonian explicitly. Hidden unremoved symmetries can imitate integrable level statistics.

4.3 Eigenstates in a physically chosen basis

Write $H = H_0 + \lambda V$, where $H_0|\alpha\rangle = \epsilon_\alpha|\alpha\rangle$, and expand

$$|n\rangle = \sum_\alpha C_\alpha^{(n)}|\alpha\rangle, \qquad \sum_\alpha |C_\alpha^{(n)}|^2 = 1. \tag{4.3}$$

The basis should answer a physical question: occupation states reveal configuration-space structure; an integrable H_0 basis reveals hybridization between its eigenstates. No basis-independent notion of participation exists, because the energy basis makes every exact eigenstate occupy a single vector.

For weights $w_\alpha = |C_\alpha^{(n)}|^2$, useful measures are the Shannon entropy and inverse participation ratio:

$$S_{\mathrm{Sh}} = -\sum_\alpha w_\alpha \ln w_\alpha, \qquad \mathrm{IPR} = \sum_\alpha w_\alpha^2, \qquad \mathrm{PR} = \mathrm{IPR}^{-1}. \tag{4.4}$$

A uniform superposition over M vectors gives $S_{\mathrm{Sh}} = \ln M$ and $\mathrm{PR} = M$. A real Haar vector has mean IPR $3/(D+2)$, so a typical participation ratio is of order $D/3$. This is a benchmark for random amplitudes, not a universal target: physical states often fill only the part of the basis consistent with their energy.

A basis state $|\alpha\rangle$ has mean final-Hamiltonian energy $E_\alpha = \langle\alpha|H|\alpha\rangle$ and energy variance

$$\sigma_\alpha^2 = \langle\alpha|H^2|\alpha\rangle - E_\alpha^2 = \sum_{\gamma\neq\alpha} |H_{\gamma\alpha}|^2. \tag{4.5}$$

Only configurations connected by the Hamiltonian appear in this exact expression. Few-body interactions make the matrix sparse and constrain the accessible energy envelope. The energy-shell picture asks whether coefficients fluctuate randomly *within* that physically allowed envelope. It does not assume equal weight across the entire spectrum. [9] uses this viewpoint to connect eigenstate structure with relaxation.

Perturbation theory offers a useful local criterion. For an unperturbed state $|\alpha\rangle$, the leading amplitude on $|\gamma\rangle$ is approximately $\lambda V_{\gamma\alpha}/(\epsilon_\alpha - \epsilon_\gamma)$. A near resonance invalidates this expansion even if λ is small. Because many-body level spacings become very small with size, weak interactions can hybridize many states. Yet the relevant denominator is the spacing among states connected by the perturbation, together with its selection rules, rather than indiscriminately the smallest spacing in the entire Hilbert space. This distinction helps explain why local interactions, sparse connectivity, and conserved quantities leave structure absent from a full random matrix.

A smooth envelope and irregular internal coefficients play different roles. The envelope determines which energies and configurations are accessible; coefficient statistics describe fluctuations within that support. Increasing participation can initially reflect a widening envelope, before the internal coefficients resemble random variables. Comparing nearby eigenstates at the same energy and in the same physical basis helps separate these effects.

4.4 Local density of states and survival probability

The local density of states (LDOS), also called the strength function, of a preparation $|\psi_0\rangle$ is

$$F_{\psi_0}(E) = \sum_n |\langle n|\psi_0\rangle|^2 \delta(E - E_n). \tag{4.6}$$

Its moments are the energy moments in the initial state. Its Fourier transform is the return amplitude,

$$\mathcal{G}(t) = \langle\psi_0|e^{-iHt/\hbar}|\psi_0\rangle = \int \mathrm{d}E\, F_{\psi_0}(E) e^{-iEt/\hbar}, \qquad P_{\mathrm{surv}}(t) = |\mathcal{G}(t)|^2. \tag{4.7}$$

An ideal Gaussian LDOS of variance σ_E^2 gives $P_{\mathrm{surv}}(t) = e^{-\sigma_E^2 t^2/\hbar^2}$. An ideal Lorentzian of width Γ gives an exponential decay after Fourier transformation. A weak-coupling golden-rule regime can produce an approximately Lorentzian central profile, while stronger mixing may produce a roughly Gaussian envelope. Actual bounded spectra have edges and discrete levels, so neither ideal profile holds at all times.

A direct expansion, independent of either approximation, gives

$$P_{\mathrm{surv}}(t) = 1 - \frac{\sigma_E^2 t^2}{\hbar^2} + O(t^4) \tag{4.8}$$

for a finite system. The early quadratic law shows why a purely exponential survival probability cannot start at $t = 0$. It also shows that fast decay of a return probability is controlled by energy spread, which alone does not establish thermalization of a local observable.

4.5 Entanglement adds a different diagnostic

Divide the chain into regions A and B. The Schmidt decomposition is $|\psi\rangle = \sum_{a=1}^{\chi} \sqrt{\lambda_a}|a\rangle_A|a\rangle_B$, where $\lambda_a \geq 0$ and $\sum_a \lambda_a = 1$. Both reduced states share the nonzero eigenvalues λ_a, and

$$S_A = -\sum_a \lambda_a \ln \lambda_a = S_B, \qquad S_A \leq \ln \min(d_A, d_B). \tag{4.9}$$

Highly excited thermal eigenstates of many local chaotic systems have entropy proportional to subsystem volume when A is small compared with the whole system. At finite energy density the coefficient is the thermodynamic entropy density, under subsystem ETH assumptions. Ground states, localized eigenstates, and constrained subspaces can behave differently.

For a Haar-random complex pure state in an unconstrained product space with $d_A \le d_B$, the Page average is

$$\mathbb{E}S_A = \sum_{k=d_B+1}^{d_A d_B} \frac{1}{k} - \frac{d_A - 1}{2d_B} \simeq \ln d_A - \frac{d_A}{2d_B}, \tag{4.10}$$

where the final expression assumes large dimensions. The exact expression was proposed by [10] and proved in subsequent work, including [11]. The unrestricted Haar average is a useful high-temperature benchmark. Fixed energy, particle number, or other constraints change the appropriate random-state ensemble and its finite-size corrections. Agreement with an unrestricted Page estimate is not required at a general temperature.

Participation entropy and entanglement entropy ask different questions. A product of local equal superpositions occupies every computational basis vector but has no entanglement. Conversely, a state can have large bipartite entropy while preserving special correlations that make some observables nonthermal. No single scalar diagnoses all aspects of chaos.

4.6 Exercises and worked answers

1. Derive the shell width. Starting from $\langle\alpha|H^2|\alpha\rangle$, establish Equation 4.5.

Answer

Insert $I = \sum_\gamma |\gamma\rangle\langle\gamma|$ between the two Hamiltonians. Hermiticity gives $\langle\alpha|H^2|\alpha\rangle = \sum_\gamma |H_{\gamma\alpha}|^2$. The diagonal term is E_α^2, which cancels in the variance.

2. Delocalized but unentangled. Evaluate the computational-basis participation and bipartite entropy of $|+\rangle^{\otimes L}$, where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

Answer

Every configuration has weight 2^{-L}, so PR $= 2^L$ and $S_{\mathrm{Sh}} = L \ln 2$. The state factorizes across every spatial cut, giving $S_A = 0$. Computational-basis delocalization alone does not imply entanglement or thermal behavior.

3. Return plateau. Assuming nondegenerate energies, find the infinite-time average of $P_{\mathrm{surv}}(t)$.

Answer

Writing $p_n = |\langle n|\psi_0\rangle|^2$ gives $P_{\mathrm{surv}}(t) = \sum_{mn} p_m p_n e^{i(E_m - E_n)t/\hbar}$. Time averaging removes terms with $m \neq n$, leaving $\sum_n p_n^2$. Its inverse is the effective number of populated energy eigenstates. Energy degeneracies require combining weights within equal-energy subspaces.

5 Eigenstate thermalization and quantum equilibration

The eigenstate thermalization hypothesis makes a striking claim: for suitable observables, an individual energy eigenstate already carries the equilibrium equation of state. Time evolution is then needed to suppress observable coherences, while the equilibrium value is encoded in the eigenstates themselves. We first establish what dephasing proves exactly, and then identify the additional hypothesis that makes equilibrium thermal.

5.1 Exact unitary dynamics and the diagonal ensemble

Let $H|n\rangle = E_n|n\rangle$ and expand a pure initial state as $|\psi_0\rangle = \sum_n c_n|n\rangle$. For a Hermitian observable A, write $A_{mn} = \langle m|A|n\rangle$ and $p_n = |c_n|^2$. Its expectation is

$$A(t) = \sum_n p_n A_{nn} + \sum_{m\neq n} c_m^* c_n e^{i(E_m - E_n)t/\hbar} A_{mn}. \tag{5.1}$$

Define the infinite-time average $\overline{X} = \lim_{\tau\to\infty} \tau^{-1} \int_0^\tau X(t)\,\mathrm{d}t$. Since the time average of $e^{i\omega t}$ vanishes for nonzero ω, nondegenerate energies imply

$$\overline{A(t)} = A_{\mathrm{DE}} = \sum_n p_n A_{nn}, \qquad \rho_{\mathrm{DE}} = \sum_n p_n |n\rangle\langle n|. \tag{5.2}$$

If energies are degenerate, the exact averaged density matrix is instead $\overline{\rho} = \sum_E P_E \rho_0 P_E$: coherences within an equal-energy subspace remain. The diagonal ensemble is thus shorthand for dephasing in the appropriate energy blocks, not a physical stochastic projection during closed-system evolution.

The diagonal weights retain memory of the preparation. There is no general reason for ρ_{DE} to equal a Gibbs state. Furthermore, an average alone does not say that the signal stays near its average. A pendulum also has a time average while continuing to oscillate with finite amplitude.

5.2 A rigorous variance bound

Assume nonresonant nonzero gaps: $E_m - E_n = E_p - E_q$ with $m \neq n$ implies $(m, n) = (p, q)$. Squaring the fluctuating term in Equation 5.1 and averaging gives the exact identity

$$\overline{|A(t) - A_{\mathrm{DE}}|^2} = \sum_{m\neq n} p_m p_n |A_{mn}|^2. \tag{5.3}$$

Define the effective dimension $d_{\mathrm{eff}} = (\sum_n p_n^2)^{-1}$. Using $p_m p_n \leq (p_m^2 + p_n^2)/2$, Hermiticity, and $\sum_n |A_{mn}|^2 = \langle m|A^2|m\rangle \leq \|A\|^2$ yields

$$\overline{|A(t) - A_{\mathrm{DE}}|^2} \leq \|A\|^2 \sum_n p_n^2 = \frac{\|A\|^2}{d_{\mathrm{eff}}}. \tag{5.4}$$

Here $\|A\|$ is the operator norm. Subtracting a constant multiple of the identity may tighten the bound without changing the fluctuations. Chebyshev's inequality then bounds the long-time fraction of instants with $|A(t) - A_{\mathrm{DE}}| > \epsilon$ by $\|A\|^2/(d_{\mathrm{eff}}\epsilon^2)$. Degenerate gaps can be accommodated by multiplying suitable bounds by their maximal degeneracy, rather than silently using Equation 5.3.

This result establishes equilibration for many populated levels and a bounded observable under its spectral assumptions. It is not a proof of a Gibbs value, monotonic relaxation, or a practical equilibration time. It is also not necessary: an energy eigenstate has $d_{\mathrm{eff}} = 1$ but is exactly stationary. Rigorous work by [12] and [13] develops broad equilibration results with carefully stated preparation, spectral, and subsystem assumptions.

5.3 The ETH ansatz

For a local or few-body observable in a chaotic many-body system, the standard form is

$$A_{mn} = \mathcal{A}(\bar{E})\delta_{mn} + e^{-S(\bar{E})/2} f_A(\bar{E}, \omega) R_{mn}, \qquad \bar{E} = \frac{E_m + E_n}{2}, \qquad \omega = \frac{E_m - E_n}{\hbar}. \tag{5.5}$$

$\mathcal{A}(E)$ is a smooth equilibrium function; $S(E)$ is a thermodynamic entropy associated with the density of states in a fixed reference energy interval; f_A is a smooth frequency envelope; and R_{mn} describes irregular order-one fluctuations after averaging within small energy-frequency bins. The diagonal variance can differ from the off-diagonal variance by a symmetry-dependent factor. Hermiticity requires compatible conjugation properties, for example $f_A(\bar{E}, -\omega) = f_A(\bar{E}, \omega)^*$ in a convention where $R_{nm} = R_{mn}^*$.

The reference interval makes e^S dimensionless; a change of convention is absorbed into f_A. The actual matrix elements of a fixed Hamiltonian are deterministic. The ansatz models their local statistical behavior and does not assert that all R_{mn} are independent. Exact identities, conservation laws, and higher correlation functions constrain them. The early proposals of [14] and [15], followed by the matrix-element formulation in [16], provide the conceptual basis of ETH.

Why the exponential? The number of states in a many-body shell is $D \sim e^S$. Random-orientation estimates in Chapter 3 make individual matrix-element fluctuations of a bounded local observable scale as $D^{-1/2}$. Conversely, the spectral sum $\sum_m |A_{mn}|^2 = \langle n|A^2|n\rangle$ remains finite because approximately e^S small contributions add. Physical locality and dynamics determine the envelope f_A; a full invariant random matrix cannot provide that frequency structure.

5.4 How diagonal ETH produces thermal values

Suppose the preparation has mean energy $E_0 = \sum_n p_n E_n$ and variance $\sigma_E^2 = \sum_n p_n (E_n - E_0)^2$. Diagonal ETH gives

$$A_{\mathrm{DE}} = \sum_n p_n \mathcal{A}(E_n) + \sum_n p_n \delta A_n = \mathcal{A}(E_0) + \frac{1}{2}\mathcal{A}''(E_0)\sigma_E^2 + \cdots + \sum_n p_n \delta A_n. \tag{5.6}$$

The linear term vanishes by definition of E_0. For an intensive local observable, write $\mathcal{A}(E) = a(E/L)$. Then $\mathcal{A}''(E) = a''(e)/L^2$. If $\sigma_E^2 = O(L)$, the curvature correction is $O(1/L)$ where a is smooth. If eigenstate deviations are uniformly small in the occupied window, their weighted sum is also small regardless of the detailed weights p_n.

The microcanonical average over a narrow window tends to the same $\mathcal{A}(E_0)$. Under ensemble equivalence, a canonical state with matching mean energy also gives it. Thus the mechanism has three distinct ingredients: narrow energy density, smooth and thermal diagonal matrix elements, and sufficiently small observable coherences for most times. The diagonal density matrices themselves need never coincide.

Off-diagonal ETH also estimates temporal fluctuations. Near a fixed energy density, $|A_{mn}|^2 \sim e^{-S}|f_A|^2$ makes Equation 5.3 small. Nevertheless, correlations between initial amplitudes and matrix elements, low-frequency hydrodynamic structure, and conserved quantities affect the approach to equilibrium. The entropy factor alone does not determine a relaxation rate.

5.5 Weak ETH, strong ETH, and observable choice

Fix a sequence of energy-density windows and an observable. **Weak ETH** says that the fraction of eigenstates whose expectation differs appreciably from the thermal value goes to zero with increasing size. **Strong ETH** requires the maximum deviation in that window to vanish. Numerical standard deviations probe typical fluctuations; testing the largest outliers is a different task.

Weak ETH does not thermalize every preparation: a state can place appreciable weight on a vanishing fraction of exceptional eigenstates. Strong diagonal ETH removes that particular obstruction for narrow-energy preparations, but dephasing and the observation time still matter. Neither form is intended to apply to all operators. The projector onto a chosen eigenstate is deliberately constructed to distinguish that state, and a conserved operator has special matrix structure. The physically relevant class is typically local, few-body, or otherwise restricted by the measurement protocol.

5.6 The quantum ergodic theorem and its limits

One can describe macroscopic observations using a coarse decomposition $\mathcal{H}_R = \bigoplus_\nu \mathcal{H}_\nu$ with projectors P_ν and dimensions d_ν. Equilibrium weights are d_ν/D_R. Von Neumann's quantum ergodic theorem, with nonresonance and suitable large macrospaces, establishes normal behavior for typical relative orientations of these macrospaces and the Hamiltonian eigenbasis: every initial vector has macrospace weights close to equilibrium for most times [17].

The quantifiers are essential. This is a rigorous typicality statement about macroscopic decompositions, not a proof that every specified local Hamiltonian has thermal eigenstates. It does not supply an experimentally useful time scale or make the fine-grained entropy increase. ETH instead asks whether physically given observables have the required structure in a physically given Hamiltonian.

Integrable systems, localized systems, special nonthermal eigenstates, phase coexistence, and approximate conservation laws delimit ETH's use. They are not reasons to discard equilibration: a system can equilibrate to a constrained or preparation-dependent value. Later chapters examine these alternatives.

5.7 Exercises and worked answers

1. A nonthermal stationary state. Does stationarity imply thermalization?

Answer

No. Any energy eigenstate is stationary. In a Hamiltonian that violates diagonal ETH, a local expectation in an eigenstate can differ from the microcanonical result at the same energy. Stationarity establishes neither the thermal value nor initial-state independence.

2. Two-level coherence. For $c_1 = c_2 = 1/\sqrt{2}$, derive the temporal variance when the gap is nonzero.

Answer

The oscillation is $\mathrm{Re}[A_{12}e^{i(E_1-E_2)t/\hbar}]$ after absorbing the relative initial phase. Its time-averaged square is $|A_{12}|^2/2$. This agrees with Equation 5.3, which includes the ordered pairs $(1,2)$ and $(2,1)$, each weighted by $1/4$.

3. A curvature error. If $a(e) = a_0 + a_1 e + a_2 e^2$ and $\sigma_E^2 = cL$, what is the leading diagonal-ensemble shift from $a(E_0/L)$ when diagonal ETH is exact?

Answer

The shift is exactly $a_2\sigma_E^2/L^2 = a_2 c/L$. Finite energy width gives a correction even when every eigenstate obeys the smooth ETH function. This correction should not be mistaken for an ETH violation.

6 Quantum quenches and numerical tests of thermalization

A quantum quench prepares an out-of-equilibrium state without maintaining contact with a reservoir. One prepares a state using an initial Hamiltonian H_i, changes a parameter rapidly, and evolves with a final Hamiltonian H_f. The state is continuous across an ideal sudden quench; the Hamiltonian and its energy eigenbasis change. This clean separation makes quenches a direct test of the mechanism developed in Chapter 5.

6.1 What a quench fixes

Suppose $|\psi_0\rangle$ is an eigenstate, ground state, or experimentally prepared state associated with H_i. Immediately after the change,

$$E_0 = \langle\psi_0|H_f|\psi_0\rangle, \qquad \sigma_E^2 = \langle\psi_0|H_f^2|\psi_0\rangle - E_0^2, \qquad p_n = |\langle n_f|\psi_0\rangle|^2. \tag{6.1}$$

Energy need not be conserved during the switch because an external agent changes the Hamiltonian. After the switch, the entire final-energy distribution p_n is constant. Unitary evolution changes relative phases. A finite switching time is a different protocol and must be modeled if it is not short compared with the relevant dynamical time scales.

For a local Hamiltonian $H_f = \sum_j h_j$ and a product state or a state with rapidly decaying connected correlations,

$$\sigma_E^2 = \sum_{jk} (\langle h_j h_k\rangle - \langle h_j\rangle\langle h_k\rangle) = O(L). \tag{6.2}$$

The argument is that each j has only a bounded sum of appreciable connected correlations with other sites. It can fail for states with sufficiently long-range correlations. When it holds, the width in energy density vanishes even though exponentially many eigenstates can be occupied. This is the natural quench setting for diagonal ETH.

6.2 Three stationary predictions

For the final Hamiltonian, the diagonal prediction is

$$A_{\mathrm{DE}} = \sum_n p_n A_{nn}. \tag{6.3}$$

It retains the preparation and is exact as an infinite-time average when energies are nondegenerate. A microcanonical comparison uses

$$A_{\mathrm{MC}} = \frac{1}{D_\Delta} \sum_{|E_n - E_0| < \Delta} A_{nn}, \qquad D_\Delta = \#\{n : |E_n - E_0| < \Delta\}. \tag{6.4}$$

The symbol Δ here denotes the half-width of the chosen window. A canonical prediction is

$$A_{\mathrm{CE}} = \frac{\sum_n e^{-\beta E_n} A_{nn}}{\sum_n e^{-\beta E_n}}, \qquad \frac{\sum_n E_n e^{-\beta E_n}}{\sum_n e^{-\beta E_n}} = E_0. \tag{6.5}$$

The last equation defines the comparison temperature, rather than assuming the initial Hamiltonian's temperature survives the quench. For bounded spectra, energies above the infinite-temperature mean can require negative β. This is mathematically consistent because the partition sum remains finite. The existence of such states does not imply that all unbounded physical systems admit negative absolute temperature.

All three predictions must enforce the same exact charges. If the state has several symmetry-sector weights, use weighted sector ensembles or establish equivalence for the observables being compared. A canonical average over an unconstrained Hilbert space can fail for the elementary reason that it permits an incorrect particle number.

Even in a chaotic system, DE, MC, and CE are not identical at finite size. They sample different energy distributions. The curvature term in Equation 5.6 predicts part of this difference. Near a critical region, a spectral edge, or a small window containing only a few levels, finite-size differences can be substantial.

6.3 What to measure in an exact-diagonalization study

A transparent test begins with the complete Hamiltonian, boundary conditions, local-operator convention, and symmetry sector. Plot A_{nn} against energy density, and show the window occupied by the initial state. A smooth curve with decreasing scatter as size grows supports diagonal ETH. A single small-system plot is an illustration, not a thermodynamic-limit result.

Useful measurements include a root-mean-square deviation within a window, the largest deviation, neighboring-eigenstate differences, and binned off-diagonal variances as functions of mean energy and frequency. These answer different questions. The window must be narrow enough to remove smooth energy dependence and wide enough to contain many states. Subtracting a local smooth fit can help measure fluctuations, but the fitting procedure must be reported because overfitting can artificially suppress them.

For a quench, evaluate $A(t)$ and compare its late-time average with A_{DE} before using a thermal ensemble. Failure of the first comparison can reveal a time window too short for dephasing, numerical errors, or unresolved degeneracy. Agreement of DE with MC tests the thermal eigenstate mechanism for the *occupied* states. Agreement of MC with CE tests ensemble equivalence for the observable and system size.

The numerical work of [18] provided a direct lattice-system test connecting eigenstate expectation values to quench outcomes. Studies across integrability breaking, including [8], also show why spectral and eigenvector diagnostics should be examined together. These are results for specified model families and finite-size sequences, not blanket proofs for arbitrary interacting Hamiltonians.

6.4 A reproducible small-chain illustration

The numerical laboratory implements an open chain with $L = 9$ and

$$H = -\sum_{j=1}^{L-1} Z_j Z_{j+1} - 1.05 \sum_{j=1}^{L} X_j - \sum_{j=1}^{L} (0.35 + 0.07j) Z_j. \tag{6.6}$$

Here X_j, Z_j are Pauli matrices, and $\hbar = 1$. The varying longitudinal field removes reflection symmetry and provides a concrete interacting real Hamiltonian. The observable is the central spin Z_5. Among computational-basis product states with central spin $+1$, the script chooses the one whose mean energy is nearest the spectral mean. Its subsequent motion is an exact unitary calculation for this finite chain.

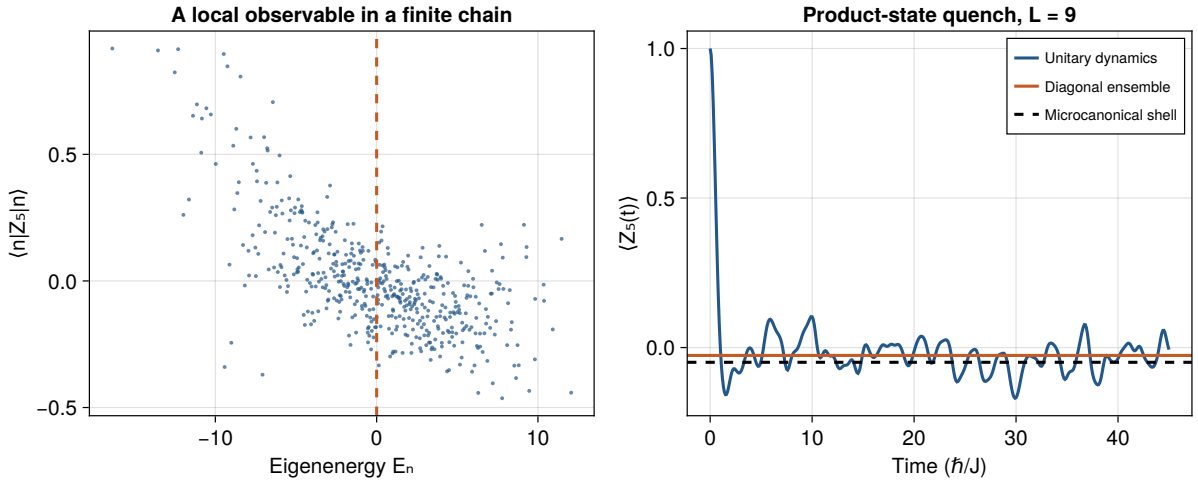


Figure 6.1: A small-chain illustration of eigenstate expectation values and quench dynamics. The Hamiltonian is Equation 6.6; the observable is the central Pauli Z. The horizontal comparisons are the exact diagonal ensemble and a microcanonical window with half-width 0.8 in the displayed energy units. The numerical laboratory gives the preparation rule, code, and verification.

Read this figure as a demonstration of the computation and of finite-size questions. To infer ETH scaling, repeat for several larger sizes with consistent energy-density windows and observables. To infer a relaxation law, specify the fitting interval and test its stability. No large-system conclusion follows from visually smooth behavior at $L = 9$.

6.5 Recurrences, times, and the order of limits

A finite isolated spectrum produces a quasiperiodic sum of phases. Arbitrarily close recurrences are possible; exact irreversible convergence of the full state cannot occur. Equilibration therefore means closeness of selected observables for most times, often after an initial transient, rather than convergence of every amplitude.

A finite-time average contains the exact kernel

$$\frac{1}{\tau} \int_0^\tau e^{i\omega t} \, \mathrm{d}t = e^{i\omega\tau/2} \frac{\sin(\omega\tau/2)}{\omega\tau/2}. \tag{6.7}$$

Very small gaps have not dephased when $\tau \ll 1/|\omega|$. However, an observable may equilibrate much earlier than the inverse smallest many-body gap because its relevant matrix elements and preparation weights need not resolve that extreme gap. Local transport, quasiparticle lifetimes, and approximate charges can dominate the experimentally relevant time.

The limits $L \to \infty$ and $t \to \infty$ can lead to different questions. Taking the infinite-time average at fixed finite L gives DE. Taking a thermodynamic limit first can support smooth relaxation laws and continuous spectra. A finite numerical study should state which quantity it estimates and avoid treating its recurrence time as a material relaxation time.

6.6 Why integrable quenches differ

At integrability, additional charges constrain p_n beyond energy. Nearby eigenstates can have different local properties even when their energy densities agree. The diagonal ensemble can then relax to a value different from MC or CE. A generalized Gibbs ensemble is designed to keep the additional locally relevant constraints; later chapters construct it explicitly.

Several caveats remain. Some observables in integrable models can still show thermal-looking values for particular preparations. A large effective dimension can make temporal fluctuations small even when the stationary value is nonthermal. Conversely, an approximately conserved quantity in a weakly nonintegrable model can produce a long prethermal plateau before eventual drift. The criterion for thermalization is a comparison of stationary local predictions under specified constraints, not whether a curve becomes visually flat.

6.7 Exercises and worked answers

1. Exact energy width in the lab preparation. For any Z-product state in Equation 6.6, find σ_E^2.

Answer

The ZZ and Z terms are diagonal in the preparation. Each transverse term flips a distinct spin, producing an orthogonal configuration with amplitude -1.05. By Equation 4.5, $\sigma_E^2 = L(1.05)^2$. The mean energy depends on the chosen spin configuration, but this variance does not.

2. Solving for temperature. Why is the canonical mean energy a monotone function of β, and when is the solution unique?

Answer

$\partial_\beta \langle H \rangle_\beta = -\operatorname{Var}_\beta(H) \leq 0$. If more than one energy has nonzero thermal weight, the variance is positive and the function is strictly decreasing. For a finite nontrivial spectrum, any energy strictly between the lowest and highest levels has a unique finite β in the unrestricted sector; endpoints require limiting temperatures.

3. A finite-window bias. Why need the mean energy of a symmetric microcanonical window around E_0 not equal E_0?

Answer

The density of states need not be symmetric. More levels may lie on one side of the center, shifting the equally weighted mean. This matters when comparing energy-dependent observables at finite width. Report both the chosen window and its actual mean energy, or reduce the width while retaining enough levels.

Part II

Thermodynamics from microscopic dynamics

7 Entropy, work, and thermodynamic structure

An isolated quantum state evolves reversibly. Thermodynamics nevertheless describes directed changes: a stirred fluid warms, an uncontrolled distribution spreads, and a cyclic engine cannot extract work from an equilibrium reservoir. The connection requires us to specify both the preparation and the information retained. This chapter first proves several finite-dimensional statements that require no chaos. It then identifies the additional assumptions needed to interpret them thermodynamically. We set Boltzmann's constant to one, so entropy is dimensionless and temperature has energy units.

7.1 From amplitudes to populations

Let a protocol change a Hamiltonian from H_i to H_f, with eigenvectors $|n,i\rangle$ and $|m,f\rangle$. Its propagator U obeys $U^\dagger U = UU^\dagger = I$. Suppose the initial density matrix is diagonal in the initial energy basis:

$$\rho_i = \sum_n p_n |n,i\rangle\langle n,i|.$$

The final energy populations are

$$q_m = \sum_n T_{mn} p_n, \qquad T_{mn} = |\langle m,f|U|n,i\rangle|^2. \tag{7.1}$$

Normalization of each column of U gives $\sum_m T_{mn} = 1$, while completeness of its rows gives $\sum_n T_{mn} = 1$. Thus T is **doubly stochastic**: every entry is nonnegative and both its row and column sums equal one. A matrix constructed this way is called unistochastic. Every unistochastic matrix is doubly stochastic; the converse fails in dimensions above two.

The assumption about the initial state matters. For a coherent preparation,

$$q_m = \sum_{n\ell} U_{mn} \rho_{n\ell} U^*_{m\ell},$$

and interference terms cannot generally be written as a fixed stochastic map acting on $p_n = \rho_{nn}$. For example, a Hadamard rotation transforms $(|0\rangle + |1\rangle)/\sqrt{2}$ into $|0\rangle$. The populations change from $(1/2, 1/2)$ to $(1, 0)$ and their Shannon entropy decreases. Nothing violates unitary dynamics: the initial phase supplied the missing information.

A more general quantum channel Φ is **unital** if $\Phi(I) = I$. For a channel that also preserves trace, the matrix

$$T_{mn} = \langle m|\Phi(|n\rangle\langle n|)|m\rangle$$

is doubly stochastic between any two orthonormal bases. Dephasing, random-unitary noise, and isolated unitary evolution are examples. Thermal amplitude damping is not unital; its preferred stationary state is generally Gibbs rather than uniform.

7.2 What a master equation assumes

A continuous population description takes the form

$$\dot{p}_n = \sum_{m \neq n} [w_{nm} p_m - w_{mn} p_n] = \sum_m L_{nm} p_m. \tag{7.2}$$

Here w_{nm} is the rate **from** m **to** n, $L_{nm} = w_{nm}$ for $n \neq m$, and $L_{mm} = -\sum_{n \neq m} w_{nm}$. Column sums of L vanish automatically. The evolution e^{Lt} is also doubly stochastic precisely when its row sums vanish:

$$\sum_{m \neq n} w_{nm} = \sum_{m \neq n} w_{mn}.$$

Symmetric rates $w_{nm} = w_{mn}$ are sufficient but stronger than necessary. One can have balanced circulation around a three-state ring without pairwise symmetry.

An exact isolated finite system does not automatically obey this equation. A Markov description discards memory and phase information. It may follow from weak perturbations, a separation between correlation and heating times, dephasing between kicks, or an explicit measurement protocol. Repeated coherent applications of U give U^r, whereas repeated population maps give T^r; these are different experiments. Before using a master equation, identify the physical mechanism making the discarded phases irrelevant.

7.3 Uniform populations: fixed point versus attractor

For D accessible levels, the vector $u_n = 1/D$ satisfies $Tu = u$. This is the infinite-temperature state in that finite Hilbert space, or within the conserved symmetry sector being considered. Being stationary does not make it an attractor.

For a fixed matrix T, convergence $T^r p \to u$ for all p follows if T is **primitive**: some positive integer power has strictly positive entries. In a finite Markov chain this is equivalent to irreducibility together with aperiodicity. Irreducibility prevents isolated communicating blocks; aperiodicity prevents persistent cycles. The swap matrix

$$T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

preserves the uniform state but alternates every nonuniform distribution forever. An identity matrix preserves every distribution. Symmetry blocks similarly retain their initial total probabilities, even if each block mixes internally. A connected continuous-time chain with finite rates avoids the discrete-time periodicity obstruction. The constructive contraction proof is in Section 13.3.

There is also a physical limit to the phrase infinite-temperature attractor. An unbounded spectrum has no normalizable uniform distribution. A particle repeatedly accelerated by a vibrating wall can heat forever. In periodically driven many-body systems, approximate conservation and long

prethermal intervals can delay heating; integrability, localization, or exact conserved quantities can obstruct the assumed mixing. The finite-matrix theorem applies only after its hypotheses are checked.

7.4 Three entropies and a rigorous inequality

The von Neumann entropy and the energy-basis diagonal entropy are

$$S_{\mathrm{vN}}(\rho) = -\operatorname{Tr}(\rho \ln \rho), \qquad S_d(\rho; H) = -\sum_n p_n \ln p_n, \quad p_n = \langle n|\rho|n\rangle. \tag{7.3}$$

The latter is the Shannon entropy of energy measurement outcomes. It depends on the chosen basis; at degeneracies one must specify whether individual states or degenerate energy subspaces are resolved. A unitary preserves S_{vN} exactly. Dephasing increases entropy, so $S_d \geq S_{\mathrm{vN}}$; equality holds when ρ is already diagonal in the specified basis. Diagonal entropy was developed as a useful thermodynamic quantity for isolated driven systems by [19].

For the initial diagonal preparations of Equation 7.1, concavity of $f(x) = -x \ln x$ gives an elementary proof:

$$\begin{aligned} S_d(q) &= \sum_m f\left(\sum_n T_{mn} p_n\right) \\ &\geq \sum_{mn} T_{mn} f(p_n) = \sum_n f(p_n) = S_d(p). \end{aligned} \tag{7.4}$$

The row sums allow Jensen's inequality; the column sums give the last equality. For an isolated protocol, this compares the endpoint populations when the initial state is diagonal. It does not establish that the instantaneous diagonal entropy is monotone at every time: coherence generated during the protocol can later be converted back into populations. Under genuinely Markovian doubly stochastic population evolution, the inequality applies to each time step and hence gives monotonicity.

A second proof uses relative entropy to the uniform vector:

$$D(p\|u) = \sum_n p_n \ln(p_n/u_n) = \ln D - S_d(p).$$

Stochastic maps contract relative entropy and preserve u, so $D(Tp\|u) \leq D(p\|u)$. This formulation extends naturally to Markov processes whose stationary distribution is nonuniform, although in that case Shannon entropy itself need not increase.

7.5 Kelvin's principle from passivity

Order the levels as $E_1 \leq \cdots \leq E_D$. A diagonal state is **passive** when $p_1 \geq \cdots \geq p_D$. Consider a cyclic protocol, so the final Hamiltonian equals the initial one. Define work delivered to the system by

$$W = \sum_n E_n(q_n - p_n).$$

Let $P_k = \sum_{n\le k} p_n$ and $Q_k = \sum_{n\le k} q_n$. For a doubly stochastic map,

$$Q_k = \sum_m a_m p_m, \quad 0 \le a_m \le 1, \quad \sum_m a_m = k.$$

Among all such coefficients the largest sum assigns unit weight to the k largest p_m. Therefore $Q_k \le P_k$. Summation by parts then yields

$$W = \sum_{k=1}^{D-1} (E_{k+1} - E_k)(P_k - Q_k) \ge 0. \tag{7.5}$$

Thus a cyclic operation cannot extract average work from a passive preparation. The conclusion needs neither ETH nor a large system. A positive-temperature Gibbs state is passive. A population-inverted state is not: swapping a more populated upper level with a less populated lower level extracts work. The stronger requirement that every number of independently prepared copies remain passive singles out equilibrium states under standard conditions; the mathematical theory is due to [20].

i Solved example: a mixed two-level rotation

Take levels $0, \epsilon$ and probabilities $(p, 1-p)$ with $p \ge 1/2$. A rotation with transition probability $r = \sin^2\theta$ has

$$T = \begin{pmatrix} 1-r & r \\ r & 1-r \end{pmatrix}.$$

The upper-level population becomes $1-p+r(2p-1)$, so $W = \epsilon r(2p-1) \ge 0$. The deviation from uniformity is multiplied by $1-2r$. Repetition converges for $0 < r < 1$, except at $r = 1$, when the two populations simply swap. Entropy increases even when a step carries the distribution through the uniform point to its opposite side.

7.6 When diagonal entropy becomes thermodynamic entropy

A thermodynamic entropy $S(E, \lambda)$ counts available states near energy E. Write the density of states as $\Omega(E, \lambda)$ and use a specified small energy width δE to define $S = \ln[\Omega \delta E]$. In a macroscopic short-range system, different reasonable shell widths change only subextensive terms.

A narrow energy distribution is necessary for a single temperature but is **not sufficient** to make S_d equal $S(E, \lambda)$. A single eigenstate has $S_d = 0$, even when ETH gives thermal local observables and $S(E, \lambda)$ is extensive. Thermodynamic entropy counts compatible states; diagonal entropy records probabilities of the actual energy measurement preparation. Do not identify them for every state.

A useful sufficient condition is a nonsparse, smooth population distribution occupying exponentially many levels in a narrow shell, with no extensive information beyond energy. Partition the spectrum into bins a with g_a states and probabilities q_a. If populations are nearly uniform inside each bin,

$$S_d = -\sum_a q_a \ln q_a + \sum_a q_a \ln g_a.$$

When the distribution has width $\sigma_E = o(N)$, the first term is subextensive, the entropy density is smooth across the occupied region, and within-bin nonuniformity has subextensive relative entropy,

this becomes $S_d = S(\bar{E}, \lambda) + o(N)$. Chaos supports equilibration of simple observables and often the required smooth population statistics after ordinary preparations; it does not logically force this condition for arbitrarily engineered states. Numerical examples of diagonal entropy after quenches are studied in [21].

7.7 Fundamental relation and first law

Define the equilibrium generalized force and temperature by

$$F = -\langle \partial_\lambda H \rangle_{\mathrm{eq}}, \qquad \beta = \frac{1}{T} = \left(\frac{\partial S}{\partial E} \right)_\lambda .$$

For a Gibbs distribution $p_n = e^{-\beta E_n}/Z$, normalization gives

$$\mathrm{d}S = -\sum_n \mathrm{d}p_n \ln p_n = \beta \sum_n E_n \mathrm{d}p_n .$$

Meanwhile,

$$\mathrm{d}E = \sum_n E_n \mathrm{d}p_n + \sum_n p_n \mathrm{d}E_n .$$

Hellmann–Feynman gives $\mathrm{d}E_n = \langle n | \partial_\lambda H | n \rangle \mathrm{d}\lambda$; hence

$$\mathrm{d}E = T\mathrm{d}S - F\mathrm{d}\lambda. \tag{7.6}$$

This is a differential identity on the equilibrium state manifold. Through ensemble equivalence and ETH, the force can also be measured in a suitable narrow-energy chaotic state. Finite systems can require corrections: surface and enclosed-volume microcanonical entropies need not give identical derivatives. For the volume $\mathcal{V}(E, \lambda) = \mathrm{Tr}\, \Theta(E - H)$, direct differentiation gives $\partial_\lambda \mathcal{V} = \Omega F_{\mathrm{mc}}$ and $\partial_E \mathcal{V} = \Omega$, making the analogous volume-entropy identity exact when its derivatives are well defined.

The **first law** instead expresses energy bookkeeping along an actual process:

$$\mathrm{d}E = \delta Q + \delta W_{\mathrm{on}}.$$

For an externally driven isolated quantum system, exact unitary dynamics gives $\dot{E} = \langle \partial_\lambda H \rangle \dot{\lambda}$ and $\delta Q = 0$. It may produce diagonal entropy while doing dissipative work. Only for a reversible quasistatic equilibrium path can one identify $\delta Q_{\mathrm{rev}} = T\mathrm{d}S$ and $\delta W_{\mathrm{on,rev}} = -F\mathrm{d}\lambda$. Substituting actual nonequilibrium work into an equilibrium identity would erase entropy production by assumption.

7.8 Exercises and solutions

Exercise 1. Show explicitly why a zero-temperature relaxation channel need not increase diagonal entropy.

Solution

For probabilities (p_0, p_1), full relaxation gives $(1, 0)$. Starting from $(1/2, 1/2)$ reduces Shannon entropy from $\ln 2$ to zero. Its population matrix has both columns equal to $(1, 0)^T$; columns sum to one but rows do not. The environment receives the discarded information and energy.

Exercise 2. A three-level passive state has energies $(0, \epsilon, 3\epsilon)$ and populations $(0.6, 0.3, 0.1)$. What work is required by the permutation exchanging levels 1 and 3?

Solution

Initial energy is 0.6ϵ. Final populations are $(0.1, 0.3, 0.6)$ and final energy is 2.1ϵ. Work into the system is 1.5ϵ. The two cumulative deficits are both 0.5; Equation 7.5 gives $\epsilon(0.5) + 2\epsilon(0.5) = 1.5\epsilon$.

Exercise 3. Explain why local ETH and zero diagonal entropy are compatible.

Solution

In a single energy eigenstate, energy measurement has one certain outcome, so $S_d = 0$. Local observables depend on a small reduced density matrix, which can be close to a thermal state because the eigenstate is entangled with the rest. Local thermality does not imply a mixed global preparation.

8 Fluctuation theorems and energy diffusion

The second law constrains averages. Fluctuation relations explain how individual realizations can return energy to a drive while the average still dissipates. Their proofs expose an important distinction: exact identities based on preparation and microscopic reversibility do not require quantum chaos, whereas replacing a thermal ensemble by one representative eigenstate requires additional structure. We retain $k_B = 1$ and count work as positive when delivered to the system.

8.1 Measuring work in a closed quantum system

For a protocol $H_F(t)$ on $0 \le t \le \tau$, let $H_i = H_F(0)$ and $H_f = H_F(\tau)$. In the **two-projective-measurement** (TPM) protocol, measure the initial energy E_n^i, evolve with U_F, and measure the final energy E_m^f. The measured work is the number

$$W_{mn} = E_m^f - E_n^i.$$

Work here labels a pair of outcomes; it is not generally the spectrum of one Hermitian operator. An initial energy measurement dephases a coherent state, so the resulting statistics describe that measurement procedure. A Gibbs initial state is already energy diagonal and is unaffected by this dephasing.

Write

$$p_F(n,m) = \frac{e^{-\beta E_n^i}}{Z_i} |\langle m, f|U_F|n, i\rangle|^2. \tag{8.1}$$

For degeneracies, one may resolve states within each energy eigenspace or work directly with spectral projectors and their multiplicities. The projector formulation makes the same identities without choosing an arbitrary basis inside a degenerate space. An experimentally broadened histogram should retain enough resolution to compare corresponding forward and reverse bins.

8.2 Microscopic reversibility and Crooks' relation

Let Θ denote the antiunitary time-reversal operator. The reversed experiment uses

$$H_R(t) = \Theta H_F(\tau - t)\Theta^{-1}, \qquad U_R = \Theta U_F^\dagger \Theta^{-1}.$$

All time-reversal-odd control parameters, including magnetic fields, must also reverse. Antiunitarity then implies equality of conditional transition probabilities:

$$|\langle m, f|U_F|n, i\rangle|^2 = |\langle \Theta n, i|U_R|\Theta m, f\rangle|^2.$$

Prepare the reverse experiment in the Gibbs state of the reversed final Hamiltonian. Dividing the two joint probabilities cancels the transition factor:

$$\frac{p_F(n,m)}{p_R(\Theta m,\Theta n)} = \frac{e^{-\beta E_n^i}/Z_i}{e^{-\beta E_m^f}/Z_f} = e^{\beta(W_{mn}-\Delta F)}, \qquad \Delta F = -\beta^{-1}\ln(Z_f/Z_i).$$

Summing over pairs with the same work gives

$$\frac{P_F(W)}{P_R(-W)} = e^{\beta(W-\Delta F)}. \tag{8.2}$$

Ratios are understood where the corresponding probabilities have support; distributional versions avoid dividing delta functions. This is the work fluctuation relation associated with [22]. Quantum TPM formulations were developed, among others, by [23]. No ETH assumption entered the proof.

Multiplication by $e^{-\beta W}$ and integration over W immediately yields

$$\langle e^{-\beta W}\rangle_F = e^{-\beta\Delta F}. \tag{8.3}$$

The nonequilibrium work equality was introduced by [24]. For an isolated quantum system it also follows directly:

$$\begin{aligned}\langle e^{-\beta W}\rangle &= \frac{1}{Z_i}\sum_{mn} e^{-\beta E_m^f}|U_{mn}|^2 \\ &= \frac{1}{Z_i}\sum_{m} e^{-\beta E_m^f} = \frac{Z_f}{Z_i}.\end{aligned}$$

The last step needs the row normalization of the transition matrix; it does not need time-reversal symmetry. Thus the equality is also valid for a trace-preserving unital channel between the two energy measurements, whereas the stated forward/reverse ratio requires a suitable reverse dynamics.

Jensen's inequality, $\langle e^X\rangle \geq e^{\langle X\rangle}$, gives $\langle W\rangle \geq \Delta F$ for $\beta > 0$. This average bound allows individual realizations with $W < \Delta F$. The exponential average weighs rare small-work events strongly, which can make estimating free-energy differences from finite samples difficult.

Solved example: sudden reversal of a two-level field

Let $H_i = (\epsilon/2)\sigma_z$ and $H_f = -H_i$, with an instantaneous switch, so $U = I$. A state initially in the lower level receives work $+\epsilon$ and the upper state receives $-\epsilon$. The partition functions agree. Their probabilities are

$$p_- = \frac{e^{\beta\epsilon/2}}{2\cosh(\beta\epsilon/2)}, \qquad p_+ = \frac{e^{-\beta\epsilon/2}}{2\cosh(\beta\epsilon/2)}.$$

Consequently $p_- e^{-\beta\epsilon} + p_+ e^{\beta\epsilon} = 1$. The mean work is $\epsilon\tanh(\beta\epsilon/2) > 0$, although negative work occurs with probability p_+. The two outcomes, including the rare one, are essential for the equality.

8.3 Work cumulants and dissipation

The logarithm of the work moment-generating function separates Gaussian and non-Gaussian contributions. If the cumulants κ_r exist in the required convergence domain, taking the logarithm of Equation 8.3 gives

$$-\beta\Delta F = -\beta\kappa_1 + \frac{\beta^2}{2}\kappa_2 - \frac{\beta^3}{6}\kappa_3 + \cdots.$$

Consequently,

$$\langle W\rangle - \Delta F = \frac{\beta}{2}\operatorname{Var} W - \frac{\beta^2}{6}\kappa_3 + \cdots.$$

The frequently used Gaussian relation between mean dissipated work and its variance is a truncation of this hierarchy. Small drive amplitude can control such a truncation over a specified protocol, but an apparently Gaussian central region of a histogram does not control its rare tails. Those tails are precisely where the exponential average can be most sensitive. For distributions whose moment-generating function is not analytic over the relevant interval, the exact equality should be used directly rather than through a formal cumulant series.

Crooks' relation also gives an information-theoretic measure of irreversibility. Multiply the logarithm of Equation 8.2 by $P_F(W)$ and integrate:

$$D(P_F(W)\|P_R(-W)) = \beta(\langle W\rangle - \Delta F) \geq 0.$$

Thus dissipated work measures how well the forward work histogram can be distinguished from the reflected reverse histogram. At the level of full trajectories, additional observed variables can retain more information than a work histogram, but under the stated work relation this particular relative entropy has exactly the displayed value. Equality means that the two work distributions coincide almost everywhere. The conclusion concerns the ensemble of outcomes, not a prohibition on any individual negative-dissipation event.

8.4 A microcanonical relation and the eigenstate question

An energy shell a of H_i has projector Π_a^i and dimension g_a^i. A shell b of H_f similarly has Π_b^f and dimension g_b^f. Uniform preparation within a shell gives

$$K_F(b|a) = \frac{1}{g_a^i}\operatorname{Tr}(\Pi_b^f U_F \Pi_a^i U_F^\dagger).$$

Cyclicity of the trace and the reversed propagator imply the exact finite-shell identity

$$g_a^i K_F(b|a) = g_b^f K_R(a|b). \tag{8.4}$$

For narrow shells of the same width, let E and $E+W$ denote their energies. With $S = \ln(\Omega\delta E)$, the continuum form is

$$\frac{P_F(W|E)}{P_R(-W|E+W)} = \exp[S_f(E+W) - S_i(E)]. \tag{8.5}$$

This is a density-of-states relation, not a canonical work theorem with a fixed temperature. The exact quantum microcanonical formulation is discussed in [25]. For a cyclic protocol and work small on the scale over which temperature varies,

$$S(E+W)-S(E)=\beta(E)W+\frac{1}{2}\beta'(E)W^2+\cdots, \qquad \beta'=-\frac{1}{T^2C}.$$

The curvature term is a finite-heat-capacity correction. Even for macroscopic E, it matters when work fluctuations become comparable to $T\sqrt{C}$. One should not claim a fixed-β Jarzynski equality for every sharply prepared finite-energy state: the reverse distribution in Equation 8.5 starts at the work-dependent energy $E+W$.

Can one replace the uniform shell by one eigenstate? Ordinary diagonal ETH controls expectation values of simple observables. A full work histogram involves $U_F^\dagger \Pi_b^f U_F$, which can be highly nonlocal. Therefore ordinary ETH alone is insufficient for arbitrary protocols. A single-eigenstate extension is justified when the **specific coarse-grained transition kernel** self-averages over nearby eigenstates, as can occur for weak, physically simple driving in a chaotic system. Under that additional assumption the shell identity holds approximately for a representative eigenstate, with errors from finite shell width, finite size, and residual fluctuations. It is a controlled expectation in specified regimes, not an exact theorem for every eigenstate and every unitary.

8.5 Why a bath produces detailed balance

Consider a small system with levels ϵ_n coupled weakly to a much larger bath. If the small system changes from m to n, the bath loses $\Delta=\epsilon_n-\epsilon_m$. Pairwise microscopic transition matrix elements are related by reversibility; the number of accessible bath final states supplies the bias:

$$\frac{w_{nm}}{w_{mn}} \simeq \frac{\Omega_B(E_B-\Delta)}{\Omega_B(E_B)} \simeq e^{-\beta_B\Delta}. \tag{8.6}$$

The second step neglects bath entropy curvature and assumes its temperature is stable. Substitution shows $w_{nm}p_m^{\mathrm{eq}}=w_{mn}p_n^{\mathrm{eq}}$ for $p_n^{\mathrm{eq}}\propto e^{-\beta_B\epsilon_n}$. This is **detailed balance**, which forbids stationary probability currents on individual links. A Gibbs stationary distribution alone is a weaker condition and does not establish pairwise balance.

In a quantum weak-coupling derivation, the same factor comes from the bath's equilibrium correlation spectrum and its KMS relation. The population master equation additionally requires suitable short bath memory and a secular approximation or other control of coherences. Strong coupling can change both the effective equilibrium state and what energy is assigned to the system.

8.6 Drift, diffusion, and a finite-size Einstein relation

Suppose many weak drive cycles produce energy increments ΔE with short memory. Per unit coarse-grained time define

$$A(E)=\lim_{\delta t\to 0}\frac{\langle \Delta E|E\rangle}{\delta t}, \qquad B(E)=\lim_{\delta t\to 0}\frac{\langle (\Delta E)^2|E\rangle}{\delta t}.$$

The limit is a coarse-grained diffusion limit, not the microscopic short-time limit of coherent Schrödinger evolution. Higher conditional moments must be negligible at the retained resolution. Expanding the gain and loss terms of an energy jump process to second order gives

$$\partial_t P = -\partial_E(AP) + \frac{1}{2}\partial_E^2(BP) = -\partial_E J, \quad J = AP - \frac{1}{2}\partial_E(BP). \tag{8.7}$$

Here $P(E)\mathrm{d}E$ is probability in an energy interval, **not** probability per eigenstate. Doubly stochastic evolution preserves equal probabilities per state, which corresponds to $P_\infty(E) \propto \Omega(E)$. If energy boundaries carry no probability current, stationarity gives

$$0 = A\Omega - \frac{1}{2}\partial_E(B\Omega), \qquad 2A(E) = B(E)\beta(E) + \partial_E B(E). \tag{8.8}$$

This is an Einstein relation for energy space. It follows within the diffusion approximation from preservation of the density-of-states measure and zero stationary current. Reversibility provides another route through the local jump relation. The derivative term must be retained for small systems or strongly energy-dependent noise. Discarding it is a thermodynamic approximation, often of relative order $1/N$ for smooth extensive systems, rather than an exact identity.

A bath instead selects $P_{\mathrm{eq}}(E) \propto \Omega(E)e^{-\beta_B E}$, leading to

$$2A = B(\beta(E) - \beta_B) + B'.$$

Thus a colder system, with larger β, tends to gain energy. Noise strength and dissipative drift cannot be chosen independently when the specified equilibrium measure must be preserved. The energy diffusion framework and its driven applications are developed in [26].

Integrating Equation 8.7 over the allowed interval gives $\dot{\mathcal{P}} = J(E_{\min}) - J(E_{\max})$. Reflecting boundaries impose $J = 0$ and conserve probability. Absorbing boundaries describe escape and must be accompanied by a loss probability. At an unbounded high-energy end, normalizability and vanishing flux at infinity replace a fictitious finite reflecting wall. Near a ground state, a diffusion expansion can fail because individual level spacings and one-sided jumps matter. Negative temperatures are possible only for appropriate bounded spectra; the same density-of-states derivative can become negative, reversing the drift toward the entropy maximum.

8.7 Two conserved quantities

Let two weakly coupled systems exchange energy q and particle number n, positive into system 1. Their initial grand-canonical weights are proportional to $e^{-\beta_j(E_j - \mu_j N_j)}$. Conservation makes the changes of system 2 equal $(-q, -n)$. Dividing the initial weight of a transition by that of its reverse gives

$$\frac{P_F(q,n)}{P_R(-q,-n)} = e^{a_E q + a_N n}, \quad a_E = \beta_1 - \beta_2, \quad a_N = -\beta_1\mu_1 + \beta_2\mu_2. \tag{8.9}$$

This assumes negligible coupling-energy corrections at measurement times and the appropriate reversed controls. Energy and particle transfer must be retained jointly: marginalizing over particles generally destroys a simple energy-only exponential ratio when $a_N \neq 0$.

For several continuous conserved coordinates x_i, define $B_{ij} = \langle \Delta x_i \Delta x_j \rangle / \delta t$. A zero-current equilibrium diffusion has

$$2A_i = \sum_j \left[B_{ij} \partial_{x_j} S + \partial_{x_j} B_{ij} \right] .$$

A nonzero stationary divergence-free current would require a separate analysis; stationarity alone in multiple dimensions does not force every current component to vanish. Off-diagonal diffusion encodes correlations such as particles carrying energy and is the starting point for thermoelectric response.

8.8 Exercises and solutions

Exercise 1. If cyclic work is Gaussian with mean a and variance b, what does canonical Jarzynski imply?

Solution

The Gaussian moment-generating function gives $\langle e^{-\beta W} \rangle = e^{-\beta a + \beta^2 b/2}$. For $\Delta F = 0$, this equals one, so $2a = \beta b$. Non-Gaussian cumulants correct this formula; it is not the complete finite-size energy-diffusion relation when the noise depends on initial energy.

Exercise 2. For $\Omega(E) \propto E^{\alpha - 1}$ and $B(E) = bE$, find the isolated drift.

Solution

Here $\beta = (\alpha - 1)/E$ and $B' = b$. Equation Equation 8.8 gives $A = b\alpha/2$. Omitting B' would incorrectly replace α by $\alpha - 1$ and can even predict zero heating for a constant density of states.

Exercise 3. Two systems have equal temperatures but different chemical potentials. Can their energy transfers have an unbiased distribution?

Solution

Equal temperatures set $a_E = 0$, but $a_N = \beta(\mu_2 - \mu_1)$ remains. The joint distribution is biased in particle transfer, and particle-energy correlations can bias the energy marginal. Equal temperatures alone do not remove chemical work.

9 Response, fluctuations, and reciprocity

A weak probe measures both how a system fluctuates and how it reacts. In equilibrium these measurements are linked by the fluctuation-dissipation theorem. The connection becomes especially interesting in an isolated chaotic system: a single energy eigenstate is stationary, yet its matrix elements can encode the same local noise and response as a thermal ensemble. This chapter derives the relations with explicit conventions, then explains what ETH adds and what it does not.

9.1 Linear response from the von Neumann equation

Take Hermitian observables A and B, a stationary initial state $[\rho_0, H_0] = 0$, and a perturbation

$$H(t) = H_0 - f(t)B.$$

The minus sign defines our convention for the applied field. In the interaction picture,

$$\dot{\rho}_I = -\frac{i}{\hbar}[-f(t)B_I(t), \rho_I(t)].$$

Keeping one power of f and integrating from the remote past gives

$$\delta\rho_I(t) = \frac{i}{\hbar}\int_{-\infty}^{t} \mathrm{d}t'\, f(t')[B_I(t'), \rho_0].$$

Taking the trace with $A_I(t)$ and moving operators cyclically inside the trace yields

$$\delta\langle A(t)\rangle = \int_{-\infty}^{\infty} \mathrm{d}t'\, \chi_{AB}(t-t')f(t'), \qquad \chi_{AB}(t) = \frac{i}{\hbar}\theta(t)\langle[A(t), B(0)]\rangle_0. \tag{9.1}$$

This is the retarded Kubo formula. It holds for any stationary state, including an individual eigenstate; thermal equilibrium is not needed for its derivation. The step function expresses causality. Kubo's general response formulation is developed in [27].

We use angular frequency and Fourier transforms

$$X(\omega) = \int_{-\infty}^{\infty} \mathrm{d}t\, e^{i\omega t}X(t), \qquad X(t) = \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t}X(\omega).$$

For a monochromatic field, $\delta A(\omega) = \chi_{AB}(\omega)f(\omega)$. If the observable itself depends explicitly on the field, its derivative contributes an additional instantaneous term. Conductivity calculations, for example, must keep the appropriate diamagnetic contribution rather than applying the commutator term to an incomplete observable.

9.2 Causality and dispersive response

A further relation follows from causality without invoking equilibrium. Because $\chi_{AB}(t)$ vanishes at negative time, its Fourier transform is analytic for $\operatorname{Im}\omega > 0$ when the response decays sufficiently fast. A small positive imaginary part of frequency is also the prescription for switching on a probe adiabatically. For a scalar response that falls off at large frequency, contour integration then gives the Kramers–Kronig relations

$$\chi'(\omega) = \frac{1}{\pi}\mathcal{P}\int_{-\infty}^{\infty}\frac{\chi''(\omega')}{\omega'-\omega}\mathrm{d}\omega', \qquad \chi''(\omega) = -\frac{1}{\pi}\mathcal{P}\int_{-\infty}^{\infty}\frac{\chi'(\omega')}{\omega'-\omega}\mathrm{d}\omega'.$$

The symbol $\mathcal{P}$ denotes the symmetric principal value around the pole. Instantaneous terms or nondecaying high-frequency pieces require a subtraction before applying these formulas. In practice the relations check whether measured absorption and dispersion can belong to one causal response.

For Hermitian $A = B$, the time-domain response is real, so $\chi(-\omega) = \chi(\omega)^*$: the real part is even and the imaginary part is odd. In particular, when the integral converges,

$$\chi'(0) = \frac{2}{\pi}\int_0^{\infty}\frac{\chi''(\omega')}{\omega'}\mathrm{d}\omega'.$$

This is a relation between response components. It is logically distinct from FDT, which relates response to fluctuations and requires the extra spectral balance condition. A stationary nonequilibrium state can obey causality and the dispersion relations while violating thermal FDT. Likewise, a vanishing absorption at one frequency says nothing by itself about the temperature or the total noise at other frequencies.

9.3 Spectral noise and detailed balance

Define the connected, ordered correlation and its spectrum by

$$C_{AB}(t) = \langle \delta A(t)\delta B(0)\rangle, \qquad C_{AB}(\omega) = \int \mathrm{d}t\, e^{i\omega t}C_{AB}(t),$$

where $\delta A = A - \langle A\rangle$. For $\rho_0 = \sum_n p_n|n\rangle\langle n|$, inserting a complete energy basis gives, away from the subtracted static piece,

$$C_{AB}(\omega) = 2\pi\sum_{nm} p_n A_{nm}B_{mn}\delta\left(\omega - \frac{E_m - E_n}{\hbar}\right). \tag{9.2}$$

With this convention positive frequency corresponds to transitions in which the system absorbs energy $\hbar\omega$. For $A = B$, the spectrum is nonnegative. It can be strongly asymmetric between positive and negative frequency: emission and absorption are different quantum processes.

In a Gibbs state $p_m/p_n = e^{-\beta(E_m-E_n)}$. Exchanging the indices in the negative-frequency spectrum therefore gives

$$C_{BA}(-\omega) = e^{-\beta\hbar\omega}C_{AB}(\omega). \tag{9.3}$$

This is the spectral detailed-balance or KMS relation. For a cross spectrum it exchanges **both** frequency and observable order. Replacing it by an evenness statement for C_{AB} would discard quantum ordering and can be wrong even at equilibrium.

9.4 The fluctuation-dissipation theorem

For a matrix of responses, define its dissipative part by

$$\chi^{\mathrm{d}}_{AB}(\omega) = \frac{\chi_{AB}(\omega) - \chi_{BA}(\omega)^*}{2i}.$$

For a single observable this reduces to $\chi''_{AA} = \Im\chi_{AA}$. The Fourier transform of the full commutator in Equation 9.1 gives

$$\chi^{\mathrm{d}}_{AB}(\omega) = \frac{C_{AB}(\omega) - C_{BA}(-\omega)}{2\hbar}.$$

Define the symmetrized spectrum

$$C^{\mathrm{sym}}_{AB}(\omega) = \frac{1}{2}\left[C_{AB}(\omega) + C_{BA}(-\omega)\right].$$

Using Equation 9.3 to eliminate the negative-frequency term yields

$$C^{\mathrm{sym}}_{AB}(\omega) = \hbar \coth\left(\frac{\beta\hbar\omega}{2}\right)\chi^{\mathrm{d}}_{AB}(\omega). \tag{9.4}$$

For scalar autocorrelations this is the familiar equilibrium fluctuation-dissipation theorem. Its low-frequency classical limit, $\beta\hbar|\omega| \ll 1$, is

$$C^{\mathrm{sym}}_{AA}(\omega) \simeq \frac{2T}{\omega}\chi''_{AA}(\omega).$$

At $T = 0$, quantum fluctuations remain and the hyperbolic cotangent tends to the sign of ω. The product has a finite positive autocorrelation spectrum. A positive-temperature passive state has $\chi''_{AA}(\omega) \geq 0$ for $\omega > 0$. A bounded-spectrum negative-temperature state can instead provide gain; its population inversion changes the sign of the absorptive response.

For $f(t) = f_0 \cos\omega t$ coupled as $-fA$, the average input power over a cycle is

$$\overline{\dot{W}} = \frac{1}{2} f_0^2 \omega \chi''_{AA}(\omega).$$

To verify the sign, write the response as $f_0[\chi' \cos\omega t + \chi'' \sin\omega t]$ and average $-\dot{f}\,\delta\langle A\rangle$. Positive χ'' therefore represents heating in our convention. Reversing either the Fourier convention or the sign of the coupling changes intermediate signs, so these definitions should always accompany a response formula.

> **i** Solved example: an oscillator spectrum
>
> For $H = p^2/(2m) + m\omega_0^2 x^2/2$, let $x = x_0(a + a^\dagger)$ with $x_0^2 = \hbar/(2m\omega_0)$. Thermal occupation is $\bar{n} = (e^{\beta\hbar\omega_0} - 1)^{-1}$. Direct calculation gives

$$C_{xx}(\omega) = 2\pi x_0^2 \left[(\bar{n}+1)\delta(\omega-\omega_0) + \bar{n}\delta(\omega+\omega_0)\right].$$

The ratio of negative- to positive-frequency weights is $\bar{n}/(\bar{n}+1) = e^{-\beta\hbar\omega_0}$. The commutator eliminates $\bar{n}$, giving

$$\chi''_{xx}(\omega) = \frac{\pi}{2m\omega_0}[\delta(\omega-\omega_0) - \delta(\omega+\omega_0)].$$

The symmetrized noise retains the factor $2\bar{n}+1 = \coth(\beta\hbar\omega_0/2)$, exactly as in Equation 9.4. This integrable example also emphasizes that canonical FDT itself does not require chaos.

9.5 How an eigenstate can satisfy thermal FDT

For one eigenstate $|n\rangle$, the exact spectrum is

$$C_{AA}^{(n)}(\omega) = 2\pi \sum_{m\neq n} |A_{nm}|^2 \delta\left(\omega - \frac{E_m - E_n}{\hbar}\right).$$

Its upward and downward transition weights are not related by Gibbs probabilities, since only one initial state is occupied. Off-diagonal ETH supplies a different mechanism. Write its smooth variance schematically as

$$\overline{|A_{nm}|^2} = e^{-S(\bar{E})}|f_A(\bar{E},\varepsilon)|^2, \quad \bar{E} = (E_n+E_m)/2, \quad \varepsilon = E_m - E_n.$$

An energy reference in S and the units of f_A must be chosen consistently; it cancels in the following ratio. Hermiticity makes the smooth variance even in ε. Replacing the sum by a density-of-states integral after suitable frequency smoothing gives

$$C_{AA}^{(n)}(\omega) \propto \Omega(E_n+\hbar\omega)e^{-S(E_n+\hbar\omega/2)}|f_A(E_n+\hbar\omega/2,\hbar\omega)|^2.$$

Set $\varepsilon = \hbar\omega$ and expand the logarithm of the ratio at fixed microscopic ε:

$$\ln\frac{C_{AA}^{(n)}(-\omega)}{C_{AA}^{(n)}(\omega)} = -\beta(E_n)\varepsilon - \varepsilon\,\partial_E \ln|f_A(E_n,\varepsilon)|^2 + O(\varepsilon^3\partial_E^3 S, \varepsilon^3\partial_E^3 \ln|f|_A^2). \tag{9.5}$$

The density of accessible final states contributes $-2\beta\varepsilon$; the ETH matrix-element envelope contributes $+\beta\varepsilon$. Their combination produces the thermal factor. For a smooth local-observable envelope depending on energy density, its logarithmic derivative is generally $O(1/N)$ at fixed frequency. Thus thermal detailed balance and FDT emerge to leading thermodynamic order with temperature fixed by the eigenstate energy. This is an ETH consequence, whereas the exact Kubo formula for that eigenstate required only stationarity. Connections between chaotic matrix elements, relaxation, and equilibrium correlation functions were developed in [16].

The order of limits matters. A finite isolated system has discrete lines and recurrences. One needs frequency windows containing many transitions, yet narrow relative to physical spectral features, or observation times below the recurrence scale. At very low frequencies, hydrodynamic modes and conservation laws can create structure that a naive smooth-envelope expansion misses. The approximation also fails when the energy difference probes a finite fraction of the total energy, or where ETH itself fails.

9.6 Two-observable ETH requires joint information

Separate ETH fits for A and B do not determine C_{AB}. One needs the smooth joint covariance

$$g_{AB}(\bar{E}, \varepsilon) = e^{S(\bar{E})}\overline{A_{nm}B_{mn}}.$$

The fluctuating parts of different observables cannot generally be treated as independent random numbers. Their covariance retains conserved overlaps, selection rules, phases, and possible time-reversal parity. For every vector of complex coefficients v_A, the matrix of positive-frequency ordered spectra obeys

$$\sum_{AB} v_A^* C_{AB}(\omega) v_B \geq 0,$$

because it is the noise spectrum of the corresponding linear combination. In particular,

$$|C_{AB}(\omega)|^2 \leq C_{AA}(\omega) C_{BB}(\omega).$$

A joint ETH description consistent with this positivity and microscopic symmetries reproduces the cross-observable version of Equation 9.4 after the same smoothing and thermodynamic limits. Autocorrelation ETH alone leaves the cross spectrum undetermined.

9.7 Onsager–Casimir reciprocity

Let x_i be slowly exchanged extensive coordinates and a_i their thermodynamic affinities, normalized so entropy production is $\dot{S} = \sum_i a_i J_i$, where $J_i = \dot{x}_i$. Near equilibrium,

$$\langle J_i \rangle = \sum_j L_{ij} a_j + O(a^2).$$

An energy affinity is an inverse-temperature difference; a particle affinity includes $-\beta\mu$. The symmetric part of L must be positive semidefinite, since $\sum_{ij} a_i L_{ij} a_j \geq 0$. The antisymmetric part produces no entropy to this order.

Suppose time reversal sends $x_i \to \epsilon_i x_i$, with $\epsilon_i = \pm 1$, and reverses magnetic field $\mathcal{B}$. Microscopic reversibility relates the corresponding equilibrium current correlations. In a classical reversible setting, or with quantum correlations replaced by the appropriate Kubo-transformed correlations, integrating this relation gives

$$L_{ij}(\mathcal{B}) = \epsilon_i \epsilon_j L_{ji}(-\mathcal{B}). \tag{9.6}$$

The Kubo transform can be written as

$$C_{ij}^K(t) = \frac{1}{\beta} \int_0^\beta \mathrm{d}u \, \langle J_j(-i\hbar u) J_i(t) \rangle_{\mathrm{eq}}.$$

For entropy-normalized affinities, a Green–Kubo expression has the form $L_{ij} = \int_0^\infty C_{ij}^K(t)\mathrm{d}t$, with volume and geometry factors determined by whether J_i is a total current or a current density. In a system without magnetic fields and for variables of the same parity, Equation 9.6 reduces

to $L_{ij} = L_{ji}$. For example, the response of particle flow to a temperature affinity equals the reciprocal energy-flow response to the corresponding particle affinity, once both forces and currents use conjugate normalizations. Comparing coefficients defined with different temperature factors can hide this equality.

9.8 Nonlinear relations from a fluctuation symmetry

Fluctuation relations contain information beyond linear response. For one integrated transfer Q in time t, assume the steady-state large-time statistics satisfy $P_a(Q)/P_a(-Q) = e^{aQ}$ at zero magnetic field, up to boundary corrections subextensive in t. Define

$$g(\lambda, a) = \lim_{t\to\infty} \frac{1}{t} \ln\langle e^{-\lambda Q}\rangle_a = \sum_{r\geq 1} \frac{(-\lambda)^r}{r!} c_r(a).$$

Here c_1 is the mean current and c_2 is the growth rate of its variance. The probability symmetry gives

$$g(\lambda, a) = g(a - \lambda, a), \qquad g(0, a) = g(a, a) = 0. \tag{9.7}$$

Expansion at $\lambda = a$ therefore implies

$$c_1(a) = \frac{a}{2} c_2(a) - \frac{a^2}{6} c_3(a) + \frac{a^3}{24} c_4(a) + \cdots.$$

At equilibrium, odd cumulants vanish. If $c_1 = La + Ma^2/2 + \cdots$ and $c_2 = C_0 + C_1 a + \cdots$, coefficient matching gives

$$L = \frac{1}{2} C_0, \qquad M = C_1. \tag{9.8}$$

The first equality is a zero-frequency fluctuation-dissipation relation. The second ties a nonlinear current coefficient to the first change of noise with affinity. Higher orders involve higher cumulants and their derivatives; there is no general license to discard them. For several affinities one differentiates the vector version of Equation 9.7. Magnetic-field reversal must be included in that symmetry before deriving nonlinear parity relations, as illustrated by [28].

9.9 Exercises and solutions

Exercise 1. At $\beta\hbar\omega = \ln 3$, what is the ratio of emission to absorption noise for a thermal Hermitian observable?

Solution

With our frequency convention, emission is the negative-frequency spectrum. Its ratio to absorption is $e^{-\beta\hbar\omega} = 1/3$. The symmetrized spectrum is not the same as either ordered spectrum.

Exercise 2. Why does causality alone not imply thermal FDT for a pure eigenstate?

Solution

Causality yields the retarded commutator response and relates its real and imaginary parts. FDT additionally requires a relation between the commutator and the symmetric fluctuations. A generic eigenstate has no Gibbs KMS identity; ETH can supply an approximate spectral balance for suitable observables and frequency windows.

Exercise 3. A current has $c_2(a) = 4 + 6a + O(a^2)$. Find its first two response coefficients under the stated fluctuation symmetry.

Solution

Equation Equation 9.8 gives $L = 2$ and $M = 6$, so $c_1(a) = 2a + 3a^2 + O(a^3)$. These numbers refer to the entropy affinity a; converting to a voltage or temperature difference may introduce additional factors.

10 Driven systems and finite reservoirs

The energy-space Einstein relation becomes useful when microscopic driving is complicated but its scaling with energy is simple. It determines the noise from a measured heating rate and a density of states, or conversely predicts heating from fluctuations. We now work through three applications: a particle accelerated by a moving chaotic boundary, a harmonic probe kicked while attached to a finite equilibrating system, and two finite bodies exchanging energy. The distributions below are consequences of stated continuum models; they are not numerical evidence that every microscopic drive realizes those models.

10.1 A particle in a moving chaotic cavity

A particle of mass m has kinetic energy $E = mv^2/2$ in a d-dimensional cavity. Assume that the fixed-wall billiard mixes its collision angles sufficiently quickly. The wall moves slowly, with normal velocity u much smaller than a typical particle speed. Locally specular reflection in the wall frame gives

$$v_n' = 2u - v_n, \qquad \Delta E = 2mu^2 - 2muv_n.$$

The fluctuating term is first order in u, while its mean vanishes for an unbiased wall velocity before accounting for collision sampling. The positive quadratic term and the bias of collision sampling both contribute to the mean. Their precise coefficient depends on geometry and wall statistics. Their **energy scaling** follows without solving trajectories: collision rate is proportional to v/L, so

$$A(E) \propto \frac{mu^2 v}{L} \propto E^{1/2}, \qquad B(E) \propto \frac{m^2u^2v^3}{L} \propto E^{3/2}.$$

The required assumptions are weak energy change per collision, short correlations on the heating time, and no slowly trapped family of trajectories dominating the statistics. A classically chaotic shape alone does not guarantee every one of these assumptions for every boundary motion. Coherent quantum interference can also invalidate a classical diffusion picture over sufficiently long times.

Weyl counting gives $\Omega(E) \propto E^{d/2-1}$ in the semiclassical regime, so $\beta = (d/2-1)/E$. Parameterize the diffusion strength by $b > 0$:

$$B(E) = bE^{3/2}.$$

Retaining the finite-system derivative in the Einstein relation gives

$$A(E) = \frac{1}{2}\left[B\beta + B'\right] = \frac{b(d+1)}{4}\sqrt{E}. \tag{10.1}$$

The derivative is essential for a single particle. In $d = 2$, the density of states is constant and $\beta = 0$, yet heating remains positive. A naive equation $2A = B\beta$ would miss it completely.

10.2 Solving the cavity diffusion model

Use the Itô stochastic equation equivalent to the Fokker–Planck description:

$$\mathrm{d}E = \frac{b(d+1)}{4}\sqrt{E}\,\mathrm{d}t + \sqrt{b}\,E^{3/4}\mathrm{d}W_t, \qquad \langle(\mathrm{d}W_t)^2\rangle = \mathrm{d}t.$$

The variable $x = \sqrt{E}$ simplifies both coefficients. Itô's rule includes the second derivative term:

$$\begin{aligned}\mathrm{d}x &= \frac{1}{2\sqrt{E}}\mathrm{d}E - \frac{1}{8E^{3/2}}(\mathrm{d}E)^2 \\ &= \frac{bd}{8}\mathrm{d}t + \frac{\sqrt{b}}{2}\sqrt{x}\,\mathrm{d}W_t.\end{aligned} \tag{10.2}$$

This square-root diffusion has a simple similarity solution. Let $p_x(x,t)$ be its density with respect to $\mathrm{d}x$. For the idealized entrance law starting at $x = 0$, substitution in its Fokker–Planck equation verifies

$$p_x(x,t) = \frac{x^{d-1}e^{-x/\vartheta}}{\Gamma(d)\vartheta^d}, \qquad \vartheta = \frac{bt}{8}, \quad x \geq 0.$$

The gamma-function identity $\int_0^\infty x^{d-1}e^{-x/\vartheta}\mathrm{d}x = \Gamma(d)\vartheta^d$ normalizes it. In an actual cavity the diffusion approximation fails sufficiently near the lowest quantum levels and at speeds comparable to u. A finite initial distribution instead approaches the displayed similarity shape once its initial energy scale becomes negligible, provided the continuum assumptions remain valid.

Changing variables back requires the Jacobian $\mathrm{d}x/\mathrm{d}E = 1/(2\sqrt{E})$:

$$P(E,t) = \frac{E^{(d-2)/2}}{2\Gamma(d)\vartheta^d}\exp\left(-\frac{\sqrt{E}}{\vartheta}\right). \tag{10.3}$$

The tail is exponential in speed, rather than in energy. This is a nonequilibrium similarity distribution; its continuing motion to higher energy rules out a stationary Gibbs interpretation. Gamma moments give

$$\langle E\rangle = d(d+1)\vartheta^2, \qquad \frac{\operatorname{Var} E}{\langle E\rangle^2} = \frac{4d+6}{d(d+1)}. \tag{10.4}$$

At the same mean energy, a canonical free particle has relative variance $2/d$. The driven distribution is therefore broader. For a rapidly internally equilibrating gas, an analogous total-energy description replaces the number of quadratic momentum coordinates by their total count. Independent particles that fail to equilibrate do not justify that replacement for the full many-particle state, even if their total energy happens to look Gaussian.

A broader driven distribution can nevertheless have the same thermodynamic entropy density as a canonical distribution. For a distribution over states represented by $P(E)$, its coarse entropy uses the measure $P(E)/\Omega(E)$:

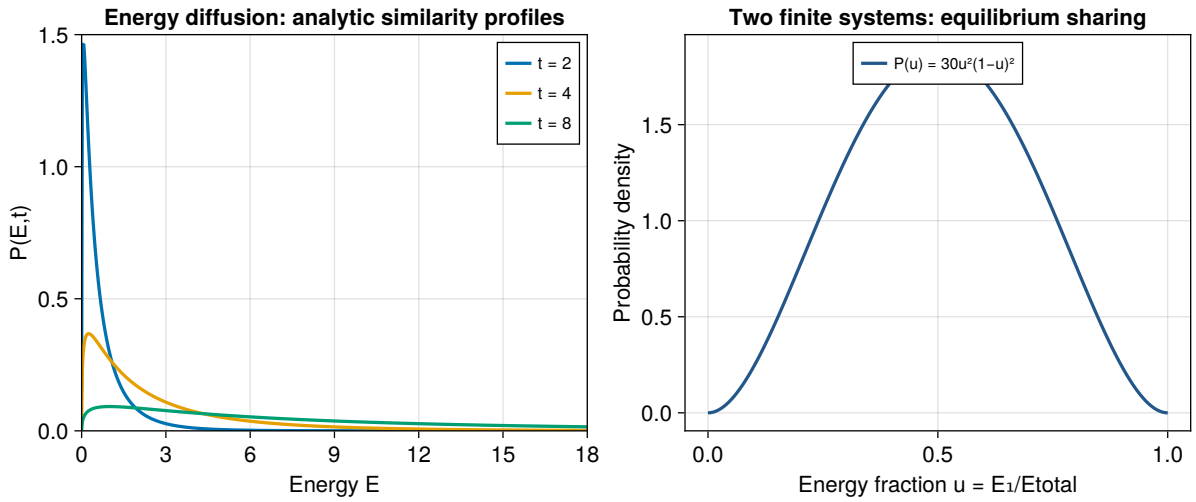


Figure 10.1: Analytic examples of energy distributions. Left: the continuum cavity model in three dimensions, with $b = 1$, showing the entrance-law solution at three times. Right: the exact equilibrium energy fraction of two finite reservoirs with density-of-states exponents $c_1 = c_2 = 3$. These are calculated model curves, not microscopic simulation data.

$$S[P] = -\int \mathrm{d}E\, P(E) \ln \frac{P(E)}{\Omega(E)}.$$

At fixed normalization and mean energy, maximizing this functional by Lagrange multipliers gives $P_{\mathrm{G}}(E) \propto \Omega(E)e^{-\beta E}$. Direct substitution then gives the exact comparison

$$S[P_{\mathrm{G}}] - S[P] = \int \mathrm{d}E\, P(E) \ln \frac{P(E)}{P_{\mathrm{G}}(E)} \geq 0.$$

The inequality is relative entropy positivity. It does not determine whether the difference is extensive; that depends on whether only a collective energy distribution differs or whether independent microscopic sectors each retain extra information.

10.3 A kicked harmonic probe inside a finite system

Consider a classical harmonic coordinate with

$$\varepsilon(x,p) = \frac{p^2}{2m} + \frac{kx^2}{2}.$$

Couple it weakly to a much larger but finite chaotic system, and call the **total** energy E. Wait between kicks long enough for the probe to equilibrate internally with that system, while the total energy changes little per cycle. Its pre-kick marginal is then approximately canonical at $T(E)$. During a very short impulse the coupling and coordinate motion are negligible. An impulse $I(x)$ changes $p \to p + I(x)$, and hence

$$\Delta E = \frac{pI(x)}{m} + \frac{I(x)^2}{2m}.$$

The equilibrium Gaussian momentum is independent of x, has zero mean, and satisfies $\langle p^2\rangle = mT$. For kicks separated by time τ, the leading drift and variance rate are consequently

$$A(E) = \frac{\langle I(x)^2\rangle}{2m\tau}, \qquad B(E) = \frac{T(E)\langle I(x)^2\rangle}{m\tau} = 2T(E)A(E). \tag{10.5}$$

In the variance, terms of order I^4 have been dropped; the retained noise is the square of the term linear in impulse. These expressions use a canonical probe marginal and thus omit finite-reservoir corrections. The exact energy-space relation restores the B' correction, consistently of the same order as corrections to that marginal.

For example, choose $I(x) = g\,\mathrm{sgn}(x)|x|^r$ with $r \geq 0$. The position variance is T/k, and a Gaussian integral gives

$$\langle I^2\rangle = g^2 \left(\frac{2T}{k}\right)^r \frac{\Gamma(r+1/2)}{\sqrt{\pi}}.$$

Thus different spatial profiles change how the heating rate grows with temperature. This is a useful concrete realization of the general energy-distribution framework of [26]; the following moment calculation is independent of the oscillator details.

10.4 A transition in fluctuation growth

Suppose over the energy range of interest

$$\beta(E) = cE^{-\alpha}, \qquad A(E) = aE^s,$$
$$a, c > 0, \qquad \alpha > 0.$$

For the harmonic impulse above, $s = \alpha r$. The mean energy μ and variance v obey the exact diffusion moment identities, when boundary terms vanish,

$$\dot\mu = \langle A\rangle, \qquad \dot v = \langle B\rangle + 2\,\mathrm{Cov}(E, A).$$

If the energy distribution remains narrow and higher cumulants are controlled, expand the smooth coefficients around μ. To leading order,

$$\dot\mu = A(\mu), \qquad \dot v = B(\mu) + 2A'(\mu)v.$$

Using the thermodynamic approximation $B \simeq 2A/\beta$ and eliminating time gives

$$\frac{\mathrm{d}v}{\mathrm{d}\mu} - \frac{2s}{\mu}v = \frac{2}{c}\mu^\alpha.$$

The integrating factor is μ^{-2s}, so

$$v(\mu) = \mu^{2s}\left[\frac{v_0}{\mu_0^{2s}} + \frac{2}{c}\int_{\mu_0}^{\mu} z^{\alpha-2s}\mathrm{d}z\right]. \tag{10.6}$$

The canonical variance at the same mean is

$$v_{\mathrm{can}} = -\frac{\mathrm{d}E}{\mathrm{d}\beta} = \frac{\mu^{1+\alpha}}{c\alpha}.$$

Let $\eta = 2s - \alpha - 1$. The elementary integral in Equation 10.6 has three qualitatively different limits:

Condition	Asymptotic comparison, for negligible initial width
$\eta < 0$	$v/v_{\mathrm{can}} \to 2\alpha/\lvert\eta\rvert$
$\eta = 0$	$v/v_{\mathrm{can}} \sim 2\alpha \ln(\mu/\mu_0)$
$\eta > 0$	v/v_{can} grows proportionally to μ^{η}

For $\eta < 0$, newly generated fluctuations dominate and the initial width becomes irrelevant. For $\eta > 0$, amplification of fluctuations generated early in the protocol dominates: the lower integration limit retains memory. At the boundary the integral is logarithmic. This is a transition in asymptotic fluctuation growth, rather than a claim of an equilibrium singularity in a partition function. Near $\eta = 0$, large $|\eta| \ln(\mu/\mu_0)$ is needed to distinguish a power from a logarithm, so long crossovers are expected.

The mean grows as

$$\mu(t) = \left[\mu_0^{1-s} + (1-s)at\right]^{1/(1-s)}$$

for $s \neq 1$, and as $\mu_0 e^{at}$ for $s = 1$. For $s > 1$ this ideal power-law equation diverges at finite time; microscopic weak-drive assumptions generally fail before interpreting that divergence literally. For the runaway-width regime, one must also monitor v/μ^2: once it is no longer small, the narrow-distribution closure cannot establish the subsequent full distribution. For a fixed-duration weak kick to remain small relative to the energy at arbitrarily high energy, its mean relative change $A(E)\tau/E \propto E^{s-1}$ must not grow; this favors $s \leq 1$. Under that restriction, the growing-ratio regime $\eta > 0$ requires $\alpha < 1$. A large system does not by itself guarantee Gaussian statistics at arbitrarily long times.

> **i** Solved example: tuning a kick profile
>
> Take $\alpha = 1/2$, for which $T \propto \sqrt{E}$, and the harmonic probe impulse exponent r. Then $s = r/2$ and $\eta = r - 3/2$. A linear impulse ($r = 1$) gives $v/v_{\mathrm{can}} \to 2$. At $r = 3/2$ the ratio grows logarithmically. At $r = 2$ it grows as $\mu^{1/2}$ within the interval where the approximations remain controlled. A feedback or force profile changes noise amplification without changing the equilibrium density of states.

10.5 Two finite bodies exchanging energy

Now switch off the drive. Two internally equilibrating bodies have total energy E_{tot}, weak coupling, and subsystem energy E in body 1. The combined isolated system assigns equal probability to states at fixed total energy, so

$$P_{\mathrm{eq}}(E) = \frac{1}{\mathcal{Z}}\Omega_1(E)\Omega_2(E_{\mathrm{tot}} - E). \tag{10.7}$$

The effective entropy derivative along the energy exchange coordinate is

$$\partial_E S_{\mathrm{tot}}(E) = \beta_1(E) - \beta_2(E_{\mathrm{tot}} - E).$$

A zero-current diffusion preserving this distribution must satisfy

$$2A(E) = B(E)[\beta_1(E) - \beta_2(E_{\mathrm{tot}} - E)] + B'(E). \tag{10.8}$$

If body 1 is colder, its larger inverse temperature gives a positive leading drift: it gains energy. At finite size, a nonconstant diffusion coefficient contributes the additional drift $B'/2$; the location where mean drift vanishes need not exactly equal the mode of the stationary probability density. Energy-exchange fluctuation dynamics for finite systems is studied in [29].

Expand the total entropy about an interior maximum E_*. To quadratic order,

$$P_{\mathrm{eq}}(E) \propto \exp\left[-\frac{(E-E_*)^2}{2v_*}\right], \qquad v_* = -\frac{1}{S''_{\mathrm{tot}}(E_*)} = T^2\frac{C_1C_2}{C_1+C_2}. \tag{10.9}$$

Both temperatures agree at the maximum. The reciprocal heat capacities add because energy entering one body leaves the other. An infinite bath, $C_2 \to \infty$, recovers $v_* = T^2C_1$. For two equal finite bodies, the fluctuations are half that bath-limit value.

When B varies negligibly across this narrow peak, linearization yields an Ornstein–Uhlenbeck process,

$$\mathrm{d}(E-E_*) = -\kappa(E-E_*)\mathrm{d}t + \sqrt{B_*}\mathrm{d}W_t, \quad \kappa = \frac{B_*}{2v_*}.$$

Its mean relaxes as $e^{-\kappa t}$ and variance satisfies

$$v(t) = v_* + [v(0) - v_*]e^{-2\kappa t}.$$

Mean relaxation and noise growth therefore provide independent measurements of the same local transport coefficient. The conditions of internal equilibration, small exchange increments, and weak coupling are crucial; a few coherent coupled levels can instead exchange energy periodically.

10.6 An exact finite-reservoir distribution

A useful model avoids the Gaussian approximation. Let $\Omega_i(E_i) \propto E_i^{c_i-1}$ for positive c_i. The energy fraction $u = E/E_{\mathrm{tot}}$ has the beta distribution

$$P(u) = \frac{u^{c_1-1}(1-u)^{c_2-1}}{\mathrm{B}(c_1,c_2)}, \quad 0<u<1, \qquad \mathrm{B}(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

Its exact moments are

$$\langle u\rangle = \frac{c_1}{c_1+c_2}, \qquad \operatorname{Var} u = \frac{c_1c_2}{(c_1+c_2)^2(c_1+c_2+1)}.$$

For $c_1, c_2 > 1$, its mode is $(c_1-1)/(c_1+c_2-2)$, which differs from its mean at finite size. The mode equalizes the derivatives of the two surface entropies. For large c_i, this distinction is relatively

small and the Gaussian heat-capacity formula follows. For $c_1 = c_2 = 3$, $P(u) = 30u^2(1-u)^2$, as plotted in Figure 10.1. The finite support and the vanishing probabilities at both endpoints are physical consequences of the available states in both bodies.

10.7 Exercises and solutions

Exercise 1. For the cavity model, verify $\mathrm{d}\langle E\rangle/\mathrm{d}t = \langle A\rangle(E)$ using the similarity distribution.

Solution

Since $\langle\sqrt{E}\rangle = d\vartheta$ and $\dot{\vartheta} = b/8$, differentiating $d(d+1)\vartheta^2$ gives $bd(d+1)\vartheta/4$. Averaging Equation 10.1 gives the same result. Replacing $\langle\sqrt{E}\rangle$ by $\sqrt{\langle E\rangle}$ would not be exact for this broad one-particle distribution.

Exercise 2. For $\alpha = 1$, $s = 0$, compare driven and canonical energy variances at large mean energy.

Solution

Here $\eta = -2$, so $v/v_{\mathrm{can}} \to 1$. Equality of the variance alone does not prove that the complete energy distribution or the quantum state is Gibbs.

Exercise 3. Two reservoirs have $c_1 = c_2 = 3$. Find their exact relative energy-fraction variance and compare with a Gaussian description.

Solution

The mean fraction is $1/2$ and variance is $9/(36 \cdot 7) = 1/28$. The distribution is visibly non-Gaussian for these small exponents. The entropy curvature at the mode gives a local Gaussian variance $1/16$, illustrating that a peak-curvature approximation need not reproduce the exact variance of a broad finite distribution.

Part III

Constraints and long-lived memory

11 Integrability and constrained equilibrium

An isolated system can relax while remembering much more than its energy. A free gas supplies the simplest example: particles redistribute in space, but each single-particle mode keeps its occupation. An interacting integrable chain has a more elaborate version of this memory. Statistical mechanics remains useful, provided we retain the conserved information that local measurements can detect.

By the end of this chapter you should be able to construct a generalized Gibbs ensemble, compute its free-particle correlations, explain why a bosonic observable can relax in a fermionized model, and identify the missing-information problem in interacting integrable systems. We set $k_B = 1$ and use $\hbar = 1$ in time evolution.

11.1 Maximum entropy with many constraints

Let Q_α be Hermitian conserved operators, $[H, Q_\alpha] = 0$, and let their initial expectation values be q_α. For now assume the charges commute with one another. We seek a density operator with these expectations and the largest von Neumann entropy. Vary

$$-\operatorname{Tr}(\rho \ln \rho) - \lambda_0(\operatorname{Tr}\rho - 1) - \sum_\alpha \lambda_\alpha(\operatorname{Tr}\rho Q_\alpha - q_\alpha).$$

Using $\delta \operatorname{Tr}(\rho \ln \rho) = \operatorname{Tr}[\delta\rho(\ln\rho + 1)]$ gives

$$\rho_{\mathrm{GGE}} = \frac{1}{Z}\exp\left(-\sum_\alpha \lambda_\alpha Q_\alpha\right), \qquad Z = \operatorname{Tr}\exp\left(-\sum_\alpha \lambda_\alpha Q_\alpha\right). \tag{11.1}$$

The multipliers satisfy $q_\alpha = -\partial_{\lambda_\alpha} \ln Z$. Their susceptibility matrix is the charge covariance matrix, $\partial_{\lambda_\alpha}\partial_{\lambda_\beta} \ln Z = \langle Q_\alpha Q_\beta\rangle - \langle Q_\alpha\rangle\langle Q_\beta\rangle$, which is positive semidefinite. Redundant charges can therefore make the multipliers nonunique even when the state is unique.

There is also a useful proof of maximality. For any other state σ satisfying the same constraints, positivity of quantum relative entropy gives

$$0 \le D(\sigma\|\rho_{\mathrm{GGE}}) = -S(\sigma) + \ln Z + \sum_\alpha \lambda_\alpha q_\alpha = S(\rho_{\mathrm{GGE}}) - S(\sigma).$$

This is a variational result, not a dynamical theorem. It does not prove that unitary evolution approaches the GGE. Nor does it say which charges suffice. Including every eigenstate projector reproduces the diagonal ensemble exactly but gives little compression. The physical question is whether local and quasilocal conserved quantities determine the state seen by every fixed finite subsystem. Early hard-core-boson calculations established the practical usefulness of this idea [30].

11.2 Free fermions: occupations, correlations, and dephasing

Consider a number-conserving quadratic Hamiltonian on L sites,

$$H = \sum_{ij} h_{ij} c_i^\dagger c_j = \sum_\mu \epsilon_\mu d_\mu^\dagger d_\mu, \qquad c_i = \sum_\mu \phi_\mu(i) d_\mu.$$

The anticommutation rules are $\{c_i, c_j^\dagger\} = \delta_{ij}$ and $\{c_i, c_j\} = 0$. Each $I_\mu = d_\mu^\dagger d_\mu$ has eigenvalues 0 and 1 and commutes with H. Constraining $n_\mu = \langle I_\mu \rangle_0$ makes the partition function factorize:

$$Z = \prod_\mu (1 + e^{-\lambda_\mu}), \qquad n_\mu = \frac{1}{1 + e^{\lambda_\mu}}, \qquad \lambda_\mu = \ln \frac{1 - n_\mu}{n_\mu}. \tag{11.2}$$

Exactly empty or full modes are allowed as limits $\lambda_\mu \to \pm\infty$. A conventional thermal state is the special case $\lambda_\mu = \beta(\epsilon_\mu - \mu_{\text{chem}})$. A general quench produces an independent function of mode label instead.

The real-space covariance follows directly:

$$C_{ij}(t) \equiv \langle c_i^\dagger(t) c_j(t) \rangle = \sum_{\mu\nu} \phi_\mu^*(i) \phi_\nu(j) e^{i(\epsilon_\mu - \epsilon_\nu)t} \langle d_\mu^\dagger d_\nu \rangle_0. \tag{11.3}$$

For nondegenerate one-particle energies its infinite-time average retains only $\mu = \nu$, exactly as predicted by the occupation GGE. In an extended system many frequencies dephase and local instantaneous correlations can settle near this value. A finite system remains quasiperiodic and can recur. Statements about stationary local states usually take the thermodynamic limit before the long-time limit.

Degeneracies require care: if $\epsilon_\mu = \epsilon_\nu$, the coherence $d_\mu^\dagger d_\nu$ is conserved too. One can diagonalize the initial covariance within each degenerate subspace, or constrain the entire stationary covariance block. Occupations in an arbitrary basis need not suffice. Pairing Hamiltonians similarly require a Nambu covariance containing both $\langle c_i^\dagger c_j \rangle$ and $\langle c_i c_j \rangle$.

For a Gaussian state, Wick's rule expresses every even correlation as sums of products of two-point functions. For example,

$$\langle c_i^\dagger c_j^\dagger c_l c_m \rangle = C_{im} C_{jl} - C_{il} C_{jm}$$

when anomalous correlations vanish. Quadratic evolution preserves Gaussianity. For initially non-Gaussian states, using Wick's rule needs an additional argument: spreading particles can dilute connected higher-order correlations in a fixed observation region. Rigorous forms of this *gaussification* require assumptions such as clustering in the initial state and delocalizing one-particle propagation [31]. Localized modes, insufficient transport, persistent degeneracies, and long-range initial correlations can obstruct the argument.

The global diagonal ensemble can retain correlations $\langle I_\mu I_\nu \rangle_0 - n_\mu n_\nu$ that Equation 11.2 discards. A GGE may nevertheless reproduce every fixed-support observable because the contribution of that information becomes negligible locally. Equality of local states is much weaker than equality of global density operators or entropies.

Solved example: density-wave melting

Take $H = -J\sum_j (c_j^\dagger c_{j+1} + \text{h.c.})$ on an even periodic chain and initially occupy every even site. Each momentum mode has $n_k = 1/2$, so its occupation GGE is the infinite-temperature state at half filling. The initial state also has $\langle c_k^\dagger c_{k+\pi}\rangle = 1/2$ with a sign fixed by the origin convention. Since $\epsilon_k = -2J\cos k$, the staggered density becomes, in the infinite chain,

$$m(t) \equiv \frac{1}{L}\sum_j (-1)^j \langle n_j(t)\rangle = \frac{1}{2}\int_{-\pi}^{\pi} \frac{\mathrm{d}k}{2\pi} e^{-4iJt\cos k} = \frac{1}{2}J_0(4Jt).$$

Here J_0 is defined by the displayed integral. Its oscillatory envelope decays as $t^{-1/2}$. The occupations never change; relaxation comes from the off-diagonal momentum coherence.

11.3 Hard-core bosons and the Jordan–Wigner string

Hard-core bosons obey $b_j^2 = 0$ and $\{b_j, b_j^\dagger\} = 1$ on one site, while operators on different sites commute. In an open chain define

$$b_j = \exp\left(i\pi \sum_{l<j} c_l^\dagger c_l\right) c_j, \qquad n_j = b_j^\dagger b_j = c_j^\dagger c_j. \tag{11.4}$$

The string supplies the minus signs needed to turn fermionic exchange into bosonic exchange. Nearest-neighbor hopping maps to fermion hopping, so the many-body energies and density dynamics are free-fermionic. On a ring the boundary hopping depends on fermion parity; calculations must select the appropriate parity sector and boundary condition.

Off-diagonal bosonic correlations contain strings. For $i < j$,

$$\langle b_i^\dagger b_j\rangle = \left\langle c_i^\dagger \prod_{i<l<j} (1-2n_l) c_j \right\rangle. \tag{11.5}$$

These can be evaluated by expanding the product and applying Wick's rule, or more efficiently as determinants/Pfaffians of a covariance matrix. At separation two, Wick's rule gives the explicit check

$$\langle b_i^\dagger b_{i+2}\rangle = C_{i,i+2}(1 - 2C_{i+1,i+1}) + 2C_{i,i+1}C_{i+1,i+2}.$$

Thus bosonic momentum occupations, obtained by Fourier transforming Equation 11.5, can change substantially although fermionic mode occupations are constants. The mapping changes the form of the observable. For fixed $j - i$, the string has finite support; local fermionic gaussification can then determine its late-time value. A momentum distribution sums all separations, so interchange of the long-distance, thermodynamic, and long-time limits needs further control.

11.4 Generalized eigenstate thermalization

Ordinary ETH says that energy density determines local properties of a typical chaotic eigenstate. Integrable eigenstates at the same energy can have different mode distributions and different local observables. The replacement is a *macrostate* containing all relevant conserved densities. Eigenstates that represent the same smooth macrostate should become locally indistinguishable as L grows. This is the useful content of generalized ETH, rather than a claim that all same-energy integrable eigenstates agree [32].

For free particles this is transparent. A momentum-occupation eigenstate has

$$\langle c_j^\dagger c_{j+r}\rangle = \frac{1}{L}\sum_k e^{ikr} n_k.$$

Divide momentum space into bins of width Δk with $L^{-1} \ll \Delta k \ll 1$. If two eigenstates have the same occupied fraction in each bin, the difference of their fixed-r correlations vanishes as the bin width shrinks. Within a bin, e^{ikr} barely changes. Wick's rule extends the result to finite-support fermionic observables. There remain exponentially many microscopic occupation patterns compatible with each smooth distribution. A generalized microcanonical ensemble samples these patterns; a GGE assigns Lagrange multipliers to their macroscopic constraints. Their local equivalence is an ensemble-equivalence statement with regularity assumptions, not an identity at finite size.

11.5 A transverse-field Ising quench

Consider

$$H(h) = -J\sum_j \left(\sigma_j^x \sigma_{j+1}^x + h\sigma_j^z\right), \qquad J > 0.$$

With $\sigma_j^z = 1 - 2c_j^\dagger c_j$, a Jordan–Wigner transformation turns the spin interaction into hopping plus pairing. Fourier transformation groups modes $k, -k$. Diagonalizing their 2×2 particle-hole matrix is a rotation with angle

$$e^{i\theta_k(h)} = \frac{h - \cos k + i\sin k}{\sqrt{1 + h^2 - 2h\cos k}}, \qquad \epsilon_k(h) = 2J\sqrt{1 + h^2 - 2h\cos k}. \tag{11.6}$$

The resulting Bogoliubov operators satisfy $H = \sum_k \epsilon_k(\gamma_k^\dagger\gamma_k - 1/2)$. If the initial state is the ground state at h_0 and time evolution uses h, composing the two rotations gives, up to an irrelevant pair phase,

$$|\psi_0\rangle = \prod_{k>0}\left[\cos\frac{\Delta\theta_k}{2} + i\sin\frac{\Delta\theta_k}{2}\gamma_k^\dagger\gamma_{-k}^\dagger\right]|0_h\rangle, \quad \Delta\theta_k = \theta_k(h) - \theta_k(h_0).$$

Consequently

$$n_k = \langle\gamma_k^\dagger\gamma_k\rangle_0 = \sin^2\frac{\Delta\theta_k}{2} = \frac{1-\cos\Delta\theta_k}{2}. \tag{11.7}$$

At each positive k, the state is a superposition of an empty and a doubly occupied pair. Its diagonal ensemble therefore has $\langle I_k I_{-k}\rangle = n_k$, whereas a product occupation GGE gives n_k^2. This illustrates the global/local distinction concretely. Oscillating anomalous terms have frequencies $2\epsilon_k$ and disappear from generic fixed-support late-time correlations when dispersive dephasing applies.

For example, the stationary transverse magnetization is

$$\langle \sigma^z \rangle_{\text{stat}} = \int_{-\pi}^{\pi} \frac{\mathrm{d}k}{2\pi} \cos\theta_k(h)\,[1-2n_k].$$

For a quench from $h_0 \to +\infty$ to $h = 1$, $\theta_k(h_0) = 0$ and $\cos\theta_k(1) = |\sin(k/2)|$. The integral is the average of $\sin^2(k/2)$, hence $\langle \sigma^z \rangle_{\text{stat}} = 1/2$. This calculation uses a covariance integral instead of a sum over 2^L spin states. At flat-dispersion points, such as $h = 0$, persistent coherent oscillations need separate treatment.

11.5.1 Truncating the charges

Mode occupations are nonlocal, but suitable combinations are local. In the Ising chain, multiplying the quadratic Hamiltonian's momentum-space matrix by $\cos(rk)$ creates a conserved quadratic charge with support extending only a distance of order r. Its Bogoliubov form is proportional to $\sum_k \epsilon_k \cos(rk)(I_k - 1/2)$. Current-type charges supply the appropriate odd-in-k family when the state requires it. Taking sufficiently many such charges is equivalent, for regular translation-invariant states, to specifying the mode distribution.

A truncated GGE retains only charges with range up to a cutoff R, chooses their multipliers from the initial expectations, and predicts a finite subsystem density matrix. Increasing R restores increasingly detailed spatial information. The practical target is convergence of the reduced state or of selected correlations as R grows. It is not exact recovery of every n_k at a small cutoff. Studies of the Ising chain show why the shortest-range charges are especially valuable for small subsystems [33]. Near criticality or for nonsmooth mode data, convergence can be slower; there is no universal finite cutoff that works for all states and observables.

11.6 Interacting integrability: the XXZ completeness problem

The spin-1/2 XXZ chain,

$$H = J\sum_j \left(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z\right), \qquad S^a = \sigma^a/2,$$

is interacting after fermionization. In the gapped regime $\Delta = \cosh\eta > 1$, its Bethe quasiparticles include bound complexes of $a = 1, 2, \ldots$ overturned spins. A homogeneous thermodynamic eigenstate is described by particle and hole densities $\rho_a(\lambda)$ and $\rho_a^h(\lambda)$ for each species, with rapidity $\lambda \in [-\pi/2, \pi/2]$. Rapidity plays the role of a momentum parameter. Scattering changes the available density of states, giving coupled constraints

$$\rho_a + \rho_a^h = a_a - \sum_b K_{ab} * \rho_b, \qquad a_a(\lambda) = \frac{\sinh(a\eta)}{\pi[\cosh(a\eta) - \cos(2\lambda)]}. \tag{11.8}$$

Here $*$ integrates over the rapidity interval. One standard kernel convention is $K_{ab} = (1 - \delta_{ab})a_{|a-b|} + 2\sum_{j=1}^{\min(a,b)-1} a_{|a-b|+2j} + a_{a+b}$. This equation is the interacting counterpart of counting allowed free momentum states. A charge density has the linear form $q_\alpha = \sum_a \int \mathrm{d}\lambda\, q_{\alpha a}(\lambda)\rho_a(\lambda)$. Its one-particle eigenvalues $q_{\alpha a}$ characterize which species and rapidities that charge can distinguish.

Counting ways to fill particle/hole slots gives the entropy density

$$s_{\mathrm{YY}} = \sum_a \int \mathrm{d}\lambda\, \left[(\rho_a + \rho_a^h)\ln(\rho_a + \rho_a^h) - \rho_a \ln \rho_a - \rho_a^h \ln \rho_a^h\right].$$

Maximizing it subject to Equation 11.8 and the charges is a generalized thermodynamic Bethe ansatz. It reduces the stationary-state problem to functions of one variable, but only if the constraints determine the root densities. An infinite list of charges can still be incomplete: infinitely many measurements of one combination need not distinguish all bound-state species.

This issue became decisive in quenches from the Néel product state $|\uparrow\downarrow\uparrow\downarrow\cdots\rangle$ (or its translation-symmetric combination). A GGE using the then-standard strictly local transfer-matrix charges disagreed with an independently obtained quench-action stationary state [34]. The quench-action method weights thermodynamic eigenstates by their overlaps with the initial state and their state-counting entropy, then finds the dominant macrostate. It provides a useful independent benchmark rather than assuming a chosen charge list is complete.

The resolution was to include additional *quasilocal* charges: extensive operators whose density has spatial tails that decay sufficiently fast, rather than stopping at a fixed range. Suitable families distinguish information about bound quasiparticle densities missed by the earlier charge set, and a complete construction recovers the tested stationary state [35]. Thus the failure of a particular GGE is a diagnosis of missing constraints, not a disproof of maximum entropy. The explicit kernels and charge families depend on the anisotropy regime; the formulas above were restricted to $\Delta > 1$ for that reason.

11.7 Exercises and worked answers

1. **Same energy, different equilibrium.** For three free modes with energies $(-1, 0, 1)$, compare occupation profiles $(1/2, 0, 1/2)$ and $(0, 1, 0)$. Do particle number and energy determine the GGE?

   Answer. Both have $N = 1$ and $E = 0$, but their central occupations are 0 and 1. A measurement of I_2 distinguishes them. Two constraints cannot determine three independent occupation expectations. Thermalization requires a mechanism that changes the extra conserved data.

2. **A global distinction.** A pair of modes has the diagonal state $\rho = (1-n)|00\rangle\langle 00| + n|11\rangle\langle 11|$. Compute its entropy and that of the product GGE with the same occupations.

   Answer. Write $h(n) = -n\ln n - (1-n)\ln(1-n)$. The diagonal state has entropy $h(n)$; the product state has entropy $2h(n)$. The missing $h(n)$ is their classical mutual information. Local equivalence in a many-mode spatial limit does not imply equal extensive entropies.

3. **Check a bosonic string.** Expand $\langle b_i^\dagger b_{i+2}\rangle$ and derive the expression in this chapter using Wick's rule.

   Answer. The string is $1 - 2n_{i+1}$. Wick's rule gives $\langle c_i^\dagger n_{i+1} c_{i+2}\rangle = C_{i,i+2}C_{i+1,i+1} - C_{i,i+1}C_{i+1,i+2}$. Subtract twice this from $C_{i,i+2}$.

4. **Design a completeness test.** What would demonstrate that a charge set is insufficient even if it contains infinitely many members?

   Answer. Find two admissible macrostates with identical values of all listed charges and different values of a local observable. Equivalently, find a nonzero change $\delta\rho_a$ satisfying the linearized Bethe constraints and $\sum_a \int q_{\alpha a}\delta\rho_a = 0$ for every listed α, while changing a physical correlation. Counting charges alone cannot exclude such a null space.

12 Prethermalization and kinetic relaxation

A small violation of integrability need not immediately erase the conserved information described in the previous chapter. Local correlations can relax rapidly to a constrained state, remain near it for a long interval, and only later move toward thermal equilibrium. This separation of stages is called *prethermalization*. It is a statement about a hierarchy of dynamical time scales, not a new exact equilibrium ensemble.

This chapter derives a useful weak-coupling description. We distinguish exact perturbative identities from the assumptions that turn them into irreversible kinetic equations. Throughout, $\hbar = k_B = 1$.

12.1 Fast relaxation and slowly changing charges

Write

$$H = H_0 + gV, \qquad [H_0, Q_\alpha] = 0, \tag{12.1}$$

where g is dimensionless and small, and V has the dimensions of energy. The exact equation is $\mathrm{d}\langle Q_\alpha\rangle/\mathrm{d}t = ig\langle[V, Q_\alpha]\rangle$. If dephasing under H_0 produces a locally stationary GGE on a time τ_{deph}, we may hope that, at later times, the state is approximately specified by slowly varying expectations $q_\alpha(t)$. The proposed local description is

$$\rho_{\mathrm{slow}}(t) = Z(t)^{-1} \exp\left[-\sum_\alpha \lambda_\alpha(t) Q_\alpha\right].$$

This ansatz concerns finite-support observables; the exact global state remains unitarily related to the initial state. If it is diagonal in a common charge basis, $\mathrm{Tr}\,\rho_{\mathrm{slow}}[V, Q_\alpha] = 0$. Indeed, the diagonal matrix elements of that commutator vanish. Dissipative charge drift then usually starts at order g^2, while order-g effects can renormalize energies or generate coherent motion within degenerate subspaces.

An observed plateau may therefore correspond to a GGE of *dressed* approximate charges, rather than the bare Q_α. Weak interaction quenches in fermionic systems give concrete examples of correlation buildup, a long-lived nonthermal distribution, and later kinetic relaxation [36]. Detailed numerical work also finds prethermal plateaux and their subsequent drift in models with weak integrability breaking [37]. These are model-dependent observations supporting the framework, not a guarantee for every perturbation.

12.2 What second-order perturbation theory actually gives

In the interaction picture, $V_I(t) = e^{iH_0 t} V e^{-iH_0 t}$ and $\dot{\rho}_I = -ig[V_I(t), \rho_I]$. Integrating once and substituting back into the expectation-value equation gives

$$\frac{\mathrm{d}\langle Q_\alpha \rangle}{\mathrm{d}t} = ig \,\mathrm{Tr}\, \rho_0 [V_I(t), Q_\alpha] - g^2 \int_0^t \mathrm{d}s \;\mathrm{Tr}\, \rho_0 [V_I(s), [V_I(t), Q_\alpha]] + O(g^3) \tag{12.2}$$

at fixed time. This expression contains memory of the preparation. It is not yet a closed equation for $q_\alpha(t)$, and its error estimate is not uniform at arbitrarily long times. Terms proportional to $g^2 t$ eventually invalidate a naive fixed-order expansion.

To obtain a kinetic equation one assumes that the correlation kernel decays on a time τ_c, while the occupations change on a much longer time τ_{kin}. One then replaces the old state inside the kernel by the slowly evolving local GGE, neglects its change during τ_c, and extends the memory integral to infinity. This is a Markov approximation with a separation of time scales. Taking $g \to 0$ while $g^2 t$ stays finite identifies the usual kinetic limit.

The real part of the oscillatory integral selects resonant transitions. For two eigenstates of H_0, first-order transition amplitudes give

$$P_{a \leftarrow b}(t) = g^2 |V_{ab}|^2 \frac{4 \sin^2[(E_a - E_b)t/2]}{(E_a - E_b)^2}.$$

In a dense spectrum, integrated against a smooth function of energy, $4\sin^2(\omega t/2)/(t\omega^2) \to 2\pi\delta(\omega)$. Therefore the transition rate is

$$W_{a \leftarrow b} = 2\pi g^2 |V_{ab}|^2 \delta(E_a - E_b). \tag{12.3}$$

Principal-value contributions from the imaginary part shift energies and should be absorbed into the effective quasiparticle dynamics when relevant. A finite isolated system has discrete lines, so replacing the sinc peak by a delta function requires an intermediate time window: long compared with microscopic correlation times, short enough that individual level discreteness and recurrences do not dominate. The thermodynamic and weak-coupling limits are part of the argument.

For a Hermitian static perturbation the leading rates are symmetric under $a \leftrightarrow b$. An approximate master equation has $\dot{p}_a = \sum_b (W_{a \leftarrow b} p_b - W_{b \leftarrow a} p_a)$. It preserves probability and redistributes weight among resonant states. Its apparent irreversibility comes from discarding coherences and memory, not from modifying the exact unitary law.

12.3 From many-body transitions to a fermionic collision integral

For weakly interacting fermions take $H_0 = \sum_k \epsilon_k n_k$. A two-body interaction contains terms $c_1^\dagger c_2^\dagger c_4 c_3$. The indices include momentum and any spin or band label. A collision transfers two occupied incoming modes to two empty outgoing modes. If the local state approximately factorizes in quasiparticle occupations, its probability contains products of n_k and $1 - n_k$.

With a nonnegative rate $\mathcal{W}_{12;34}$ that includes the golden-rule factor, the momentum and energy constraints, the antisymmetrized matrix element, and consistent counting of equivalent channels, one obtains

$$\dot{n}_1 = \sum_{234} \mathcal{W}_{12;34} \left[(1-n_1)(1-n_2)n_3 n_4 - n_1 n_2 (1-n_3)(1-n_4) \right]. \tag{12.4}$$

We use the ordered-index convention in which the rate is symmetric under $1 \leftrightarrow 2$, $3 \leftrightarrow 4$, and $(12) \leftrightarrow (34)$; any identical-particle counting factor belongs in $\mathcal{W}$. The two terms are gain and loss. Pauli blocking is essential: a collision cannot fill an already occupied fermion mode. Bosonic collisions instead contain stimulated-emission factors $1 + n$.

The energy constraint is $\epsilon_1 + \epsilon_2 = \epsilon_3 + \epsilon_4$. Crystal momentum is conserved modulo a reciprocal lattice vector. Summing Equation 12.4 over all modes and pairing gain with loss proves $\dot{N} = 0$. Multiplication by ϵ_1 and symmetrization proves conservation of $\sum_k \epsilon_k n_k$. This is the leading kinetic energy; conservation of the exact interacting energy may require quasiparticle shifts and correlation-energy corrections at higher order.

The closure assumes that correlations between incoming occupations can be neglected at the accuracy of interest. It can fail during initial correlation buildup, near singular resonances, or when additional slow variables are omitted. Integrable interacting quasiparticles require their own scattering and charge structure; a bare free-particle equation should not be transplanted without a derivation.

12.4 An entropy theorem and the destination of collisions

Define the fermionic kinetic entropy

$$S_{\mathrm{kin}} = -\sum_k [n_k \ln n_k + (1-n_k)\ln(1-n_k)].$$

It is the entropy of a product occupation ensemble, not the conserved entropy of the exact isolated density matrix. Put $X = (1-n_1)(1-n_2)n_3 n_4$ and $Y = n_1 n_2 (1-n_3)(1-n_4)$. Differentiating and using the rate symmetries gives

$$\dot{S}_{\mathrm{kin}} = \frac{1}{4} \sum_{1234} \mathcal{W}_{12;34} (X - Y) \ln \frac{X}{Y} \geq 0. \tag{12.5}$$

The inequality follows because $\ln x$ is increasing. If $X > Y$, both factors are positive; if $X < Y$, both are negative. Zero occupations are handled by continuous limits. The factor $1/4$ comes from averaging over exchanging the two members of each pair and exchanging incoming with outgoing states.

Equality requires $X = Y$ for every active channel. Let $\varphi_k = \ln[n_k/(1-n_k)]$. The stationarity condition is

$$\varphi_1 + \varphi_2 = \varphi_3 + \varphi_4$$

on all allowed collisions. Thus every additive collision invariant produces a stationary distribution. If particle number and energy are the only independent invariants and the collision network connects the relevant states, then

$$\varphi_k = \beta\mu - \beta\epsilon_k, \qquad n_k = \frac{1}{e^{\beta(\epsilon_k - \mu)} + 1}.$$

If ordinary momentum is conserved without Umklapp processes, a drifting distribution can include a term proportional to momentum. If channels are disconnected, additional invariants survive and a nonthermal stationary state is possible. The entropy theorem proves monotonicity within the kinetic approximation; convergence to a unique thermal state needs these extra connectivity assumptions.

> **i** A one-dimensional kinematic obstruction
>
> For identical particles with $\epsilon_k = k^2/(2m)$, two-body collisions conserve $P = k_1 + k_2$ and $k_1^2 + k_2^2$. Since $k_1 k_2 = [P^2 - (k_1^2 + k_2^2)]/2$, the outgoing momenta solve the same quadratic equation as the incoming ones. Hence $\{k_3, k_4\} = \{k_1, k_2\}$. Such collisions only exchange labels and cannot relax the distribution. Thermalization then needs another channel: three-body scattering, a lattice dispersion with suitable phase space, additional species, or another integrability-breaking process. A formal g^2 prefactor does not establish a nonzero relaxation rate.

12.5 Time scales and the interpretation of a plateau

If a typical resonant rate is $g^2 J$ for microscopic energy J, dimensional analysis suggests $\tau_{\mathrm{kin}} \sim (g^2 J)^{-1}$. The prethermal window is $\tau_{\mathrm{deph}} \ll t \ll \tau_{\mathrm{kin}}$. This estimate assumes nonvanishing second-order phase space and nonsingular matrix elements. Selection rules can eliminate the leading channel. Low-temperature Pauli blocking can add strong temperature factors. Approximate symmetries, constrained dynamics, and higher-order processes can produce other powers or much longer times. High-frequency driven prethermalization is another mechanism and need not obey this weak-static-perturbation estimate.

A useful numerical diagnosis follows several observables. First establish fast decay of oscillations. Then test whether intermediate values agree with an appropriate GGE or dressed-charge ensemble. Finally vary g and check a predicted rescaling of the slow drift, while controlling finite size and accessible time. A flat trace by itself cannot distinguish a true stationary state from an unresolved slow evolution.

The hierarchy also clarifies what "thermalization" means operationally. A system may have nearly thermal densities while its momentum distribution is still nonthermal, or short-range correlators may settle while a transport mode relaxes slowly. The chosen observables and tolerance belong in any quoted equilibration time.

12.6 Exercises and worked answers

1. **Verify charge conservation.** Show directly that Equation 12.4 conserves a quantity $Q_f = \sum_k f_k n_k$ whenever $f_1 + f_2 = f_3 + f_4$ for every active collision.

   Answer. After the same symmetrization used in the entropy proof, $\dot{Q}_f = \frac{1}{4}\sum \mathcal{W}(X - Y)(f_1 + f_2 - f_3 - f_4) = 0$. Choosing $f = 1$ and $f = \epsilon$ gives number and energy.

2. **A single reaction channel.** Consider four distinct modes with only $12 \leftrightarrow 34$ allowed. Why need its stationary state not be determined by number and energy alone?

   Answer. Each event changes occupations by a multiple of $(-1, -1, +1, +1)$. Every vector f orthogonal to this vector is invariant, a three-dimensional space. For example $n_1 - n_2$

and $n_3 - n_4$ are separately conserved. Detailed balance fixes only one relation among four occupations. More channels are needed for thermal uniqueness.

3. **Scaling test.** Suppose the same local observable obeys $A(t) - A_{\mathrm{th}} \simeq A_0 e^{-cg^2 Jt}$. What happens when g is halved?

   Answer. The slow decay time increases by four. Curves should approximately collapse when plotted against $g^2 Jt$ after the initial transient. Failure of collapse can indicate higher-order processes, a changed plateau, finite-size effects, or the absence of the assumed kinetic regime; it is not by itself evidence against unitary dynamics.

4. **Entropy accounting.** How can Equation 12.5 coexist with constant exact von Neumann entropy?

   Answer. S_{kin} uses only single-mode occupations and discards intermode correlations. Unitary evolution can move information into correlations while the entropy of the maximum-entropy product state consistent with the retained data rises. The two entropies are functions of different information.

Part IV

Work through it

13 Mathematical tools and two revealing examples

This chapter collects the seven auxiliary topics behind the book's main argument. The kicked rotor illustrates the distinction between classical diffusion and quantum interference. Zeta zeros illustrate spectral universality and the limits of numerical evidence. The remaining sections provide proofs and continuum derivations that can be reused independently. A probability vector is a column vector throughout; a doubly stochastic matrix has both row sums and column sums equal to one.

13.1 The kicked rotor: from a map to localization

13.1.1 Classical stroboscopic dynamics

Choose dimensionless time with one kick per unit interval and Hamiltonian

$$H(\theta, P, t) = \frac{P^2}{2} + K\cos\theta \sum_{n\in\mathbb{Z}} \delta(t-n), \qquad \theta \equiv \theta + 2\pi.$$

Let (θ_n, P_n) denote the state just before kick n. Integrating Hamilton's equation through the impulse and then through free rotation gives the *standard map*:

$$P_{n+1} = P_n + K\sin\theta_n, \qquad \theta_{n+1} = \theta_n + P_{n+1} \pmod{2\pi}. \tag{13.1}$$

Its tangent matrix in the order (θ, P) is $\left(\begin{smallmatrix} 1+K\cos\theta & 1 \\ K\cos\theta & 1 \end{smallmatrix}\right)$, with determinant one. Thus phase-space area is preserved even when nearby trajectories separate exponentially. For small kicks many invariant curves survive; for stronger kicks large chaotic regions appear, with regular islands often remaining.

The fixed points at $P = 0$ are $\theta = 0$ and $\theta = \pi$. Their tangent eigenvalues obey

$$\lambda^2 - (2 + K\cos\theta_*)\lambda + 1 = 0.$$

For our positive-cosine kick convention, $\theta_* = \pi$ is the potential minimum. Its trace is $2 - K$, so for $0 < K < 4$ the eigenvalues are a complex-conjugate pair of modulus one: the fixed point is elliptic at linear order. At $K > 4$ they become real reciprocal eigenvalues, one with modulus larger than one, giving a hyperbolic instability. At $\theta_* = 0$, the trace is $2 + K > 2$, so that point is hyperbolic for every $K > 0$. The endpoints $K = 0, 4$ are marginal and need nonlinear analysis. Near the elliptic point, writing $\theta = \pi + q$ gives $q_{n+1} - 2q_n + q_{n-1} \simeq -Kq_n$; for $K \ll 1$ this approaches a pendulum's small-oscillation equation. Hyperbolicity of a fixed point is local information: it does not prove that the entire map is chaotic or determine the threshold for unrestricted momentum transport.

If successive angles were independent and uniform, each momentum increment would have mean zero and variance $K^2/2$. Then $\langle (P_n - P_0)^2 \rangle \simeq D_P n$, with $D_P \simeq K^2/2$. This random-phase estimate is useful in a strongly chaotic regime. Correlations between kicks, trapping near islands, and accelerator modes can change the diffusion coefficient or even the diffusion law. Classical chaos alone does not justify treating every kick as independent noise.

13.1.2 Floquet evolution

Quantization gives momentum states $|m\rangle$ with $P|m\rangle = \hbar_{\text{eff}} m|m\rangle$, $m \in \mathbb{Z}$. Wavefunctions are periodic in θ; equivalently, $[\theta, P] = i\hbar_{\text{eff}}$ holds in the usual local angular-coordinate sense. Evolution from just before one kick to just before the next is

$$U_F = \exp\left(-\frac{iP^2}{2\hbar_{\text{eff}}}\right) \exp\left(-\frac{iK\cos\theta}{\hbar_{\text{eff}}}\right). \tag{13.2}$$

Using the Fourier expansion $e^{-iz\cos\theta} = \sum_l (-i)^l J_l(z) e^{il\theta}$ gives

$$\langle m|U_F|n\rangle = e^{-i\hbar_{\text{eff}} m^2/2} (-i)^{m-n} J_{m-n}(K/\hbar_{\text{eff}}).$$

A numerical simulation can alternate a diagonal kick in angle representation with a diagonal free phase in momentum representation, using Fourier transforms. A momentum cutoff must be large enough that the spreading wavepacket never reaches its artificial boundary.

For generic nonresonant parameters in a classically diffusive regime, interference can stop the quantum momentum spreading. Floquet eigenstates become localized in momentum, with envelopes schematically $|\psi_m|^2 \sim e^{-2|m-m_0|/\ell}$. The relation between the rotor eigenvalue problem and a one-dimensional Anderson-type lattice problem explains this *dynamical localization* [38]. The phase sequence is deterministic, so one must not silently replace every parameter choice by independent random disorder.

A useful scaling estimate compares classical spreading in mode number, $\langle (\Delta m)^2 \rangle \sim (D_P/\hbar_{\text{eff}}^2) n$, with a break time of order the number ℓ of appreciably occupied Floquet components. Setting $\ell^2 \sim (D_P/\hbar_{\text{eff}}^2)\ell$ gives $\ell \sim D_P/\hbar_{\text{eff}}^2$ and $n_{\text{break}} \sim \ell$, up to convention-dependent numerical factors. This is a scaling argument, not an exact formula across all K.

Resonances are an essential exception. When $\hbar_{\text{eff}}/(4\pi)$ is rational, the free phases are periodic in momentum and resonance or antiresonance may replace localization. For $\hbar_{\text{eff}} = 4\pi$, the free operator is the identity. Starting from $m = 0$, after n kicks the angle wavefunction is proportional to $e^{-inK\cos\theta/\hbar_{\text{eff}}}$, so $\langle P^2 \rangle = K^2 n^2/2$: ballistic growth. At $\hbar_{\text{eff}} = 2\pi$, the free evolution translates the angle by π, successive cosine kicks cancel, and $U_F^2 = 1$. For a kicked particle rather than a rotor, conserved quasimomentum introduces another resonance condition. Weak noise or decoherence can also destroy interference and restore long-time diffusion.

13.2 Zeta zeros: unfolding, evidence, and conjecture

For $\Re s > 1$, the Riemann zeta function is $\zeta(s) = \sum_{n \geq 1} n^{-s}$. Analytic continuation defines it elsewhere, apart from its pole at $s = 1$. Its nontrivial zeros lie in the critical strip $0 < \Re s < 1$. The Riemann hypothesis asserts that all lie on $\Re s = 1/2$. It is separate from statements about spacing correlations.

Write zeros on the critical line as $1/2 + i\gamma_n$ with positive increasing γ_n. In a range in which all zeros are known to lie on that line, or conditionally on the Riemann hypothesis, a smooth counting function is

$$\overline{N}(T) = \frac{T}{2\pi} \ln \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8}.$$

The local density changes slowly with height: $\overline{N}'(T) = (2\pi)^{-1} \ln(T/2\pi)$. Raw spacings at different heights are therefore incomparable. Define unfolded positions $x_n = \overline{N}(\gamma_n)$ and spacings $s_n = x_{n+1} - x_n$. Their local mean approaches one. In a narrow window near T, the equivalent approximation is $s_n \simeq (\gamma_{n+1} - \gamma_n) \ln(T/2\pi)/(2\pi)$.

The conjectured distinct-pair correlation is the bulk GUE expression

$$R_2(s) = 1 - \left(\frac{\sin \pi s}{\pi s} \right)^2 . \tag{13.3}$$

The self-pair delta function at $s = 0$ is excluded from this convention. Expanding the sine shows $R_2(s) \sim \pi^2 s^2/3$ near zero, a form of quadratic repulsion. Pair correlation is not the same statistic as the nearest-neighbor distribution, although both can be compared with random matrices.

Montgomery obtained conditional results for a restricted Fourier range and proposed the fuller pair-correlation statement [39]. Odlyzko's high-precision computations found detailed numerical agreement with random-matrix predictions [40]. This is strong numerical evidence for spectral universality. It is neither a proof of the full correlation conjecture nor a proof of the Riemann hypothesis.

The Hilbert–Pólya idea asks whether a suitable self-adjoint operator could encode the zeros as spectral data, so that its real eigenvalues would force their real parts to be 1/2. A successful proof needs a rigorously established correspondence with all relevant zeros; merely constructing an operator with selected imaginary parts as eigenvalues achieves nothing by itself. No such operator or proof is supplied by matching a spacing histogram. The physics lesson is methodological: a universal statistic can reveal structure without identifying its microscopic mechanism.

13.2.1 Prime modes and complex-temperature zeros

Unique prime factorization supplies another connection to statistical mechanics. For $\Re s > 1$, absolute convergence permits expanding

$$\prod_{p \text{ prime}} (1 - p^{-s})^{-1} = \prod_{p} \sum_{n_p \geq 0} p^{-s n_p} = \sum_{n \geq 1} n^{-s} = \zeta(s).$$

Each choice of nonnegative integers n_p corresponds to exactly one integer $n = \prod_p p^{n_p}$. Introduce independent bosonic modes with normal-ordered energies $H = E_* \sum_p (\ln p) N_p$, where $E_* > 0$ sets the energy unit and the vacuum energy is zero. A Fock state then has energy $E_n = E_* \ln n$, with one state per positive integer. Its partition function is

$$Z(\beta) = \prod_p \frac{1}{1 - e^{-\beta E_* \ln p}} = \zeta(\beta E_*), \qquad \Re(\beta E_*) > 1.$$

The convergence restriction is physical: at high enough temperature the increasing number of accessible logarithmic-energy levels makes the trace diverge. Analytically continuing ζ beyond this

domain does not make that divergent trace a thermal state. In particular, the critical-line zeros are outside the domain of this literal infinite-mode Gibbs ensemble. This distinction prevents the partition-function analogy from becoming a purported physical proof about the zeros.

For any Hamiltonian with a convergent Gibbs trace, its energy probability distribution has characteristic function

$$\chi_\beta(u) = \langle e^{iuH} \rangle_\beta = \frac{Z(\beta - iu)}{Z(\beta)}.$$

Here u has units of inverse energy. Thus zeros of the characteristic function are complex-temperature zeros of the partition function on the corresponding vertical line. Zeros in complex inverse temperature are called Fisher zeros. A finite-level cutoff makes Z a finite sum, so its complex zeros are unambiguous; interpreting their accumulation as a thermodynamic singularity requires a specified limit. Neither analytic continuation nor removal of a cutoff can be interchanged with that limit without justification. For the untruncated prime-mode model, the Euler product has no zeros in its convergent half-plane, so its physical characteristic function does not sample the critical-line zeros.

A related Loschmidt amplitude is obtained by purifying the Gibbs state: $|\Psi_\beta\rangle = \sum_n \sqrt{p_n}|n\rangle|n\rangle$. Evolving only one copy gives overlap $Z(\beta + it/\hbar)/Z(\beta)$. This establishes a precise relation to return amplitudes for that preparation, not for an arbitrary quench. Proposed broader links between complex-zero patterns and ergodicity remain conjectural. The prime-mode Hamiltonian itself has conserved occupation numbers; a partition-function identity alone does not establish chaotic dynamics.

13.3 When the uniform state is an attractor

Let D be a $d \times d$ doubly stochastic matrix and $u_i = 1/d$. Since its rows sum to one, $Du = u$. Its columns summing to one ensure normalization of Dp. These facts establish a stationary uniform state, but they do not establish attraction. The identity matrix preserves every distribution, and a cyclic permutation rotates distributions forever.

A sufficient condition is *primitivity*: for some integer m, every entry of D^m is positive. Set $a = \min_{ij}(D^m)_{ij} > 0$ and $\delta = da \leq 1$. If $\delta < 1$, write

$$D^m = \delta J + (1 - \delta)R, \qquad J_{ij} = 1/d,$$

where R is again doubly stochastic and nonnegative. If $\delta = 1$, $D^m = J$ and mixing is already exact. For any vector x, column stochasticity and the triangle inequality give

$$\|Rx\|_1 \leq \sum_{ij} R_{ij}|x_j| = \|x\|_1.$$

Because $J(p - u) = 0$,

$$\|D^{mn}p - u\|_1 \leq (1 - \delta)^n \|p - u\|_1. \tag{13.4}$$

Intermediate applications of D cannot increase this norm, so $D^n p \to u$. The bound is constructive, although often far from optimal. Spectrally, primitivity means that the eigenvalue one is simple and all other eigenvalues have modulus smaller than one. For a finite Markov chain it is equivalent to

irreducibility plus aperiodicity. Reducible dynamics can retain independent conserved sectors, and a uniform state over the whole Hilbert space is then the wrong target. Time-dependent products require their own mixing conditions; primitivity of an unrelated average matrix is insufficient.

13.4 A constructive proof of Birkhoff's theorem

The theorem. Every finite doubly stochastic matrix is a convex combination of permutation matrices: $D = \sum_a w_a P_a$, $w_a \geq 0$, $\sum_a w_a = 1$.

First regard the nonzero entries of D as edges of a bipartite graph connecting rows to columns. For a subset S of rows, let $N(S)$ be the columns connected to them. Their total row weight is $|S|$, and it all lies in $N(S)$. The total weight in those columns is $|N(S)|$, so $|N(S)| \geq |S|$.

This condition guarantees a matching covering all rows. To see why, start with a matching of maximal size. If a row is unmatched, follow alternating paths: an unused edge from a row to a column, then the matching edge back to a row. If an unmatched column is reached, reversing the path increases the matching, contradicting maximality. Otherwise every reached column is matched to a reached row, while the initial unmatched row supplies one additional reached row. The reached rows then outnumber their neighboring columns, contradicting the condition above. Hence a perfect matching exists.

Choose its permutation matrix P and put $w = \min_i D_{i,P(i)} > 0$. Then $D - wP$ is nonnegative with row and column sums $1 - w$. If $w = 1$ we are done. Otherwise

$$D = wP + (1 - w)D_1, \qquad D_1 = \frac{D - wP}{1 - w}$$

and D_1 is doubly stochastic. At least one previously positive entry has become zero, and no zero has become positive. Repeating the procedure must terminate after finitely many steps. Expanding the nested decompositions gives the desired positive weights, which sum to one.

For example, $\left(\begin{smallmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{smallmatrix}\right) = 0.6I + 0.4\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. This is an algebraic representation of population mixing. It does not assert that a particular unitary evolution physically tosses a classical coin between permutations. Quantum transition probabilities $D_{ij} = |U_{ij}|^2$ form a subset of doubly stochastic matrices, and Birkhoff's theorem applies to that subset as well.

13.5 A rigorous passive-work inequality

Consider a cyclic process: the final Hamiltonian equals the initial one. Order its energies as $E_1 \leq \cdots \leq E_d$ and let the initial state be diagonal with $p_1 \geq \cdots \geq p_d$. Such a state is *passive*. A unitary process gives final energy populations $q = Dp$, where $D_{ij} = |U_{ij}|^2$ is doubly stochastic. We prove

$$\langle W \rangle = \sum_i E_i (q_i - p_i) \geq 0. \tag{13.5}$$

For each $k < d$, define $r_j = \sum_{i \leq k} D_{ij}$. Then $0 \leq r_j \leq 1$ and $\sum_j r_j = k$. Since the p_j decrease with j, the largest possible value of $\sum_j r_j p_j$ is obtained by giving weight one to the first k entries. This can also be seen by repeatedly moving available weight from a larger index to a smaller one, which never lowers the sum. Therefore

$$\sum_{i\le k} q_i = \sum_j r_j p_j \le \sum_{i\le k} p_i.$$

Set $R_k = \sum_{i\le k}(q_i - p_i) \le 0$, with $R_0 = R_d = 0$. Discrete summation by parts yields

$$\langle W\rangle = \sum_i E_i(R_i - R_{i-1}) = \sum_{k=1}^{d-1}(E_k - E_{k+1})R_k \ge 0.$$

Each factor in each product is nonpositive. No chaos assumption, large-system limit, or canonical ensemble was used. The proof works for every doubly stochastic population map. Degenerate energies cause no problem because the corresponding energy gaps vanish.

A Gibbs distribution with $\beta \ge 0$ is passive because its populations decrease with energy. Conversely, a population inversion $p_j > p_i$ for $E_j > E_i$ permits work extraction by swapping levels: the change is $(E_j - E_i)(p_i - p_j) < 0$. Coherences also matter: a general passive state must commute with H, since an appropriate unitary can otherwise lower its energy. With degeneracies one can diagonalize within each energy subspace.

Passivity for one copy is weaker than *complete passivity*, which requires that no number of copies allow work extraction by a joint cycle. The latter singles out equilibrium Gibbs states and ground-state cases under the appropriate hypotheses [20,41]. For a noncyclic protocol with different initial and final spectra, Equation 13.5 is not the relevant comparison without additional terms.

13.6 Deriving the energy drift–diffusion relation

Let $P(E,t)$ be a density per unit energy and $w(E,\omega)\mathrm{d}\omega$ the rate for a jump from E to $E+\omega$. The gain–loss equation is

$$\partial_t P(E) = \int \mathrm{d}\omega\, [w(E-\omega,\omega)P(E-\omega) - w(E,\omega)P(E)]\,.$$

If jumps are small compared with the scale on which the distribution and rates vary, Taylor expansion of the gain term produces the first two Kramers–Moyal terms:

$$\partial_t P = -\partial_E(AP) + \frac{1}{2}\partial_E^2(BP), \qquad A(E) = \int \mathrm{d}\omega\, \omega w(E,\omega), \quad B(E) = \int \mathrm{d}\omega\, \omega^2 w(E,\omega). \quad (13.6)$$

A has units energy/time and B has units energy squared/time. On an infinitesimal interval they are the mean energy increment and its variance divided by elapsed time; the distinction between the second raw moment and variance is higher order in that interval. Neglecting third and higher jump moments needs a controlled small-jump regime, rather than an assumption that every finite-time work distribution is Gaussian.

Write the probability current as $j = AP - \frac{1}{2}\partial_E(BP)$. For energy-space evolution induced by a unital or doubly stochastic microscopic process at fixed Hamiltonian, the state uniform over individual levels is stationary. Its coarse-grained energy density is proportional to the density of levels $\Omega(E)$, which is generally not constant. For a bounded spectrum with reflecting endpoints, the stationary current vanishes. Substituting $P_\infty \propto \Omega$ gives

$$2A = B\,\partial_E \ln \Omega + \partial_E B = \beta(E)B + \partial_E B, \qquad \beta(E) = \partial_E S(E), \quad S(E) = \ln[\Omega(E)\delta E]. \tag{13.7}$$

The arbitrary fixed coarse-graining width δE drops out of β. The derivative term is required for spatially varying diffusion in the Itô/Fokker–Planck convention used here. Dropping it is an additional slowly-varying-coefficient approximation. This argument uses stationarity and a zero-current boundary condition; a doubly stochastic matrix need not satisfy pairwise detailed balance.

For a system weakly coupled to a thermal reservoir at inverse temperature β_b, the equilibrium energy density is $\Omega(E)e^{-\beta_b E}$. The corresponding zero-current relation is $2A = (\beta(E) - \beta_b)B + B'$. For two finite isolated subsystems exchanging energy, the stationary marginal is proportional to $\Omega_1(E)\Omega_2(E_{\mathrm{tot}} - E)$, so the entropy gradient is $\beta_1(E) - \beta_2(E_{\mathrm{tot}} - E)$. In the macroscopic limit, the drift therefore drives energy toward the subsystem of higher inverse temperature (the colder body), with the finite-size derivative term supplying the correction.

For several exchanged quantities x_i, the same calculation gives

$$A_i = \frac{1}{2}\sum_j \left(B_{ij}\partial_{x_j} S + \partial_{x_j} B_{ij}\right) \tag{13.8}$$

when the stationary diffusion has zero vector current. B_{ij} is a symmetric covariance-rate matrix. Nonzero circulation currents or time-reversal breaking require a more general treatment.

13.7 Deriving Onsager reciprocity

Take two large bodies exchanging extensive quantities x_i, each even under time reversal for the moment. Define the thermodynamic affinities $F_i = \partial S_{\mathrm{tot}}/\partial x_i$, with x_i oriented into the first body. For energy exchange $F_E = \beta_1 - \beta_2$. A microscopic reversibility relation for an exchange Δx over a mesoscopic interval τ has the near-equilibrium form

$$\frac{p_F(\Delta x)}{p_F(-\Delta x)} = e^{F\cdot\Delta x},$$

when the reversed experiment coincides with the forward one, affinities vary negligibly during the interval, and the coarse-grained exchange description is valid. At equilibrium $p_0(\Delta x) = p_0(-\Delta x)$. Linearizing the ratio gives $p_F(\Delta x) - p_F(-\Delta x) = p_0(\Delta x)\sum_j F_j \Delta x_j + O(F^2)$. Hence

$$J_i \equiv \frac{\langle \Delta x_i\rangle}{\tau} = \sum_j L_{ij}F_j + O(F^2), \qquad L_{ij} = \frac{\langle \Delta x_i \Delta x_j\rangle_0}{2\tau}. \tag{13.9}$$

The symmetry $L_{ij} = L_{ji}$ follows from the symmetry of a covariance. Positivity also follows: for real v_i, $v_i L_{ij} v_j = \langle (v\cdot\Delta x)^2\rangle_0/(2\tau) \geq 0$. Thus entropy production $\dot{S} = F\cdot J = F^{\mathsf{T}} L F$ is nonnegative to quadratic order. Reciprocity constrains cross effects, for example the relation between heat transport induced by a particle affinity and particle transport induced by a temperature affinity. It is the content of microscopic reversibility beyond a mere positivity condition [42].

There is a useful Green–Kubo version. For classical equilibrium currents, $\Delta x_i = \int_0^\tau I_i(t)\mathrm{d}t$. If their correlations decay and τ is long compared with that decay time but short compared with the relaxation of the bodies, stationarity gives

$$L_{ij} = \frac{1}{2}\int_{-\infty}^{\infty} \mathrm{d}t\, \langle I_i(t) I_j(0)\rangle_{\mathrm{eq}}.$$

For equal time-reversal parities, reversibility converts this into the familiar integral over positive times. In a quantum calculation the linear-response version uses the canonical Kubo correlation, with imaginary-time averaging, rather than assuming that noncommuting current products are classical random variables. In a thermal state it can be written $\langle A; B\rangle_{\mathrm{eq}} = \beta^{-1}\int_0^\beta \mathrm{d}\lambda\, \mathrm{Tr}[\rho_{\mathrm{eq}} e^{\lambda H} A e^{-\lambda H} B]$ for zero-mean operators. Forces must be normalized as entropy affinities, as above, to compare prefactors.

If a magnetic field B_{mag} or another time-reversal-odd control is present, reversal changes that control too. If variable x_i has time-reversal parity ε_i, the appropriate relation is Onsager–Casimir reciprocity, $L_{ij}(B_{\mathrm{mag}}) = \varepsilon_i\varepsilon_j L_{ji}(-B_{\mathrm{mag}})$. Symmetry at the same field is then not generally required. Ballistic currents or additional conserved quantities can prevent the correlation integral from converging to a finite diffusive coefficient. Far from equilibrium, derivatives of the covariance and higher cumulants enter; the linear relation proved here should not be asserted unchanged.

13.8 Exercises and worked answers

1. **Area preservation.** Compute the determinant of the standard map's tangent matrix. Does this rule out exponential separation?

   Answer. The determinant is $1 + K\cos\theta - K\cos\theta = 1$. One expanding tangent direction can coexist with one contracting direction, keeping their area product fixed. Volume preservation is compatible with chaos.

2. **A mixing counterexample.** The matrix $D = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ is doubly stochastic and irreducible. Why does it not approach the uniform state?

   Answer. It has period two, $D^2 = I$, and an eigenvalue -1. No power has all positive entries, so it is not primitive. A distribution $(p, 1-p)$ alternates with $(1-p, p)$.

3. **Compute passive work.** Let $E = (0, 1, 3)$ and $p = (0.6, 0.3, 0.1)$. A unitary swaps the first and last levels. Find the work.

   Answer. The final populations are $(0.1, 0.3, 0.6)$. The initial energy is 0.6 and the final energy 2.1, so $\langle W\rangle = 1.5$. The swap supplies energy to the system, as passivity requires.

4. **Check a diffusion law.** Suppose $\Omega(E) \propto E^\alpha$ and $B(E) = bE^r$ on a positive-energy interval. Find the zero-current drift.

   Answer. $\beta(E) = \alpha/E$, so Equation 13.7 gives $A(E) = b(\alpha + r)E^{r-1}/2$. Both the state-counting contribution α and the diffusion-gradient contribution r matter. Boundary conditions and validity of the small-jump approximation still have to be supplied.

5. **Reciprocity versus positivity.** Does positive entropy production by itself imply $L_{12} = L_{21}$?

   Answer. No. $F^{\mathsf{T}} L F$ depends only on the symmetric part of L. An antisymmetric part contributes zero to entropy production. Reciprocity needs microscopic reversibility and the appropriate parity/field reversal, not positivity alone.

14 A reproducible numerical laboratory

This chapter turns the distinctions developed throughout the book into calculations you can inspect and change. It uses Julia for the numerical work and CairoMakie for plotting. The checked-in figures are sufficient to read the chapter; a Julia installation is needed only to regenerate or extend them. All results in this chapter are newly generated examples, rather than reproductions of data in the review.

14.1 Reproduce before extending

From the project directory run:

```
julia --project=. -e 'using Pkg; Pkg.instantiate()'
julia --project=. scripts/generate_figures.jl
```

The project specifies Julia 1.12 and records package versions in `Manifest.toml`. The script uses seed `20261019` and sets the BLAS thread count to one. The seed fixes the random samples in this environment; exact floating-point output can still vary slightly across platforms and library versions. Both PNG and vector PDF versions are saved. The HTML book uses PNG and the combined PDF uses the vector sibling.

The complete Julia script is part of the published book, along with the project dependencies, manifest, and diagnostics. Read the Hamiltonian construction before trusting a plot: a wrong basis convention can produce plausible-looking dynamics.

14.2 Experiment 1: level statistics with a known ensemble

Start with the Gaussian orthogonal ensemble, so the statistical model is specified independently of any many-body conjecture. Draw a real matrix X with independent standard normal entries and set

$$H = \frac{X + X^{\mathsf{T}}}{\sqrt{2D}}, \qquad D = 350. \tag{14.1}$$

Its off-diagonal variance is $1/D$ and its diagonal variance is $2/D$. At large D its mean density has the semicircle form

$$\rho(E) = \frac{D}{2\pi}\sqrt{4 - E^2}, \qquad |E| < 2.$$

Unfolding transforms energy to mean level count. An antiderivative of the normalized density is

$$F(E) = \frac{1}{2} + \frac{1}{\pi}\left[\arcsin\frac{E}{2} + \frac{E}{4}\sqrt{4 - E^2}\right].$$

Check this by differentiation: the two derivative terms combine to $\sqrt{4-E^2}/(2\pi)$. Define unfolded coordinates $x_n = DF(E_n)$ and spacings $s_n = x_{n+1} - x_n$. The calculation retains eigenvalues 71 through 280 in each of 25 matrices, avoiding the spectral edges, and normalizes the pooled spacings to mean one. This ensemble-density unfolding is appropriate because the ensemble is known. An arbitrary spin chain generally needs its own smooth counting function or an unfolding-free statistic.

The comparison sample consists of independent unit-mean exponential random numbers. That is the exact nearest-neighbor spacing law for a homogeneous Poisson point process. No matrix diagonalization is needed to generate this benchmark. The spacing data include both samples.

The GOE Wigner surmise, $P(s) = (\pi s/2)e^{-\pi s^2/4}$, is exact for a suitably normalized two-level model and an excellent approximation to bulk large-GOE spacings. It is not the exact large-D distribution. The suppression at $s = 0$ is the useful qualitative distinction from e^{-s}.

Checkpoint. Plot the ratio $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$. Avoid forming a ratio across the boundary between two independent matrix samples; the checked-in pooled file is intended for the spacing histogram. For a new ratio calculation, keep each spectrum separate.

Worked answer. Form `diff(E)` within each retained spectrum, then compute neighboring ratios before pooling. A slowly varying local density approximately cancels. The Poisson mean is $2\ln 2 - 1$, obtained by integrating r against $2/(1+r)^2$ on $[0, 1]$. Ratios reduce unfolding sensitivity but still require symmetry resolution in a physical Hamiltonian.

14.3 Experiment 2: exact dynamics of a small spin chain

We use an open chain with $L = 9$ spin-1/2 sites and $D = 512$ states,

$$H = -\sum_{i=1}^{L-1} Z_i Z_{i+1} - 1.05 \sum_{i=1}^{L} X_i - \sum_{i=1}^{L} (0.35 + 0.07i) Z_i. \tag{14.2}$$

Here X, Z are Pauli matrices, not spin operators $\sigma/2$. Energies are in units of the Ising coupling and times in its inverse with $\hbar = 1$. The longitudinal field breaks the global spin flip, its spatial variation breaks reflection, and the transverse field mixes magnetization sectors. The model is real, so its remaining antiunitary symmetry is ordinary complex conjugation. These observations make the example useful for studying a generic finite chain; they do not prove chaos or ETH in an infinite system.

A computational state is an integer $s = 0, \ldots, 2^L - 1$. Site 1 is the least significant bit. A zero bit has $Z = +1$, a one bit $Z = -1$, and X_i flips bit $i - 1$. Thus the off-diagonal entry is `H[xor(s, 1 << (i-1)) + 1, s+1] = -1.05`. The extra one converts zero-based bit labels to Julia array indices. All ZZ and longitudinal-field terms are diagonal. This construction needs no tensor-library convention.

The observable is $A = Z_5$. Among computational states with $Z_5 = +1$, choose the one whose diagonal energy is nearest the mean eigenenergy. This deterministic rule selects a nonstationary product state with energy in the central region. It is a sudden preparation in the final Hamiltonian, equivalently a quench from a strong pinning Hamiltonian whose product ground state is the chosen configuration. The initial energy width remains finite; no narrow-shell limit is presumed.

Diagonalize $H = V\mathrm{diag}(E_n)V^\dagger$. If the chosen initial basis index is j, then $c_n = V_{jn}^*$. In this real model conjugation changes nothing, but it matters for complex Hamiltonians. Compute

$$
\begin{aligned}
A_{nn} &= \sum_s Z_5(s)|V_{sn}|^2, \\
\psi_s(t) &= \sum_n V_{sn} c_n e^{-iE_n t}, \\
A(t) &= \sum_s Z_5(s)|\psi_s(t)|^2, \\
A_{\mathrm{DE}} &= \sum_n |c_n|^2 A_{nn}.
\end{aligned}
\tag{14.3}
$$

The comparison microcanonical value is an unweighted average over $|E_n - E_0| < 0.8$, with $E_0 = \sum_n |c_n|^2 E_n$. Its window is an explicit finite-size choice. A better comparison study varies that width and system size instead of choosing a window after looking at agreement.

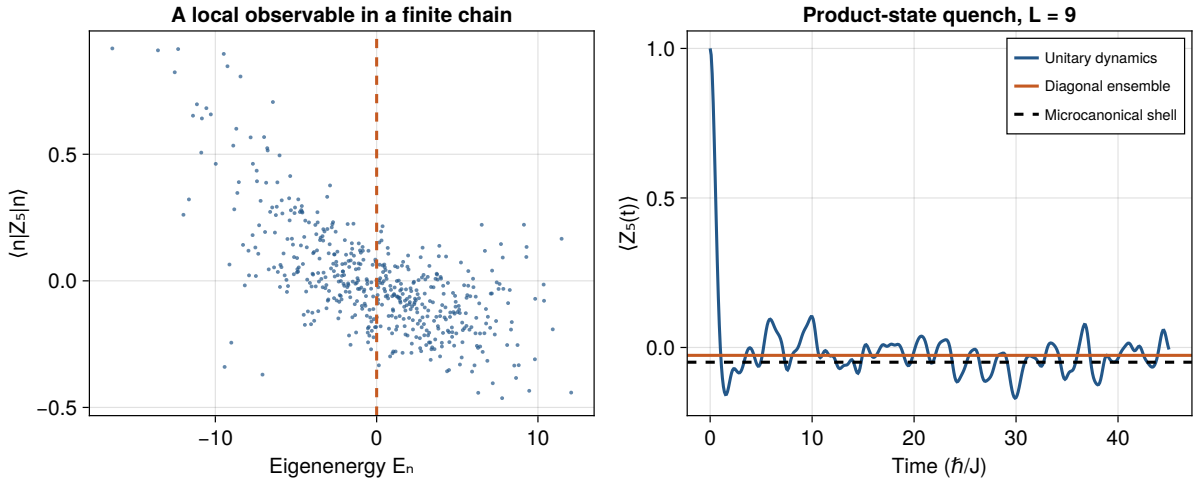


Figure 14.1: Independent finite-chain illustration: diagonal matrix elements of the central spin and exact product-state dynamics under the Hamiltonian specified in the text. The lines compare unitary dynamics, the diagonal ensemble, and an energy-window ensemble.

The raw eigenstate table contains energies, diagonal observable values, and initial probabilities; the time series contains the dynamics. The diagnostic file reports the chosen bit state, shell count, energy width, effective dimension $d_{\mathrm{eff}} = 1/\sum_n p_n^2$, and the gap-nondegeneracy expression for the long-time variance. The code does not certify that all energy gaps are nondegenerate; that expression is identified by its assumption.

The script checks Hermiticity, the eigensystem residual, normalization, agreement of the energy computed in both bases, and $A(0) = 1$. These checks catch basis, eigenvector, and propagation errors independently of whether the dynamics look thermal.

Checkpoint. Why can the diagonal and microcanonical lines differ even when the eigenstate scatter is fairly smooth?

Worked answer. The diagonal ensemble weights the full energy distribution p_n, while the microcanonical average uses a particular shell. If the smooth curve $a(E)$ bends across the initial distribution, then $\sum_n p_n a(E_n) - a(E_0) \simeq a''(E_0)\sigma_E^2/2$. Finite eigenstate fluctuations and finite shell sampling add other errors. A broad distribution or a spectral-edge state need not be represented by a single temperature.

14.4 Experiment 3: classical phase-space structure

The script iterates the standard map

$$p_{n+1} = p_n + K \sin\theta_n \pmod{2\pi}, \qquad \theta_{n+1} = \theta_n + p_{n+1} \pmod{2\pi}.$$

Nine initial points are evolved for 1,800 steps at $K = 0.2$ and $K = 5$. The angle-momentum area is preserved: the kick and free-drift Jacobians each have determinant one. Putting momentum on a torus makes the plot compact; a physical kicked rotor can instead have unbounded momentum on a cylinder. Diffusion on that cylinder cannot be measured from momentum reduced modulo 2π.

Invariant curves, resonant islands, and broad chaotic regions can coexist. Visual coverage of a region by finitely many iterates does not prove ergodicity. To estimate a Lyapunov exponent, evolve a tangent vector with the map Jacobian, periodically renormalize it, and accumulate the logarithms of its growth factors. Compare trajectories started in different visible regions.

Worked extension. In the order (θ, p) the Jacobian is

$$M_n = \begin{pmatrix} 1 + K\cos\theta_n & 1 \\ K\cos\theta_n & 1 \end{pmatrix}, \qquad \det M_n = 1.$$

With $v_{n+1} = M_n v_n$, record $\ell_n = \ln\|v_{n+1}\|$, rescale to unit norm, and estimate $\lambda = \sum_n \ell_n/n$. Discard a transient and vary both trajectory length and initial conditions. Positive finite-time growth is a numerical diagnostic, not a proof that every trajectory is chaotic.

14.5 Experiment 4: distributions from a continuum model

The heating figure in Chapter 10 illustrates a specified Fokker–Planck model and an equilibrium sharing law. It does not simulate wall collisions. For the first panel choose $S(E) = \frac{1}{2}\ln E$, $B(E) = E^{3/2}$, and $A(E) = \sqrt{E}$, satisfying $2A = B\partial_E S + \partial_E B$. In the equivalent Itô equation,

$$\mathrm{d}E = \sqrt{E}\,\mathrm{d}t + E^{3/4}\mathrm{d}W_t.$$

Itô's rule with $x = \sqrt{E}$ gives

$$\mathrm{d}x = \frac{3}{8}\mathrm{d}t + \frac{1}{2}\sqrt{x}\,\mathrm{d}W_t.$$

The entrance-law solution from $x = 0$ is a gamma density of shape 3 and scale $t/8$. Transforming back produces

$$P(E,t) = \frac{\sqrt{E}}{4(t/8)^3} e^{-8\sqrt{E}/t}, \qquad E > 0.$$

Normalization follows by $E = x^2$, $\mathrm{d}E = 2x\mathrm{d}x$: the integral becomes the normalized gamma integral. Its mean is $\langle E\rangle = 3 \cdot 4(t/8)^2$. The continuum law becomes questionable near energies comparable to the microscopic driving scale; its entrance solution is a mathematical similarity profile that illustrates late-time scaling, rather than a microscopic derivation for a particle initially at rest.

The second panel takes densities of states $\Omega_1(E) \propto E^2$ and $\Omega_2(E) \propto E^2$. At fixed total energy, $u = E_1/E_{\mathrm{tot}}$ has $P(u) = 30u^2(1-u)^2$. Integrating the polynomial proves normalization; symmetry gives $\langle u \rangle = 1/2$. The sharing table permits a direct numerical check. Equal mean energy here follows from choosing identical densities of states. In general equilibrium equalizes inverse temperature.

14.6 A project that tests a claim

Extend the spin-chain calculation to $L = 7, 8, 9, 10$ and compare the central part of the spectrum. Specify the window and statistic before examining the result. One useful quantity is the variance of A_{nn} after subtracting a local smooth energy trend. Track the maximum deviation separately from the average deviation: these address different strengths of ETH. Another is the scale of off-diagonal matrix elements inside a narrow mean-energy bin, resolved by frequency.

Dense exact diagonalization uses memory of order $D^2 = 4^L$ and work of order D^3. Doubling a visible chain length is therefore far more expensive than doubling an ordinary array. Keep the first project small, preserve the data, and state the accessible sizes. A careful finite-size conclusion is more informative than an unsupported assertion of universal thermalization.

Guide to the review

This map uses the numbered topic inventory of D'Alessio *et al.* [1]. Each row identifies where to learn the corresponding subject here. The book supplies independent explanations and examples; the original review remains the source for its particular historical discussion and numerical datasets. All seven appendices are included in the topic map.

Introduction, chaos, and eigenstates

Review section	Topic	Book location
1	Thermalization problem and isolated experiments	Foundations, Quenches
2.1	Classical chaos, integrability, and phase space	Classical chaos
2.2	Random matrix ensembles and spectral statistics	Random matrices
2.2.1	Chaotic eigenfunctions	Random matrices
2.2.2	Matrix elements of operators	Random matrices, ETH
2.3	Berry–Tabor conjecture	Random matrices
2.4	Semiclassical limit and Berry random waves	Random matrices
3.1	Wigner–Dyson and Poisson examples	Many-body chaos, Numerical lab
3.2	Many-body eigenstate structure	Many-body chaos
3.3	Quantum chaos and entanglement	Many-body chaos
3.4	Delocalization in energy space	Many-body chaos
4.1	Meaning of quantum thermalization	Foundations, ETH
4.2	ETH ansatz	ETH
4.2.1	ETH implies thermalization under conditions	ETH
4.2.2	Quantum ergodic theorem	ETH
4.3.1	Diagonal and off-diagonal matrix elements in lattices	ETH, Quenches
4.3.2	Quench dynamics and stationary predictions	Quenches, Numerical lab

Thermodynamic laws and response

Review section	Topic	Book location
5.1	Setup and doubly stochastic transitions	Entropy and work
5.1.1	Master equations	Entropy and work
5.2.1	Infinite-temperature attractor and convergence conditions	Entropy and work, Proofs
5.2.2	Increase of diagonal entropy	Entropy and work
5.2.3	Kelvin formulation and passive states	Entropy and work, Proofs
5.3.1	Fundamental thermodynamic relation	Entropy and work
5.3.2	Fundamental relation versus energy accounting	Entropy and work
6.1.1	Gibbs-initial fluctuation theorems	Fluctuation theorems
6.1.2	Microcanonical and eigenstate versions	Fluctuation theorems
6.2	Open-system detailed balance	Fluctuation theorems
6.3	Energy drift and diffusion	Fluctuation theorems, Driven systems
6.4	Heating Fokker–Planck equation	Driven systems
6.5	Two or more conserved quantities	Fluctuation theorems, Response
6.6	Linear response and Onsager reciprocity	Response, Proofs
6.7	Nonlinear response coefficients	Response
6.8	Single-eigenstate fluctuation–dissipation relation	Response
6.9	Two-observable correlations	Response
7.1	Fluctuating chaotic cavity	Driven systems
7.2	Harmonic driving and energy-distribution transition	Driven systems
7.3	Energy exchange between finite systems	Driven systems

Integrability and auxiliary derivations

Review section	Topic	Book location
8.1	Maximum entropy with conserved charges	Integrability
8.1.1	Free spinless fermions	Integrability
8.1.2	Hard-core bosons	Integrability
8.2	Generalized eigenstate thermalization	Integrability
8.2.1	Ising chain and truncated charge ensembles	Integrability

Review section	Topic	Book location
8.3	XXZ quenches and charge completeness	Integrability
8.4	Prethermalization and kinetic equations	Prethermalization
A	Kicked rotor and dynamical localization	Mathematical appendix
B	Riemann zeta zeros and GUE	Mathematical appendix
C	Infinite-temperature attractor proof	Mathematical appendix
D	Birkhoff's theorem	Mathematical appendix
E	Passive-state work inequality	Mathematical appendix
F	Continuous drift–diffusion derivation	Mathematical appendix
G	Onsager derivation	Mathematical appendix

The numerical lab adds runnable finite-system calculations. The bibliography provides direct primary-source citations beyond the organizing review. Exercises and solutions are placed with the concepts they test.

References

The review motivates the topic map; primary research papers support individual developments. Bibliographic links point to the published source or an author preprint. All figures in this book are generated for this project.

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